# QED on $S^3$: Spectral Geometry of Perturbative Transcendentals

GeoVac Project

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**Abstract**

We compute perturbative QED spectral sums on the three-sphere $S^3$ using the Camporesi–Higuchi Dirac spectrum and the Hodge-1 (photon) spectrum with SO(4) vertex selection rules. Eight structural results emerge. (1) **T9 theorem:** the spectral zeta function of the squared Dirac operator $D^2$ is a two-term polynomial in $\pi^2$ with rational coefficients at every integer $s$, with no odd-zeta content there (half-integer $s \geq 5/2$ carries odd zeta values; Theorem 1). (2) **Even/odd parity discriminant:** operator order (2nd vs 1st) is the primary transcendental discriminant (conditional on the period conjecture, $\zeta(2k{+}1) \notin \mathbb{Q}(\pi)$)—even $s$ produces $\pi^{\text{even}}$, odd $s \geq 5$ produces $\zeta(3)$, $\zeta(5)$, etc. (Theorem 2). (3) $\chi_{-4}$ **identity:** $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s{-}2))$ at every integer $s \geq 2$, a concrete spectral-zeta to Dirichlet-$L$ bridge (Theorem 3). (4) **Self-energy structural zero:** the one-loop electron self-energy on $S^3$ vanishes identically at the ground state $n_{\text{ext}} = 0$ as a consequence of vertex parity (Theorem 4). (5) **Anomalous magnetic moment:** the one-loop vertex correction with a $q = 1$ probe photon yields $F_2/[\alpha/(2\pi)] = 1.084$ at the lowest external mode ($n_{\text{ext}} = 1$), approximately 8.4% above the flat-space Schwinger value, matching the Parker–Toms leading curvature correction within 0.4%. The next-order coefficient $c_3 = -5.946(3) \times 10^{-7}$ is nonzero at $200.9\,\sigma$, so the curvature expansion does not terminate at second order (Sec. 7.7). (6) **Depth-$k$ tower:** loop order bounds the motivic depth of the level-2 (alternating-MZV) content; it does not automatically mint new irreducibles. The two-loop $S_{\min}$ decomposes exactly into Hoffman double $t$-values (classical periods of $\mathcal{MT}(\mathbb{Z}[1/2])$; the prior "PSLQ-irreducible" claim was a basis-coverage artifact); the three-loop $S^{(3)} = 31.5726\ldots$ (rigorous bracket $[31.57063, 31.57300]$) has realized depth $\leq 2$: the catalogued depth-2 generators $t(5,1)$, $t(5,3)$, $t(7,1)$ enter the components at coefficients $\pm\frac{1}{2}$ and cancel in the total, where the sole surviving depth-2 constant is $\zeta(5,3) \in \mathcal{MT}(\mathbb{Z})$ (Proposition 4, Sec. 10.4). (7) $\zeta(3)$ **complementarity:** the degeneracy-weighted fine-structure shell sum $\mathcal{S}(n) = n(5n-4)/2$ and the Dirac mode degeneracy $g_n^{\text{Dirac}} = 2n(n+1)$ satisfy $\mathcal{S}(n) + g_n^{\text{Dirac}} = 9n^2/2$ exactly, so the Dirichlet series identity $D_{\mathcal{S}}(s) + D_g(s) = \frac{9}{2}\,\zeta(s{-}2)$ holds for all $s$: the $\zeta(s{-}1)$ terms—all odd-zeta content—cancel identically (Theorem 7). (8) **Fine-structure Euler sum cancellation and product survival rule:** the degeneracy-weighted Dirichlet sums $D_p$ at order $(Z\alpha)^{2p}$ in the exact Dirac hydrogen spectrum, computed through $D_6$ ($(Z\alpha)^{12}$, weight 11), satisfy a product survival rule: $\zeta(2) \times \zeta(\text{odd})$ cancels algebraically at every order, while exactly $\max(0, \lfloor(2p{-}5)/2\rfloor)$ products survive (Observations 1, 2). The cancellation mechanism is the $n^2$ Fock degeneracy — the same weighting that produces $F = \zeta(2) = D_{n^2}(4)$ in Paper 2. However, PSLQ finds no small-coefficient $\mathbb{Q}$-linear relation between Paper 2's $K = \pi(B + F - \Delta)$ and $D_2, \ldots, D_6$: $K$ lies in $\mathbb{Q}[\pi]$ ($B$ a rational Casimir trace, $F = \zeta(2)$, $\Delta$ rational), while the Sommerfeld sums carry odd-zeta content from the Euler–Zagier MZV algebra. A complementary *graph-native* formulation (Sec. 15) computes all one-loop diagrams as finite matrix traces on the GeoVac Fock graph, requiring no infinite sums, regularization, or renormalization. All graph-native QED quantities are $\pi$-free (algebraic); the projection exchange constant relating graph and continuum VP is *rational* at every finite truncation, confirming Paper 18's calibration-tier classification. The continuum self-energy structural zero (Theorem 4) is broken on the finite graph: the pendant-edge theorem (Proposition 5) proves $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$ exactly, never healing in the graph limit. The root cause

is that the graph photon is a scalar 1-cochain, lacking the spin-1 vector structure whose SO(4) selection rules enforce the continuum zero. The graph-native anomalous moment $F_2(t)$ is an even rational function of the hopping coupling $t$ over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$, with $F_2(\kappa)$ decreasing as a power law $\sim 3.5\, n_{\max}^{-0.57}$ ($F_2 = 2.353, 1.873, 1.589, 1.396, 1.253$ at $n_{\max} = 2, \ldots, 6$). A systematic selection rule census shows that only 1 of 8 continuum QED selection rules survives on the scalar Fock graph (Gaunt/CG sparsity). Building QED on the native Dirac graph with $(n, \kappa, m_j)$ node labels recovers 4 of 8 rules, establishing a three-tier partition: 1 always survives, 3 are spinor-recoverable (intrinsic to spinor quantum numbers), and 4 require vector photon structure (calibration tier). A five-track vector QED sprint (VQ-1–5) confirms this partition from a complementary direction: neither Pauli spin matrices at the vertex, nor direction-resolved edge Hodge decomposition, nor their combination recovers any of the 4 vector-required rules. However, the transverse photon propagator $G_T = (d_1 d_1^\top)^+$ built from Paper 30's plaquette boundary operator *does* recover the ground-state structural zero ($\Sigma_T(\mathrm{GS}) = 0$ exactly), refining the partition to 1+3+1+3: the plaquette-recoverable tier is topological (pendant-edge exclusion from closed walks), requiring no photon quantum numbers. Adding explicit vector photon quantum numbers $(q, m_q)$ with Clebsch–Gordan vertex coupling recovers 7 of 8 continuum selection rules for scalar electrons and **8 of 8 for Dirac electrons** (Sec. 15.12). The calibration cost is exactly one transcendental per loop: $\pi$, entering as the $S^2$ Weyl exchange constant $1/(4\pi)$ from Paper 18's taxonomy. For scalar electrons, the sole failure is Furry's theorem; for Dirac electrons, the diagonal vertex coupling $V(a, a, q, m_q) = 0$ vanishes identically by the *Dirac spinor phase constraint*: the $\alpha$-matrix is off-diagonal in large/small component space, and for $a = b$ the cross terms combine, via Hermiticity and the $i$-factor on the small component, into an imaginary part that $m_l$ conservation forces to vanish. This is a kinematic identity of the single-particle vertex, not the field-theoretic charge conjugation mechanism. For Dirac electrons, the 8 rules partition as 1–7–8: graph topology → angular momentum conservation → Dirac spinor phase constraint (zero additional cost). The VP projection constant $C$ does *not* serve as a universal graph-to-continuum projector: $C \times F_2$ diverges from $\alpha/(2\pi)$, confirming that different diagram topologies require independent projection factors. Comprehensive transcendental tables for $D(s)$, $D_{\mathrm{even}}(s)$, $D_{\mathrm{odd}}(s)$, and the one-loop self-energy are presented, all backed by the test suite.

# 1 Introduction

The GeoVac framework [1, 2] computes quantum-mechanical observables from the spectral data of operators on compact manifolds, on a discrete lattice of their quantum-number labels (on $S^3$ the scalar spectrum is the diagonal $SO(4)$ Casimir on those labels; the lattice's graph Laplacian does not converge to the $S^3$ Laplacian, Paper 7). On the three-sphere $S^3$, the Dirac operator has a discrete spectrum with known eigenvalues and degeneracies [12], and the framework produces structurally sparse qubit Hamiltonians [2] whose Pauli count is exactly linear in $Q$ across molecules at fixed basis ($N_{\mathrm{Pauli}} = 27.90\, Q$, exact global-$M_L$ rule).

Paper 18 [3] introduced an exchange-constant taxonomy that classifies where transcendental numbers enter quantum-mechanical computations: intrinsic (from angular structure), calibration (from the $S^3$ projection), and embedding (from the electron–electron cusp). The present paper extends this taxonomy to perturbative QED by computing spectral sums that probe the interaction structure of quantum electrodynamics on $S^3$.

All calculations in the paper use four modules in the `geovac` codebase: `qed_vacuum_polarization` (one-loop heat kernel), `qed_vertex` (two-loop sunset with vertex parity), `qed_self_energy` (one-loop self-energy and vertex correction), and `qed_three_loop` (three-loop chain topology with $O(N^3)$ factorization). Every stated numerical value is backed by a test in `tests/test_qed_*.py` (the four named modules' suites collect 150 tests; the full glob collects 645 as of 2026-07-04 — all passing). Two scope notes (2026-07-04): the Sommerfeld product-survival closed forms $D_4$–$D_6$ are pinned at value level in `tests/test_paper28_gap_closure_pins.py` (their Euler-sum derivation pins are a

named follow-up), and the flat-space consistency checks are frozen in `tests/test_qed_flat_limit.py`.

The central finding is that the $S^3$ spectral geometry produces the odd zeta values that appear in flat-space perturbative QED—$\zeta(3)$, $\zeta(5)$—together with vertex-parity content specific to $S^3$ (Catalan's $G$, Dirichlet $\beta(4)$, which do not appear in the flat-space $a_e$ through three loops), through a purely geometric mechanism (operator order plus vertex topology), together with multi-loop constants that climb the depth filtration of the classical level-2 (alternating) MZV ring: the two-loop $S_{\min}$ is identified in closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ (Sec. 9).

# 2 Spectral Data on $S^3$

## 2.1 Dirac spectrum

On the unit three-sphere, the Dirac operator has eigenvalues and degeneracies [12]

$$|\lambda_n| = n + \tfrac{3}{2}, \qquad g_n = 2(n+1)(n+2), \qquad n = 0, 1, 2, \ldots \tag{1}$$

The Dirac Dirichlet series is $D(s) = \sum_{n=0}^{\infty} g_n / |\lambda_n|^s$. Via Hurwitz zeta,

$$D(s) = 2\,\zeta(s-2, \tfrac{3}{2}) - \tfrac{1}{2}\,\zeta(s, \tfrac{3}{2}). \tag{2}$$

Using $\zeta(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$, this becomes

$$D(s) = 2(2^{s-2} - 1)\,\zeta_R(s-2) - \tfrac{1}{2}(2^s - 1)\,\zeta_R(s). \tag{3}$$

We define the even-$n$ and odd-$n$ sub-sums:

$$D_{\text{even}}(s) = \sum_{\substack{n=0 \\ n\,\text{even}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \qquad D_{\text{odd}}(s) = \sum_{\substack{n=1 \\ n\,\text{odd}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \tag{4}$$

so that $D(s) = D_{\text{even}}(s) + D_{\text{odd}}(s)$.

## 2.2 Hodge-1 (photon) spectrum

The transverse vector Laplacian (Hodge-1 operator) on $S^3$ has eigenvalues and degeneracies

$$\mu_q = q(q+2), \qquad d_q^T = q(q+2), \qquad q = 1, 2, 3, \ldots \tag{5}$$

(Labeling-convention note, 2026-07-04: this is the framework's $q$-labeling as implemented and documented in `geovac/hodge1_s3.py`; the literature's coclosed/transverse family — Ikeda–Taniguchi 1978, Camporesi 1990 — is indexed so that the eigenvalue reads $(k+1)^2$ with multiplicity $2k(k+2)$. Reconciling the two labelings at the operator level is the named hodge1/P28 follow-up, deferred as its own item; the module's anchor test documents the divergence.) The Ricci shift $+2$ relative to the scalar Laplacian eigenvalues $\ell(\ell+2)$ follows from the Bochner–Weitzenböck identity $\Delta_1 = \nabla^*\nabla + \mathrm{Ric}$ on $S^3$ (Ricci scalar $R = 6$ on unit $S^3$, giving $\mathrm{Ric} = 2g$). There are no zero modes ($\beta_1 = 0$ for $S^3$).

### 2.3 Vertex selection rule

The QED vertex on $S^3$ couples spinor harmonics at levels $n_1, n_2$ to a vector harmonic at level $q$ with the SO(4) selection rule:

$$|n_1 - n_2| \le q \le n_1 + n_2, \qquad n_1 + n_2 + q \text{ is odd.} \tag{6}$$

The parity constraint arises from the $\gamma^\mu$ coupling, which flips the chirality representation of $\mathrm{SO}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$.

The SO(4) Clebsch–Gordan channel count $W(n_1, n_2, q) \in \{0, 1, 2\}$ weights each allowed triple by the number of independent coupling channels (the two components of the transverse vector harmonic under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$).

## 3 Theorem 1: Squared Dirac Spectral Zeta (T9)

**Theorem 1** (T9: $D^2$ spectral zeta is $\pi^{\text{even}}$)**.** *The spectral zeta function of the squared Dirac operator on unit $S^3$ satisfies*

$$\zeta_{D^2}(s) = 2^{2s-1}\big[\lambda(2s{-}2) - \lambda(2s)\big], \tag{7}$$

*where $\lambda(2k) = (1 - 2^{-2k})\,\zeta_R(2k) = \text{rational} \times \pi^{2k}$. At every integer $s \ge 1$, $\zeta_{D^2}(s)$ is a two-term polynomial in $\pi^2$ with rational coefficients; no odd-zeta content appears at any integer $s \ge 1$. The statement is an integer-$s$ one: at half-integer $s \ge 5/2$ the same formula (7) carries odd zeta values (e.g. $\zeta_{D^2}(5/2) = 14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$), and $s = 1/2, 3/2$ are simple poles with rational residues.*

*Proof.* The squared Dirac eigenvalues are $\lambda_n^2 = (n + 3/2)^2 = (2n + 3)^2/4$, so $\zeta_{D^2}(s) = \sum_{n\ge 0} g_n \cdot (4/(2n + 3)^2)^s = 4^s \sum_{n\ge 0} 2(n + 1)(n + 2)/(2n + 3)^{2s}$. Substituting $m = 2n + 3$ (which runs over odd integers $\ge 3$), the sum becomes $\sum_{m\ge 3, m\,\text{odd}} (m^2 - 1)/2 \cdot 4^s/m^{2s}$. The $(m^2 - 1)/2$ factor splits as $(m^{2-2s} - m^{-2s})/2$, giving two Dirichlet-type sums over odd integers. Each odd-integer sum is $\lambda(2k) = (1 - 2^{-2k})\zeta_R(2k)$ with the first three odd values ($m = 1$) added or subtracted to correct the lower limit. By the Euler–Bernoulli formula, $\zeta_R(2k) = (-1)^{k+1}(2\pi)^{2k} B_{2k}/(2 \cdot (2k)!)$, so $\lambda(2k) = \text{rational} \times \pi^{2k}$. At integer $s$ both arguments $2s - 2$ and $2s$ are even, so odd-zeta values $\zeta_R(2k + 1)$ cannot appear; at half-integer $s \ge 5/2$ both arguments are odd and $\ge 3$, so $\lambda(2s - 2)$, $\lambda(2s)$ are rational multiples of odd zeta values (at $s = 1/2, 3/2$ an argument equals 1, a pole; at negative half-integer $s$ the values $\zeta(1 - 2k) = -B_{2k}/(2k)$ are rational). □

**Verification:** Symbolic identity confirmed in `sympy` for $s = 1, \ldots, 10$. Numerical agreement to $10^{-60}$ at $s = 2, \ldots, 20$. The integer-$s$ closed forms at $s = 2, 3$ and the half-integer value $\zeta_{D^2}(5/2) = 14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$ are checked against the direct spectral sum in `tests/test_paper28_t9_half_integer.py`.

Table 1: Squared Dirac spectral zeta $\zeta_{D^2}(s)$.

| $s$ | $\zeta_{D^2}(s)$ |
|---|---|
| 1 | $-\pi^2/4$ |
| 2 | $\pi^2 - \pi^4/12$ |
| 3 | $\pi^4/3 - \pi^6/30$ |
| 4 | $2\pi^6/15 - 17\pi^8/1260$ |

**Remark 1** (Where $\sqrt{\pi}$ lives: heat-trace coefficients, not $\zeta_{D^2}$ (Sprint Mixed-Tate Test, June 2026; corrected 2026-09-30))**.** *The $\sqrt{\pi}$ met in heat-kernel-side analyses of $\zeta_{D^2}$ does not live in the spectral zeta function at any $s$. It lives only in the heat-trace coefficients: the short-time expansion $\operatorname{Tr} e^{-tD^2} \sim \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2}$ carries the $(4\pi t)^{-3/2}$-type prefactor of an odd-dimensional heat kernel, and the Mellin transform $\zeta_{D^2}(s) = \Gamma(s)^{-1} \int_0^\infty t^{s-1} \operatorname{Tr} e^{-tD^2}\, dt$ divides each coefficient by $\Gamma(s)$ at its pole. At $s = 3/2$ this gives the residue $(\sqrt{\pi}/2)/\Gamma(3/2) = 1$, and at $s = 1/2$ the residue $(-\sqrt{\pi}/4)/\Gamma(1/2) = -1/4$: both rational. Away from the poles, the values at half-integer $s \geq 5/2$ are odd-zeta combinations (Theorem 1), not multiples of $\sqrt{\pi}$. At integer $s \geq 1$ the values are pure $\pi^{2k} \cdot \mathbb{Q}$ (Table 1 above). After the standard Connes–Chamseddine volume normalization (multiplication of the heat-trace coefficients by $(4\pi)^{d/2}$ with $d = 3$), the coefficients also cleanly enter $\mathbb{Q}[\pi^2]$: Paper 51 [8] Cor. 2.1 gives $a_0^{D^2} = 4\pi^2$, $a_1^{D^2} = -2\pi^2$, $a_k = 0$ for $k \geq 2$. At integer $s$ the Geo-Vac $S^3$ M2 sub-mechanism is therefore mixed-Tate over $\mathbb{Q}$ in the F–M sense (Paper 32 [7] § VIII Corollary), in the strictly smaller pure-Tate sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$.*

**Remark 2** (No functional equation for $\zeta_{D^2}$: the quadratic multiplicity is the obstruction (Sprint GB-2, 2026-05-29))**.** *Sprint RH-O (Apr. 2026) reported, by a 13,080-template numerical search, that the Dirac spectral zeta $D(s)$ admits no spectral-triple functional equation (best residual* 0.594 *vs. classical anchors at $10^{-51}$). The closed form* (7) *upgrades this to an* analytic *statement and pinpoints the mechanism. Since $\zeta_{D^2}(s) = D(2s)$, both $\zeta_{D^2}(s)$ and its reflection $\zeta_{D^2}(c-s)$ lie in the two-dimensional space* $\operatorname{span}\{\zeta_R(2s),\, \zeta_R(2s-2)\}$ *over the field of elementary (gamma/exponential/trigonometric) functions; a functional equation $\zeta_{D^2}(s) = \chi(s)\, \zeta_{D^2}(c-s)$ with $\chi$ elementary exists iff the two coefficient vectors are parallel, i.e.* $\det = AD - BC = 0$.

*The Riemann reflection $z \mapsto 1 - z$ matches the argument set $\{2s-2,\, 2s\}$ to its reflected partner only at the natural axis $c = 3/2$ (equivalently $c = 1/2$), the Dirac ground-state value $|\lambda_0| = 3/2$ — so the axis is* not *the problem. The obstruction is the multiplier: the two terms acquire the gamma ratio $\Gamma(2s)/\Gamma(2s-2) = (2s-1)(2s-2)$, which forces* $\det \neq 0$ *(verified* $|\det| = 2.8,\, 8.4,\, 60.3,\, 8.2$ *at $s = 1+i,\, \frac{3}{2}+3i,\, 2+5i,\, \frac{1}{2}+10i$; the candidate-multiplier ratio $\chi_1/\chi_2$ tracks $(2s-1)(2s-2)$). The exponent gap of* 2 *between the two $\lambda$-terms is produced* precisely *by the quadratic Weyl multiplicity $g_n = 2(n+1)(n+2)$: a constant-multiplicity (single-power) spectrum lies in a one-dimensional span, is trivially parallel, and* does *inherit a functional equation. The reflection algebra is cross-checked against direct evaluation of* (7) *to $10^{-41}$.*

Consequence (unification of two RH walls). *The "wrong Weyl law" wall (RH-N: $\gamma_n \sim \sqrt{n}$, harmonic-oscillator density) and the "no functional equation" wall (RH-O) are the* same *structural fact: both are consequences of the $S^3$ quadratic degeneracy. The very multiplicity $n^2$ that places the framework's even-$s$ values on the Bernoulli ladder (Paper 32's Bernoulli-ladder remark: $F = \zeta(2) = \pi^2 B_2$, etc.) is the multiplicity that denies $\zeta_{D^2}$ a critical line. This is consistent with — and sharpens — WH6: the framework's zeta carries $\zeta_R$'s special values but not its zero-controlling functional equation. Source:* `debug/sprint_gb2_zeta_dsq_functional_eq.py`, `debug/sprint_gb2_zeta_dsq_memo.md`.

## 4 Theorem 2: Even/Odd Parity Discriminant

**Theorem 2** (Parity discriminant)**.** *At even integer $s$, the Dirac Dirichlet series $D(s)$ is purely $\pi^{\text{even}}$. At odd integer $s$, $D(s)$ involves $\zeta_R(s-2)$ and $\zeta_R(s)$ (odd-zeta values; $s \geq 5$, since $\zeta_R(1)$ is a pole). Operator order (2nd vs 1st) is the primary transcendental discriminant, conditional on the period conjecture $\zeta(2k+1) \notin \mathbb{Q}(\pi)$: the second-order operators $\Delta_{\text{LB}}$ and $D^2$ produce $\pi^{\text{even}}$ exclusively at integer $s$ (this holds for $\Delta_{\text{LB}}$ and $D^2$, with spectra $n^2 - 1$ and $(n + 3/2)^2$, whose partial fractions have rational poles; it is not a property of every second-order operator — the spinor connection*

*Laplacian* $\nabla^*\nabla = D^2 - 3/2$ *has* $\zeta_{\nabla^*\nabla}(2) = \frac{1}{12}\big(14\,\pi\tan(\frac{\pi\sqrt{6}}{2})/\sqrt{6} + 5\pi^2\sec^2(\frac{\pi\sqrt{6}}{2})\big)$*, outside* $\mathbb{Q}[\pi^2]$*) (their spectral determinants are not covered:* $\zeta'_{D^2}(0) = 3\zeta_R(3)/(4\pi^2) + \frac{1}{2}\ln 2$*); first-order operators produce odd-zeta values at odd* $s$*.*

*Proof.* From Eq. (3), $D(s) = 2(2^{s-2} - 1)\,\zeta_R(s{-}2) - \frac{1}{2}(2^s - 1)\,\zeta_R(s)$. At even $s$, both $\zeta_R(s{-}2)$ and $\zeta_R(s)$ are Bernoulli-type: $\zeta_R(2k) = (-1)^{k+1}(2\pi)^{2k}B_{2k}/(2\cdot(2k)!) = \text{rational} \times \pi^{2k}$. At odd $s$, both $\zeta_R(s{-}2)$ and $\zeta_R(s)$ are genuine odd-zeta values ($\zeta(3)$, $\zeta(5)$, ...), conjecturally $\mathbb{Q}$-linearly independent of all $\pi^{2k}$ (open: Apéry's theorem gives the irrationality of $\zeta(3)$ and Rivoal's infinitely many irrational odd zeta values [27, 28]; neither gives independence from $\pi$). □

Table 2: Dirac Dirichlet series $D(s)$: PSLQ identifications at 80 dps.

| $s$ | $D(s)$ | Class |
|---|---|---|
| 4 | $\pi^2 - \pi^4/12$ | $\pi^{\text{even}}$ |
| 5 | $14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$ | odd-zeta |
| 6 | $\pi^4/3 - \pi^6/30$ | $\pi^{\text{even}}$ |
| 7 | $62\,\zeta(5) - \frac{127}{2}\,\zeta(7)$ | odd-zeta |
| 8 | $2\pi^6/15 - 17\pi^8/1260$ | $\pi^{\text{even}}$ |
| 9 | $254\,\zeta(7) - \frac{511}{2}\,\zeta(9)$ | odd-zeta |
| 10 | $17\pi^8/315 - 31\pi^{10}/5670$ | $\pi^{\text{even}}$ |

The appearance of $\zeta(3)$ at $s = 5$ is the $S^3$ analog of the $\zeta(3)$ that enters flat-space two-loop QED through the Schwinger–Rosner vertex correction [13].

## 5 Theorem 3: The $\chi_{-4}$ Identity

**Theorem 3** ($\chi_{-4}$ identity)**.** *For every integer* $s \geq 2$ *(the even/odd spectral sums converge for* $s \geq 4$*; at* $s = 2, 3$ *by Hurwitz continuation),*

$$D_{\text{even}}(s) - D_{\text{odd}}(s) \;=\; 2^{s-1}\big(\beta(s) - \beta(s{-}2)\big), \tag{8}$$

*where* $\beta(s) = L(s, \chi_{-4})$ *is the Dirichlet beta function.*

*Proof.* Express $D_{\text{even}}(s)$ and $D_{\text{odd}}(s)$ via Hurwitz zeta at quarter-integer shifts. With $m = n + 3/2$ and full degeneracy $g_n = 2(n+1)(n+2) = 2(m^2 - 1/4)$, the even-$n$ eigenvalues are $m = 2(k + 3/4)$ and the odd-$n$ ones $m = 2(k + 5/4)$, $k \geq 0$, so $D_{\text{even}}(s) = 2^{-(s-3)}\zeta(s{-}2, 3/4) - 2^{-(s+1)}\zeta(s, 3/4)$ and $D_{\text{odd}}(s) = 2^{-(s-3)}\zeta(s{-}2, 5/4) - 2^{-(s+1)}\zeta(s, 5/4)$. Two elementary Hurwitz identities: $\zeta(s, 5/4) = \zeta(s, 1/4) - 4^s$ (shift recurrence) and $\zeta(s, 3/4) - \zeta(s, 1/4) = -4^s\beta(s)$ (the Dirichlet beta definition). Together, $\zeta(s, 3/4) - \zeta(s, 5/4) = 4^s(1 - \beta(s))$; substituting into the difference, $D_{\text{even}} - D_{\text{odd}} = 2^{s-1}(1{-}\beta(s{-}2)) - 2^{s-1}(1{-}\beta(s)) = 2^{s-1}(\beta(s){-}\beta(s{-}2))$, which is (8). The Hurwitz decompositions and the closed form are verified bit-exactly against direct spectral sums at $s \in \{4, 5, 6, 7\}$ to 40 digits. □

**Verification:** Eq. (8) verified symbolically (`sympy` residual zero) at $s \in \{2, 3, \ldots, 10, 15\}$ and by PSLQ at 100-digit precision at every integer $s$ in the tested range. At $s = 4$, $D_{\text{even}}(4) - D_{\text{odd}}(4) = 8(\beta(4) - G)$ (since $G = \beta(2)$), consistent with the explicit decompositions (9)–(10) below.

### 5.1 Vertex-parity decomposition at $s = 4$

The two-loop sunset diagram couples spinor pairs $(n_1, n_2)$ subject to the vertex parity constraint. The even-$n$ and odd-$n$ sub-sums of $D(4)$ decompose as:

$$D_{\text{even}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} - 4G + 4\beta(4), \tag{9}$$

$$D_{\text{odd}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} + 4G - 4\beta(4), \tag{10}$$

$$D(4) = \pi^2 - \tfrac{\pi^4}{12} \qquad \text{(Catalan and } \beta(4) \text{ cancel)}. \tag{11}$$

Here $G = \beta(2) = \sum_{n=0}^{\infty}(-1)^n/(2n+1)^2 \approx 0.9159656$ is Catalan's constant and $\beta(4) \approx 0.9889446$ is the Dirichlet beta function at $s = 4$. The mechanism: vertex parity acts as the Dirichlet character $\chi_{-4}$ on the mode sum, and the quarter-integer Hurwitz shifts expose $L(s, \chi_{-4})$ in the even/odd difference. Verified by PSLQ at 80-digit precision with basis $\{1, \pi^2, \pi^4, G, \beta(4)\}$.

**Corollary 1.** *The $\chi_{-4}$ identity is a structural realization of the vertex-parity-as-$\chi_{-4}$ mechanism, but does not constrain the zeros of $\beta(s)$: a zero of $\beta$ at $s = \rho$ propagates to zeros of $D_{\text{diff}}$ at both $\rho$ and $\rho + 2$, consistent with but not enforcing $\Re\rho = 1/2$.*

**Remark 3** (Motivic home of Theorem 3)**.** *Each of $D_{\text{even}}(s)$, $D_{\text{odd}}(s)$, and their difference is a period of mixed Tate motives at level 4 over $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$, by direct application of Deligne's classification at $N = 4$ and the explicit basis of Glanois (Sprint M3 Cyclotomic Mixed-Tate, June 2026, memo `debug/sprint_m3_cyclotomic_mixed_tate_memo.md`; Paper 32 § VIII Corollary `cor:m3_cyclotomic_mixed_tate`; Paper 18 § III.7). The vertex-parity-as-$\chi_{-4}$ correspondence IS the Deligne–Glanois Galois descent from level 4 to level 2 on the framework's natural QED observables: the un-restricted $D(s) = D_{\text{even}} + D_{\text{odd}}$ lives in the smaller level-$\leq 2$ ring (Theorem 1 and Theorem 2); the $\zeta(s, 1/4)$ pieces in $D_{\text{even}}, D_{\text{odd}}$ individually cancel in the sum but survive in the difference, exposing the level-4 Dirichlet L-value content. Catalan $G = \beta(2)$ and $\beta(4)$ are periods in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at depth 1 but conjecturally not in $\mathcal{MT}(\mathbb{Z}[1/2])$ at any depth — the level-4 lift is structural to the vertex-parity mechanism.*

### 5.2 Three-axis taxonomy

The transcendental content of any QED spectral sum on $S^3$ is classified by three independent axes:

1. **Operator order.** Second-order $(D^2) \to \pi^{\text{even}}$ at integer $s$ (Theorem 1); first-order $(|D|) \to$ odd-zeta at odd $s \geq 5$ (Theorem 2).
2. **Bundle type.** Scalar vs spinor modulates coefficients but not the transcendental class (Paper 18 Sec IV).
3. **Diagram topology.** Vertex parity introduces Dirichlet $L$-function characters ($G$, $\beta(4)$) that are absent from unrestricted sums (Theorem 3).

## 6 Theorem 4: Self-Energy Structural Zero

The one-loop electron self-energy on $S^3$ (Feynman gauge) is:

$$\Sigma(n_{\text{ext}}) = \sum_{n_{\text{int}}, q} \frac{W(n_{\text{ext}}, n_{\text{int}}, q)\; g_{n_{\text{int}}}\; d_q^T}{|\lambda_{n_{\text{int}}}|^{2s_e}\; \mu_q^{s_\gamma}}, \tag{12}$$

where the sum is over internal electron mode $n_{\text{int}}$ and photon mode $q$ subject to the vertex selection rule (6), $s_e = 2$ (so $|\lambda|^4$ in the denominator), $s_\gamma = 1$, and $W$ is the SO(4) channel count.

**Theorem 4** (Self-energy structural zero)**.** $\Sigma(0) = 0$ *exactly. The one-loop self-energy vanishes identically at the Dirac ground state* $n_{\text{ext}} = 0$.

*Proof.* At $n_{\text{ext}} = 0$, the vertex selection rule requires $|0 - n_{\text{int}}| \leq q \leq 0 + n_{\text{int}}$ and $0 + n_{\text{int}} + q$ odd. The first condition forces $q = n_{\text{int}}$. The second then requires $2\,n_{\text{int}}$ to be odd, which is impossible for any integer $n_{\text{int}}$. Therefore $W(0, n_{\text{int}}, q) = 0$ for all $(n_{\text{int}}, q)$, and the sum is empty. □

This is the $S^3$ analog of the fact that the ground state of a compact system cannot emit into a vacuum mode: the vertex parity constraint is a spectral selection rule, not a dynamical cancellation.

### 6.1 Self-energy for excited states

For $n_{\text{ext}} \geq 1$, the self-energy is nonzero and convergent. We compute $\Sigma(n_{\text{ext}})$ at physical propagator powers ($s_e = 2$, $s_\gamma = 1$) for $n_{\text{ext}} = 0, 1, \ldots, 5$ at cutoff $n_{\max} = 30$.

Table 3: One-loop self-energy $\Sigma(n_{\text{ext}})$ on $S^3$, $n_{\max} = 30$.

| $n_{\text{ext}}$ | $\Sigma(n_{\text{ext}})$ |
|---|---|
| 0 | 0 (exact, Theorem 4) |
| 1 | 1.2854 |
| 2 | 1.7862 |
| 3 | 2.0843 |
| 4 | 2.2884 |
| 5 | 2.4398 |

The self-energy increases monotonically with $n_{\text{ext}}$, consistent with stronger coupling at higher momentum shells. At $n_{\text{ext}} = 1$, PSLQ at 50 dps identifies $\Sigma(1)$ as containing $\pi^{\text{even}}$ and Catalan $G$ components (vertex parity active on the asymmetric coupling to the $n_{\text{ext}} = 1$ shell).

### 6.2 Vertex correction

The one-loop vertex correction involves two vertex insertions:

$$\Lambda(n_{\text{ext}}) = \sum_{n_{\text{int}}, q_1, q_2} \frac{W(n_{\text{ext}}, n_{\text{int}}, q_1)\; W(n_{\text{int}}, n_{\text{ext}}, q_2)\; g_{n_{\text{int}}}\; d^T_{q_1}\; d^T_{q_2}}{|\lambda_{n_{\text{int}}}|^{2s_e}\; \mu^{s_\gamma}_{q_1}\; \mu^{s_\gamma}_{q_2}}. \tag{13}$$

This is $O(N^3)$ in the cutoff. Both $\Lambda(n_{\text{ext}})$ and $\Sigma(n_{\text{ext}})$ are backed by 35 tests in `tests/test_qed_self_energy.py`.
**Note:** The shell-summed vertex correction $\Lambda(n_{\text{ext}})$ defined above uses the CG-weight channel count $W$ and is $m_j$-independent (it gives $B = 0$ by the Wigner–Eckart theorem). The anomalous magnetic moment extraction requires the $m_j$-resolved three-point function developed in the next section.

## 7 Anomalous Magnetic Moment on $S^3$

### 7.1 Why the two-point vertex correction gives $B = 0$

The two-point vertex correction $\Lambda(n_{\text{ext}})$ defined in Eq. (13) sums over all internal modes and photon polarizations at *fixed* external quantum numbers $(n_{\text{ext}}, j_{\text{ext}}, m_j)$. On $S^3$, the vertex amplitude at each junction involves a product of four $\text{SU}(2)_L \times \text{SU}(2)_R$ Clebsch–Gordan coefficients. When the

photon polarizations $(m_{g,L}, m_{g,R})$ are summed over at both vertices, the SO(4) Wigner–Eckart theorem forces the result to be independent of $m_j$:

$$\Lambda(n_{\text{ext}}, j, m_j = +\tfrac{1}{2}) = \Lambda(n_{\text{ext}}, j, m_j = -\tfrac{1}{2}). \tag{14}$$

Therefore the spin-dependent piece $B_{2\text{pt}} = \Lambda(+\frac{1}{2}) - \Lambda(-\frac{1}{2}) = 0$ identically. The two-point vertex correction is a scalar under SO(4) rotations and cannot distinguish spin orientations. To extract the anomalous magnetic moment ($F_2$ form factor), an external probe must break the rotational symmetry.

### 7.2 Three-point vertex correction with probe photon

The anomalous magnetic moment is extracted from a three-point function in which an external probe photon of mode $q_{\text{probe}} = 1$ (the lowest transverse vector harmonic, the $S^3$ analog of a uniform magnetic field) is inserted between the two loop-photon vertices. The physical topology is:

$$\langle \text{ext}, m_j |\, \Gamma^{1-\text{loop}}_{\text{probe}} \,| \text{ext}, m_j \rangle = \sum_{\substack{n_{\text{int}},\, j_{\text{int}} \\ m_{j,\text{int}},\, m'_{j,\text{int}} \\ q_{\text{loop}}}} \frac{V_1 \;\cdot\; V_{\text{probe}} \;\cdot\; V_3}{|\lambda_{n_{\text{int}}}|^4 \; \mu_{q_{\text{loop}}}}, \tag{15}$$

where:

- $V_1$: amplitude for the external electron $(m_j)$ to emit a loop photon and transition to the internal state $(m_{j,\text{int}})$;
- $V_{\text{probe}}$: amplitude for the probe photon at $q = 1$ to scatter the internal electron from $m_{j,\text{int}}$ to $m'_{j,\text{int}}$, summed over probe polarizations;
- $V_3$: amplitude for the internal state $(m'_{j,\text{int}})$ to reabsorb the loop photon and return to the external state $(m_j)$.

Each vertex amplitude is a product of four Clebsch–Gordan coefficients coupling the $\text{SU}(2)_L \times \text{SU}(2)_R$ representations of source, target, and photon:

$$V = \sum_{m_{L,1}} C^{j_s}_{j_{s,L}\, j_{s,R}}(m_{L,1},\, m_j - m_{L,1};\, m_j) \cdot C^{j_t}_{j_{t,L}\, j_{t,R}}(m_{L,2},\, m_{R,2};\, m'_j) \cdot C^{j_{t,L}}_{j_{s,L}\, j_{g,L}}(m_{L,1},\, m_{g,L};\, m_{L,2}) \cdot C^{j_{t,R}}_{j_{s,R}\, j_{g,R}}(m_{R,1},\, m$$

(16)

where $m_{L,2} = m_{L,1} + m_{g,L}$ and $m_{R,2} = m_{R,1} + m_{g,R}$, and the sum runs over all $m_{L,1} \in \{-j_{s,L}, \ldots, +j_{s,L}\}$.

The probe photon at $q = 1$ has channels $(j_{g,L}, j_{g,R}) = (1, 0)$ or $(0, 1)$, corresponding to the two components of the transverse vector harmonic. The probe couples internal states $m_{j,\text{int}} \to m'_{j,\text{int}}$, enabling spin-dependent amplitudes that the polarization-summed two-point function cannot access.

The spin-dependent piece of the vertex correction is:

$$B = \Gamma^{1-\text{loop}}_{\text{probe}}(m_j = +\tfrac{1}{2}) - \Gamma^{1-\text{loop}}_{\text{probe}}(m_j = -\tfrac{1}{2}). \tag{17}$$

### 7.3 Tree-level normalization

The tree-level (zeroth-order) coupling of the external electron to the $q = 1$ probe photon provides the normalization for $F_2$ extraction. At the external state $n_{\text{ext}} = 1$, $j = 1/2$ (Camporesi–Higuchi

level $n_{\mathrm{CH}} = 1$, with $j_L = 1$, $j_R = 1/2$):

$$V_{\mathrm{tree}}(m_j = +\tfrac{1}{2}) = -V_{\mathrm{tree}}(m_j = -\tfrac{1}{2}) = 0.2790, \tag{18}$$

$$V_{\mathrm{magnetic}} = V_{\mathrm{tree}}(+\tfrac{1}{2}) - V_{\mathrm{tree}}(-\tfrac{1}{2}) = 0.5579. \tag{19}$$

The charge piece $[V_{\mathrm{tree}}(+\frac{1}{2}) + V_{\mathrm{tree}}(-\frac{1}{2})]/2 = 0$ vanishes—the $q = 1$ probe photon couples purely to the magnetic moment at tree level, as expected for the $S^3$ analog of a uniform magnetic field.

Analytically, $V_{\mathrm{magnetic}} = 2(\sqrt{2}/3 - \sqrt{3}/9)$, a pure algebraic number from the Clebsch–Gordan coefficients.

The Pauli form factor is then:

$$F_2 = \frac{B}{V_{\mathrm{magnetic}}}, \tag{20}$$

and the anomalous magnetic moment on $S^3$ is $a_e^{S^3} = F_2$.

### 7.4 Convergence and results

The three-point vertex correction converges rapidly. Table 4 shows the per-level contributions for the external state at $n_{\mathrm{ext}} = 1$.

Table 4: Per-level contributions to the anomalous magnetic moment on $S^3$. External state $n_{\mathrm{ext}} = 1$, $j = 1/2$. The $n_{\mathrm{int}} = 0$ contribution vanishes by the self-energy structural zero (Theorem 4). Contributions drop by approximately 100× per successive allowed level pair.

| $n_{\mathrm{int}}$ | $\Delta B$ | $B_{\mathrm{cumul}}$ | $F_2/[\alpha/(2\pi)]$ | Status |
|---|---|---|---|---|
| 0 | 0 | 0 | — | $\Sigma(0) = 0$ structural zero |
| 1 | $6.96 \times 10^{-4}$ | $6.96 \times 10^{-4}$ | 1.074 | 99% of final |
| 2 | $3.76 \times 10^{-6}$ | $7.00 \times 10^{-4}$ | 1.080 | |
| 3 | $2.85 \times 10^{-6}$ | $7.02 \times 10^{-4}$ | 1.084 | |
| 4 | $1.34 \times 10^{-7}$ | $7.025 \times 10^{-4}$ | 1.084 | |
| 5 | $1.25 \times 10^{-7}$ | $7.027 \times 10^{-4}$ | 1.084 | |
| 6 | $1.25 \times 10^{-8}$ | $7.027 \times 10^{-4}$ | 1.084 | |
| 7 | $1.36 \times 10^{-8}$ | $7.027 \times 10^{-4}$ | 1.084 | |
| 8 | $2.00 \times 10^{-9}$ | $7.027 \times 10^{-4}$ | 1.084 | |

Three structural features are visible:

1. **Rapid convergence.** The $n_{\mathrm{int}} = 1$ level contributes 99% of the final value. Each subsequent allowed level pair contributes roughly 100× less than the previous one.

2. **Vertex parity filtering.** The $n_{\mathrm{int}} = 0$ contribution vanishes by Theorem 4 (the loop-photon vertex at $n_{\mathrm{ext}} = 1$, $n_{\mathrm{int}} = 0$ requires $q_{\mathrm{loop}} = 1$ and $1 + 0 + 1 = 2$ is even, forbidden). The dominant contribution comes from $n_{\mathrm{int}} = 1$ (the nearest allowed level to the external state). Higher even-$n_{\mathrm{int}}$ levels contribute but are suppressed relative to odd-$n_{\mathrm{int}}$ neighbors.

3. **Curvature correction.** The ratio $F_2/[\alpha/(2\pi)]$ converges to 1.084, not to 1.

### 7.5 Headline result

At $n_{\max} = 12$ (converged to 5 significant figures; the value is already stable at $n_{\max} = 8$):

$$\frac{F_2}{\alpha/(2\pi)} = 1.084, \tag{21}$$

with $B = 7.027 \times 10^{-4}$ and $V_{\text{magnetic}} = 0.5579$.

The anomalous magnetic moment on unit $S^3$ exceeds the flat-space Schwinger value by approximately 8.4%. This excess is a curvature correction: on a three-sphere of radius $R$, the electron propagator samples a compact geometry where the shortest-wavelength virtual photon modes are modified relative to flat space. In the flat-space limit $R \to \infty$, the curvature correction vanishes and $F_2/[\alpha/(2\pi)] \to 1$, recovering the Schwinger result [10] exactly.

### 7.6 Mode dependence

The anomalous magnetic moment depends strongly on the external electron mode. Table 5 shows $F_2/[\alpha/(2\pi)]$ computed at $n_{\text{ext}} = 1, \ldots, 5$.

Table 5: Mode-dependent anomalous magnetic moment on unit $S^3$. Each row reports the tree-level magnetic coupling $V_{\text{mag}}$, the one-loop correction $B$, and the normalized form factor ratio. The $n_{\text{ext}} = 1$ result uses $n_{\max} = 12$ (converged to 5 digits); $n_{\text{ext}} = 2$–5 use $n_{\max} = 5$ (converged to 3 digits via $n^{-7}$ power-law tail). Exact algebraic forms for $V_{\text{mag}}$ and the dominant $B(n_{\text{int}} = 1)$ are available at each $n_{\text{ext}}$; all expressions are sums of square roots with rational coefficients (no transcendentals).

| $n_{\text{ext}}$ | $\lambda_{\text{ext}}$ | $V_{\text{mag}}$ | $B$ | $F_2/[\alpha/(2\pi)]$ | vs. Schwinger |
|---|---|---|---|---|---|
| 1 | 5/2 | 0.5579 | $7.027 \times 10^{-4}$ | 1.084 | +8.4% |
| 2 | 7/2 | 0.3893 | $2.772 \times 10^{-4}$ | 0.613 | −39% |
| 3 | 9/2 | 0.3001 | $1.296 \times 10^{-4}$ | 0.370 | −63% |
| 4 | 11/2 | 0.2445 | $6.940 \times 10^{-5}$ | 0.244 | −76% |
| 5 | 13/2 | 0.2064 | $4.139 \times 10^{-5}$ | 0.173 | −83% |

The physical mechanism is that the loop correction $B$ drops faster with $n_{\text{ext}}$ than the tree-level coupling $V_{\text{mag}}$. Across the five modes, $B$ spans a factor of 17× while $V_{\text{mag}}$ spans only 2.7×. The tree-level amplitude follows $V_{\text{mag}} \to (4/3)/|\lambda|$ asymptotically (Richardson-extrapolated to 4-digit agreement with 4/3), while the loop correction decays as a steeper power of $|\lambda|$. Higher-mode external electrons are "heavier" relative to the curvature scale ($\lambda = n + 3/2$ sets the effective mass), and the loop integral is progressively suppressed.

**Comparison with Parker–Toms.** The leading curvature correction to the anomalous magnetic moment in a general curved spacetime is [11]:

$$\frac{F_2}{\alpha/(2\pi)} = 1 + \frac{R_{\text{scalar}}}{12\,|\lambda|^2} + O(|\lambda|^{-4}), \tag{22}$$

where $R_{\text{scalar}} = 6$ on unit $S^3$ and $|\lambda|$ is the Dirac eigenvalue of the external state. At $n_{\text{ext}} = 1$ ($|\lambda| = 5/2$), this predicts $1 + 6/(12 \cdot 25/4) = 1 + 2/25 = 1.080$, which matches the computed 1.084 within 0.4%. The expansion parameter is $\xi = R_{\text{scalar}}/(12\,|\lambda|^2) \approx 0.08$ at $n_{\text{ext}} = 1$, small enough for the leading term to dominate.

However, the perturbative expansion in $1/|\lambda|^2$ diverges badly for the full mode-dependent curve: a two-term fit gives $c_1 \approx -20$, far from the Parker–Toms $c_1 = 1/2$. The discrepancy arises because the expansion (22) applies to a *single external mode at fixed curvature*, treating $R/\lambda^2$ as the small parameter. When we vary $n_{\text{ext}}$ at fixed $R = 1$, we change not only $|\lambda|$ but also the angular structure of *every* CG coefficient in the vertex amplitude and the internal sum. The resulting mode dependence is non-perturbative in $1/|\lambda|^2$.

The five computed values are well described by a power law:

$$\frac{F_2}{\alpha/(2\pi)} = (5.91 \pm 0.36)\ |\lambda|^{-1.84\pm0.06}. \tag{23}$$

The fitted exponent is $2.8\sigma$ from $-2$; the forced $p = -2$ fit has twice the RMS residual. Analysis of the dominant internal-mode contribution ($n_{\text{int}} = 1$) shows its local log-log slope converging toward $-2$ ($-2.57 \to -2.26 \to -2.16 \to -2.11$ across $n_{\text{ext}} = 1, \ldots, 5$), but higher internal modes contribute an increasing fraction at larger $n_{\text{ext}}$ (99% at $n_{\text{ext}} = 1$ vs. 69% at $n_{\text{ext}} = 3$), pulling the effective exponent above $-2$. Whether the true asymptotic exponent is exactly $-2$ (with finite-curvature corrections) or a distinct value remains an open question.

The power law implies $F_2 \to 0$ as $|\lambda| \to \infty$ on unit $S^3$. This is not inconsistent with the flat-space Schwinger limit: the flat-space limit is $R \to \infty$ at fixed physical mass (requiring $n \sim mR \to \infty$ simultaneously), not $|\lambda| \to \infty$ on a fixed-size sphere. Taking $|\lambda|$ large at fixed $R = 1$ corresponds to a heavy particle on a small sphere, where form-factor suppression is expected.

**Verification:** The computation uses exact `sympy` Clebsch–Gordan coefficients at every vertex, with no numerical approximation in the angular algebra. The propagator denominators are exact rationals ($|\lambda_n|^4$ and $\mu_q$ from Eqs. (1) and (5)). Computation scripts: `debug/alpha_g_minus_2_vertex.py`, `debug/alpha_g_minus_2_curvature_fit.py`.

### 7.7 Curvature expansion coefficients

The Parker–Toms expansion (22) at $n_{\text{ext}} = 1$ provides a high-precision extraction of the curvature coefficients. Writing

$$\frac{F_2}{\alpha/(2\pi)} = 1 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots, \qquad x \equiv \frac{1}{|\lambda|^2} = \frac{4}{25} \tag{24}$$

at $n_{\text{ext}} = 1$ ($|\lambda| = 5/2$), the leading coefficient $c_1 = R_{\text{scalar}}/12 = 1/2$ follows from the Parker–Toms theorem [11] for a Dirac field on unit $S^3$.

The converged $n_{\text{ext}} = 1$ spectral sum ($n_{\text{int}} = 0 \ldots 25$ with power-law tail correction, 13 significant figures) gives a corrected ratio of 1.0844529226552, hence $\delta \equiv F_2/[\alpha/(2\pi)] - 1 = 0.0844529226552$. Subtracting the leading term $c_1 x = 0.08$ and dividing by $x^2$:

$$c_2^{\text{apparent}} = \frac{\delta - c_1 x}{x^2} = 0.173942291220598 \pm 10^{-12}. \tag{25}$$

We take $c_2 = a + b\pi^2$ for rationals $a$, $b$ as a PSLQ ansatz. (T9, Theorem 1, constrains only the integer-$s$ values of $\zeta_{D^2}$; it does not constrain every one-loop quantity on $S^3$ — the one-loop determinant $\zeta'_{D^2}(0) = 3\zeta(3)/(4\pi^2) + \frac{1}{2}\ln 2$ is a counterexample, and so (by a PSLQ identification, Sec. 6; no regression test pins it) is the one-loop self-energy $\Sigma(1)$, reported to carry Catalan $G$; beyond one loop, so are the vertex-parity-restricted sums of Theorem 3 (two-loop sunset), which carry $G$ and $\beta(4)$ — so the ansatz is not a T9 consequence.)

PSLQ identification against the Paper 2 spectral invariants $B = 42$, $F = \pi^2/6$, $\Delta = 1/40$ [9] yields

$$\boxed{c_2 = \frac{2 - B\Delta - F\Delta - F/B}{5} = \frac{19}{100} - \frac{41\pi^2}{25200}}, \tag{26}$$

numerically 0.173942310300. The match to (25) is 8 digits (residual $1.9 \times 10^{-8}$), compared with 5.5 digits for the next-best candidate $(6\pi^2 - 58)/7$ and 3 digits for the purely rational 25/144.

**Structural decomposition.** The three Paper 2 invariants enter (26) through three bilinear combinations:

$$B\Delta = \frac{42}{40} = \frac{21}{20}, \qquad F\Delta = \frac{\pi^2}{240}, \qquad \frac{F}{B} = \frac{\pi^2}{252}. \tag{27}$$

The numerator $2 - B\Delta - F\Delta - F/B = \frac{19}{20} - \frac{41\pi^2}{5040}$ is rational plus a single $\pi^2$ term, as the ansatz assumes. The denominator 5 is a simple integer.

**High-precision $c_3$ extraction.** Extending the $n_{\text{ext}} = 1$ spectral sum to $n_{\text{int}} = 50$ with interleaved even/odd power-law tail correction (Hurwitz zeta analytic tail; $p_{\text{even}} \approx 7.62$, $p_{\text{odd}} \approx 7.71$) gives a converged ratio of 1.0844529207082, hence $\delta = 0.0844529207082$. The residual after subtracting $c_1 x + c_2 x^2$ is $-2.44 \times 10^{-9}$, yielding

$$c_3 = \frac{\delta - c_1 x - c_2 x^2}{x^3} = -5.946(3) \times 10^{-7}, \tag{28}$$

nonzero at $200.9\,\sigma$ (uncertainty from the tail-correction power-law fit). The convergence is slow: the estimate drifts by $\sim 10^{-8}$ per five internal levels at $n_{\text{int}} = 50$, so the true asymptote lies near $-5.95$ to $-5.97 \times 10^{-7}$. The ratio $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$ confirms the expansion is rapidly converging but *does not* terminate at second order. PSLQ identification against the Paper-2 invariant basis $\{1, \pi^2, \pi^4, B\Delta, F\Delta, F/B, c_2, c_2^2, c_1 c_2, \Delta, \Delta^2, B\Delta^2, F\Delta^2, F^2/B\}$ at $n_{\text{int}} = 50$ finds no relation with integer coefficients bounded by $10^4$; $c_3$ is not expressible in these invariants at current precision.

**Multi-$n_{\text{ext}}$ obstruction.** The curvature coefficients in (24) are defined for the $j = 1/2$ sector of the external Dirac mode. At $n_{\text{ext}} \geq 2$, the Dirac level carries multiple total angular momentum sectors ($j = 1/2, 3/2, \ldots, n_{\text{ext}} + 1/2$) with distinct form factors. The $j = 1/2$ component is a $2/[(n_{\text{ext}} + 1)(n_{\text{ext}} + 2)]$ fraction of the total degeneracy, shrinking as $\sim 1/n^2$; extracting universal curvature coefficients from a full trace requires Wigner–Eckart decomposition into multipole form factors, which is not yet implemented. Consequently, the high-precision extraction is currently restricted to $n_{\text{ext}} = 1$, where $j = 1/2$ is the unique sector.

**Paper 2 invariants in $c_2$: a numerical coincidence, not a derivation.** Since $\alpha$ cancels in the ratio $F_2/[\alpha/(2\pi)]$, $c_2$ is a pure geometric invariant of the Dirac spectral sum on $S^3$, fitted in the ansatz form $a + b\pi^2$ with rationals $a, b$ (an ansatz, not a T9 consequence; see above). The PSLQ identification (Eq. 26) writes $a$ and $b$ as a small-integer combination of $B$, $F$, $\Delta$, but this should be read with caution.

First, the form $(c_0 - c_1 B\Delta - c_2 F\Delta - c_3 F/B)/c_4$ has more free small-integer coefficients than the two real numbers $(a, b)$ it explains. Finding *some* small-integer combination matching $a + b\pi^2$ to high precision is expected by chance at this search size; the cross-invariant beats the next-best PSLQ candidate $(6\pi^2 - 58)/7$ by roughly 2.4 digits, which is suggestive but not conclusive.

Second, and decisively: the *same* form does not predict $c_3$. Eq. (28) measures $c_3 = -5.946(3) \times 10^{-7}$ at $200\sigma$, and the extended Paper-2 invariant basis $\{1, \pi^2, \pi^4, B\Delta, F\Delta, F/B, c_2, c_2^2, c_1 c_2, \Delta, \Delta^2, B\Delta^2, F\Delta^2, F^2/B$ admits no PSLQ relation at integer coefficients bounded by $10^4$. A genuine structural identity

should extend. This single-term failure is consistent with $c_2$'s identification being a numerical coincidence between two natural Tier-A spectral invariants of the same manifold, both built from $|\lambda_n|$ and $g_n$, rather than evidence of shared structural origin.

A genuine bridge between the Paper 2 invariants and the curvature expansion would require deriving $c_2$ (and ideally $c_3$) from the Schwinger–DeWitt heat-kernel expansion of the spinor on $S^3$, with $B$, $F$, $\Delta$ appearing as identifiable terms in the expansion rather than via post-hoc PSLQ matching. This derivation is not done in this work and is flagged as an open question.

Computation scripts: `debug/g2_c2_verification.py`, `debug/g2_c2_diagnostic.py`.

# 8 One-Loop Vacuum Structure

The one-loop QED effective action involves traces of $D^2$. The Seeley–DeWitt heat kernel expansion on unit $S^3$ (four-component spinors; the printed values include the $(4\pi)^{-3/2}$ prefactor) gives

$$a_0 = \sqrt{\pi}, \qquad a_1 = -\sqrt{\pi}/2, \qquad a_k = 0 \ (k \geq 2) \tag{29}$$

(exact in `sympy`, from the Vassilevich curvature polynomials with the Lichnerowicz endomorphism $E = -R/4$; twice the two-component closed form (31) below). An earlier version printed $a_0 = a_1 = \sqrt{\pi}$, $a_2 = \sqrt{\pi}/8$, produced by a sign error in the production coefficient function (corrected 2026-10-02, with the function; `tests/test_qed_vacuum_polarization.py`). The standard flat-space one-loop vacuum polarization coefficient and QED beta function are

$$\Pi = \frac{1}{48\pi^2}, \qquad \beta(\alpha) = \frac{2\alpha^2}{3\pi}. \tag{30}$$

These are the standard flat-space values, entered here rather than computed from the $S^3$ data (an earlier version said they were reproduced from $S^3$ spectral data alone; the code returns the standard constants). They are consistent with the flat-space limit, since the gauge-field term of the four-dimensional $a_2$ is local.

## 8.1 SD two-term exactness for the Dirac heat kernel

The Dirac heat kernel on unit $S^3$, $K(t) = \mathrm{Tr}\, e^{-tD^2} = \sum_{n=0}^{\infty} 2(n+1)(n+2)\, e^{-(n+3/2)^2 t}$, has a Seeley–DeWitt expansion that terminates at exactly two terms:

$$K(t) = \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2} + O(e^{-\pi^2/t}). \tag{31}$$

All higher SD coefficients $a_k = 0$ identically for $k \geq 2$. The proof uses the Jacobi theta modular identity $\theta_2(0, e^{-t}) = \sqrt{\pi/t}\ \theta_4(0, e^{-\pi^2/t})$, which maps the small-$t$ expansion to a large-$\pi^2/t$ regime where $\theta_4 \to 1 + O(e^{-\pi^2/t})$. Numerical verification at 80 dps (50,000 terms) confirms $a_2 = -3.88 \times 10^{-74}$ (numerical noise, not a physical coefficient), and the residual does not scale as $t^2$.

Setting $\mathrm{Tr}\, e^{-D^2/\Lambda^2} = K/\pi$ gives the depressed cubic

$$2\Lambda^3 - \Lambda = \frac{4(K/\pi)}{\sqrt{\pi}}, \tag{32}$$

with Cardano closed form $\Lambda = \sqrt[3]{A + \sqrt{A^2 - 1/216}} + \sqrt[3]{A - \sqrt{A^2 - 1/216}}$, where $A = (B + F - \Delta)/\sqrt{\pi} \approx 24.610$ (an earlier version printed a spurious factor 2; the value $\Lambda_\infty$ below used the correct

$A$). This yields $\Lambda_\infty = 3.7102454679060528505$ (50 dps). At this cutoff, a sharp spectral truncation includes exactly $40 = \Delta^{-1}$ modes ($n = 0, 1, 2$), since $|\lambda_2| = 3.5 < 3.71 < 4.5 = |\lambda_3|$.

The SD ratio $a_0(D^2)/a_0(\Delta_{\mathrm{LB}}) = 2 = \mathrm{rank}(\text{spinor bundle})$ clarifies the supertrace sprint F1 result (ratio = 4): the discrepancy is the Dirac vs. Weyl counting convention ($4 = 2 \times 2$).

*Structural conclusion:* the Connes–Chamseddine framework provides an exact 2-term functional form for the spectral action but does *not* provide a mechanism to fix $\Lambda$. The value $\Lambda = 3.710\ldots$ is determined by inverting the spectral action at $K/\pi$, not selected autonomously.

### 8.2 Parametric spectral action on $S^3_R$ and $\zeta_{\mathrm{unit}}(-k) = 0$

The two-term exactness theorem (31) has an equivalent formulation on the spectral-zeta side which makes the spectral action on the family $\{S^3_R : R > 0\}$ *exactly* two-term for any reasonable cutoff function $f$.

**The $\zeta_{\mathrm{unit}}(-k) = 0$ identity.** The spectral zeta of $D^2$ on unit $S^3$ admits a Hurwitz decomposition:

$$\zeta_{\mathrm{unit}}(s) = \sum_n 2(n{+}1)(n{+}2)\,(n+\tfrac{3}{2})^{-2s} = 2\,\zeta_H(2s{-}2,\,\tfrac{3}{2}) - \tfrac{1}{2}\,\zeta_H(2s,\,\tfrac{3}{2}). \tag{33}$$

At non-positive integers $s = -k$ ($k = 0, 1, 2, \ldots$), using $\zeta_H(-n, a) = -B_{n+1}(a)/(n+1)$:

$$\zeta_{\mathrm{unit}}(-k) = -\frac{2\,B_{2k+3}(3/2)}{2k+3} + \frac{B_{2k+1}(3/2)}{2(2k+1)}. \tag{34}$$

**Theorem 5** ($\zeta_{\mathrm{unit}}(-k) = 0$ for all $k \geq 0$)**.** *The right-hand side of* (34) *vanishes identically for every non-negative integer $k$. Equivalently,*

$$4(2k+1)\,B_{2k+3}(3/2) = (2k+3)\,B_{2k+1}(3/2) \qquad \forall\, k \geq 0. \tag{35}$$

*Proof sketch.* Via the Mellin transform $\mathrm{Tr}\, e^{-tD^2} = (1/2\pi i)\int \Gamma(s)\,\zeta_{\mathrm{unit}}(s)\,t^{-s}\,ds$, shifting the contour through the right half-plane picks up contributions from poles of $\Gamma(s)\zeta_{\mathrm{unit}}(s)$. The two-term exactness theorem (31) states that the asymptotic expansion at small $t$ has only the leading $t^{-3/2}$ and $t^{-1/2}$ terms, with no Taylor terms $t^k$ ($k \geq 0$). The Taylor coefficients would otherwise be sourced by poles of $\Gamma(s)$ at $s = -k$ with residue $(-1)^k/k!$, giving contributions proportional to $\zeta_{\mathrm{unit}}(-k)$. Their absence forces $\zeta_{\mathrm{unit}}(-k) = 0$. Verified symbolically in `sympy` for $k = 0, 1, 2, 3, 4, 5$. □

**Exact two-term spectral action on $S^3_R$.** For any cutoff $f$ with Mellin transform $\phi(s) = \int_0^\infty f(x)\,x^{s-1}\,dx$ that is meromorphic with poles only at non-positive integers (e.g. Gaussian $e^{-x}$, polynomial $e^{-x^2}$, sharp $\Theta(1-x)$), the spectral action on $S^3_R$ is

$$S(R, \Lambda) := \mathrm{Tr}\, f(D^2/\Lambda^2) = \phi(3/2)\,(\Lambda R)^3 - \tfrac{1}{4}\,\phi(1/2)\,(\Lambda R) + O(\text{exp small}) \tag{36}$$

modulo exponentially small corrections in $(\Lambda R)^2$. The two-term form is *exact*: there are no power-law corrections at any order. Corrections from poles of $\phi(s)$ at non-positive integers vanish identically because each one is proportional to $\zeta_{\mathrm{unit}}(-k) = 0$.

This is a clean structural consequence of two-term exactness: on the family of round $S^3_R$ at fixed cutoff function, the spectral action is EXACTLY of Einstein–Hilbert plus cosmological-constant form ($\phi(3/2)(\Lambda R)^3$ is the cosmological-constant slot via $\mathrm{Vol}(S^3_R) \propto R^3$; $-\frac{1}{4}\phi(1/2)\Lambda R$ is the Einstein–Hilbert slot via $R_{\mathrm{scalar}} \cdot V \propto R$).

**Formal extremum and the CC spectral-action reading.** Three natural cutoffs give:

| Cutoff | $\phi(3/2)$ | $\phi(1/2)$ | $u_{\rm crit} = \sqrt{\phi(1/2)/(12\,\phi(3/2))}$ |
|---|---|---|---|
| Gaussian $e^{-x}$ | $\sqrt{\pi}/2$ | $\sqrt{\pi}$ | $1/\sqrt{6} \approx 0.408$ |
| Polynomial $e^{-x^2}$ | $\frac{1}{2}\Gamma(3/4)$ | $\frac{1}{2}\Gamma(1/4)$ | $\approx 0.497$ |
| Sharp $\Theta(1-x)$ | $2/3$ | $2$ | $1/2$ |

The asymptotic spectral action $S_{\rm asymp}(u) = Au^3 - Bu$ with $A = \phi(3/2) > 0$, $B = \frac{1}{4}\phi(1/2) > 0$ has a unique minimum at $u_{\rm crit} = \sqrt{B/(3A)} = O(1)$. Under the Connes–Chamseddine spectral action principle (Chamseddine–Connes 1997), $S_{\rm asymp}$ IS the effective classical action for the geometry; this gives a formal de-Sitter-like preferred radius $R_{\rm crit} \sim 1/\Lambda$, the GeoVac analog of CC's spectral-action-selects-metric mechanism.

**Honest scope.** The literal QM action $S_{\rm QM}(u) = \sum_n g_n f(\lambda_n^2/\Lambda^2)$ is monotonically increasing in $u$ for all three cutoffs and has no literal extremum: at $u \to 0$ all modes are Boltzmann-suppressed by the cutoff and $S_{\rm QM} \to 0$, while at $u \to \infty$ many modes pass the cutoff and $S_{\rm QM} \to Au^3$. The asymptotic $S_{\rm asymp} = Au^3 - Bu$ agrees with $S_{\rm QM}$ to machine precision for $u \gtrsim 2$ (Gaussian); the formal extremum at $u_{\rm crit} \sim 0.4$–$0.5$ sits in the IR regime where the asymptotic does NOT faithfully represent the QM exact sum.

Under the CC spectral-action reading (asymptotic = classical action), GeoVac on $S^3_R$ reproduces CC-style Einstein–Hilbert + cosmological constant with a stronger structural result: *the asymptotic terminates at two terms with no higher-curvature corrections at any order*, by Theorem 5. CC's continuum spectral action on a generic 4-manifold has higher-curvature corrections at every order; the GeoVac substrate eliminates them structurally. Sprint G1 driver: `debug/g1_spectral_action_S3_radius.py` and memo `debug/g1_spectral_action_S3_radius_memo.md`.

### 8.3 Extension to $S^3_R \times S^1_\beta$: factorization and 4D CC structure

The two-term exactness theorem (31) and the $\zeta_{\rm unit}(-k) = 0$ identity (34) PROPAGATE from $S^3$ to the thermal product $S^3_R \times S^1_\beta$ (Euclidean compactified time with antiperiodic boundary conditions for fermions). On the product, the spectral action acquires the exact 4D Connes–Chamseddine form: Einstein–Hilbert + cosmological constant with no higher-curvature corrections at any order.

**Factorization.** The Dirac operator on $S^3_R \times S^1_\beta$ is $D = \gamma^0 \otimes \partial_\tau + I \otimes D_{S^3}$ with $\{\gamma^0, D_{S^3}\} = 0$, so $D^2 = D^2_{S^3} \otimes I + I \otimes (-\partial_\tau^2)$. The heat kernel factorizes:

$$\operatorname{Tr} e^{-tD^2}\big|_{S^3_R \times S^1_\beta} = K_{S^3}(t, R) \cdot K_{S^1}(t, \beta). \tag{37}$$

$K_{S^3}(t,R) = K_{\rm unit}(t/R^2)$ has the two-term form (31). $K_{S^1}(t,\beta) = \sum_k e^{-t\omega_k^2}$ with Matsubara frequencies $\omega_k = 2\pi(k+1/2)/\beta$ admits Poisson resummation:

$$K_{S^1}(t,\beta) = \frac{\beta}{2\sqrt{\pi t}}\,\theta_4\Big(0, e^{-\beta^2/(4t)}\Big) = \frac{\beta}{2\sqrt{\pi t}} + O\Big(e^{-\beta^2/(4t)}\Big)\,. \tag{38}$$

**Two-term exactness on the product.** Both factors have clean asymptotic expansions modulo exp-small. Multiplying:

$$\operatorname{Tr} e^{-tD^2}\big|_{S^3_R \times S^1_\beta} = \frac{\beta R^3}{4}\,t^{-2} - \frac{\beta R}{8}\,t^{-1} + O\big(\text{exp small in } \min(R^2/t, \beta^2/t)\big)\,. \tag{39}$$

EXACTLY two power-law terms, no Taylor terms in $t$, no higher curvature corrections. The cross-product of "exp small" with "power-law" remains exp small.

**Corollary 2** ($\zeta_{S^3\times S^1}(-k)=0$)**.** *For the product space $S^3_R\times S^1_\beta$ with antiperiodic boundary conditions for fermions, $\zeta_{S^3\times S^1}(-k)=0$ for all non-negative integers $k$.*

*Proof.* $\zeta_{S^3\times S^1}(-k)$ corresponds to the Taylor coefficient of $\operatorname{Tr} e^{-tD^2}$ at $t=0$ via Mellin inversion (cf. proof of Theorem 5). By (39), $\operatorname{Tr} e^{-tD^2}$ has no Taylor terms in $t$, only power-law $t^{-2}, t^{-1}$ and exp-small corrections. □

**4D Connes–Chamseddine form.** For Gaussian cutoff $f(x)=e^{-x}$, the spectral action factorizes and matches the asymptotic to bit-exact precision at large $\Lambda$:

$$S(R,\beta,\Lambda)=\operatorname{Tr} e^{-D^2/\Lambda^2}=\frac{\beta R^3}{4}\,\Lambda^4-\frac{\beta R}{8}\,\Lambda^2+O(\text{exp small}). \tag{40}$$

This is the standard 4D CC form: $\Lambda^4$ cosmological constant + $\Lambda^2$ Einstein–Hilbert.

- $\Lambda^4$ coefficient: $\beta R^3/4=\operatorname{Vol}(S^3_R\times S^1_\beta)/(8\pi^2)$ where $\operatorname{Vol}=2\pi^2R^3\beta$.
- $\Lambda^2$ coefficient: $-\beta R/8=-\frac{1}{96\pi^2}\int R_{\text{scalar}}\sqrt{g}\,d^4x$ where $\int R_{\text{scalar}}\sqrt{g}\,d^4x=12\pi^2R\beta$ (only $S^3$ contributes; $S^1$ is flat).
- Higher $\Lambda$ powers: exactly zero by Corollary 2.

**Formal extremum and de Sitter vacuum.** The asymptotic $S(R,\beta,\Lambda)=(\beta R^3/4)\Lambda^4-(\beta R/8)\Lambda^2$ has $\partial S/\partial R=0$ at $R_{\text{crit}}\Lambda=1/\sqrt{6}$ (identical to the one-parameter G1 extremum). At $R=R_{\text{crit}}$, $\partial S/\partial\beta<0$, so the joint minimum is at $\beta\to\infty$. This corresponds to the decompactified $S^1=\mathbb{R}_\tau$ limit, i.e., the **zero-temperature de Sitter vacuum** with radius $R_{\text{crit}}\sim 1/\Lambda$. Same standard de Sitter prediction (and same cosmological-constant-problem gap) as CC's continuum spectral action.

**Stefan–Boltzmann lives in the exp-small.** The thermal free energy $F_{\text{SB}}(T)=-7\pi^2/180\cdot VT^4$ for free fermions does NOT appear in the UV asymptotic (40). It lives in the exp-small corrections from higher Matsubara modes ($m\geq 1$ in Poisson resum). The CC spectral-action expansion (UV / asymptotic) and Stefan–Boltzmann thermal physics (IR / convergent) are structurally distinct contributions, as already known in CC's continuum literature.

Sprint G2 driver: `debug/g2_spectral_action_S3_x_S1.py` and memo `debug/g2_spectral_action_S3_x_S1_mem`

### 8.4 Spinor-bundle specificity: scalar and TT-tensor heat traces

The two-term exactness theorem (31), which underwrites both the parametric spectral action on $S^3_R$ (§8.2) and the 4D thermal extension (§8.3), is *specific to the spinor bundle on $S^3$*. The scalar Laplacian and the Lichnerowicz Laplacian on transverse-traceless symmetric rank-2 tensors (the graviton sector) have heat traces with infinitely many Seeley–DeWitt coefficients, the standard CC continuum expansion.

This sharpens the structural reading of two-term exactness: it reflects the half-integer-shifted Dirac spectrum, NOT the geometry of $S^3$ per se. We derive closed-form heat traces for the three operators of interest.

**Three closed forms. Dirac** (Theorem (31), half-integer spectrum $(n+3/2)^2$ with multiplicity $2(n+1)(n+2)$, Jacobi $\theta_2$ inversion):

$$K_{D^2}(t)\big|_{S^3} = \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2} + O(e^{-\pi^2/t}). \tag{41}$$

**Scalar Laplacian** (integer spectrum $n(n+2)$ with multiplicity $(n+1)^2$, $n \geq 0$, Jacobi $\theta_3$ inversion + exponential prefactor):

$$K_{\Delta}(t)\big|_{S^3} = \frac{\sqrt{\pi}}{4}\, \frac{e^t}{t^{3/2}} + O(e^{-\pi^2/t}). \tag{42}$$

**TT-tensor Lichnerowicz** (integer spectrum $n(n+2)-2$ for $n \geq 2$ with multiplicity $2(n-1)(n+3)$, Jacobi $\theta_3$ inversion + discrete corrections from missing $n = 0, 1$ levels):

$$K_{\Delta_L^{TT}}(t)\big|_{S^3} = e^{3t}\left[\frac{\sqrt{\pi}}{2}\, t^{-3/2} - 4\sqrt{\pi}\, t^{-1/2} + 4\right] + 6e^{2t} + O(e^{-\pi^2/t}). \tag{43}$$

All three closed forms are verified bit-exactly (relative error $< 10^{-50}$ at $t \leq 0.1$) numerically in `debug/g3_graviton_lichnerowicz_S3.py`.

**Seeley–DeWitt coefficient comparison.** Expanding into standard form $K(t) = (4\pi t)^{-3/2} \sum_k a_k t^k$:

**Theorem 6** (Scalar SD closed form on $S^3$)**.** *For the scalar Laplacian on unit $S^3$, all Seeley–DeWitt coefficients are nonzero and given by*

$$a_k^{\Delta} = \frac{2\pi^2}{k!} \qquad \forall k = 0, 1, 2, \ldots \tag{44}$$

*Proof.* From (42), expanding $e^t = \sum_k t^k/k!$ gives $K_\Delta(t) = \sum_k (\sqrt{\pi}/(4k!))\, t^{k-3/2}$. Matching to $K_\Delta = (4\pi t)^{-3/2} \sum_k a_k t^k$ yields $a_k = (4\pi)^{3/2}\sqrt{\pi}/(4k!) = 2\pi^2/k!$. □

For the TT-tensor, the heat trace (43) contains BOTH half-integer-power terms (from the smooth Weyl asymptotic with $e^{3t}$ prefactor) AND integer-power terms (from the constant 4 and the discrete correction $6e^{2t}$). The standard SD form captures only the half-integer-power content; the integer-power content is "non-SD," reflecting the truncation of the spectrum to $n \geq 2$.

Half-integer-power coefficients from $e^{3t}[(\sqrt{\pi}/2)\, t^{-3/2} - 4\sqrt{\pi}\, t^{-1/2}]$:

$$\begin{aligned} a_0^{TT} &= 4\pi^2, \\ a_1^{TT} &= -20\pi^2 = (\sqrt{\pi}/2)(3) - 4\sqrt{\pi} \text{ scaled by } (4\pi)^{3/2}, \\ a_2^{TT} &= -78\pi^2, \quad \text{etc.} \end{aligned}$$

All nonzero, with $a_k^{TT} \propto \pi^2 \cdot p_k(3)$ for explicit polynomials $p_k$ from the $e^{3t}$ expansion.

**Structural distinction.**

| Operator | Spectrum shift | Two-term exact? | Higher SD coefficients |
|---|---|---|---|
| $D^2$ Dirac | half-integer | YES (Paper 28) | $a_k = 0$ for $k \geq 2$ |
| $\Delta$ scalar | integer | NO | $a_k = 2\pi^2/k!$ (Thm 6) |
| $\Delta_L^{TT}$ Lichnerowicz | integer (truncated) | NO | all nonzero + non-SD pieces |

**Implication.** The clean CC Einstein–Hilbert + cosmological constant structure on $S^3$ that emerges from the spectral action (36) (Sprint G1) and (40) (Sprint G2) is a feature of the *spinor bundle*, not of $S^3$ as a manifold. Gravitons (spin-2 TT modes) on $S^3$ have a heat trace with infinite higher-curvature SD coefficients plus non-SD discrete corrections from the spectrum's $n \geq 2$ cutoff. The graviton sector on the GeoVac substrate, were it constructed via CC-style metric variation, would inherit the standard CC continuum expansion with corrections at every order, NOT the clean two-term form of the Dirac sector.

This is consistent with the framework's structural-skeleton scope: clean structure lives in the Dirac sector sourced by the Fock projection (Paper 23 rigidity), with gravity-side coupling constants remaining external calibration data.

Sprint G3 driver: `debug/g3_graviton_lichnerowicz_S3.py` and memo `debug/g3_graviton_lichnerowicz_S3`

### 8.5 Graviton diagnostic: physical $(1,1)$ modes on the truncated substrate

The structural-test phase of G1–G3 establishes that the spectral action on the GeoVac substrate has EXACT Einstein–Hilbert + cosmological constant form on the Dirac sector but inherits standard CC continuum behavior on the spin-2 sector. The natural follow-up question: does the discrete substrate host graviton-like perturbations of $D$ at the operator level, or are gravitons structurally blocked?

**Setup.** Perturb $D = D_0 + \epsilon V$ where $V \in \mathrm{Herm}(\mathcal{H}_{n_{\max}})$. Expand the spectral action: $S[D_0+\epsilon V] = S[D_0] + \epsilon S^{(1)}[V] + \frac{1}{2}\epsilon^2 S^{(2)}[V,V] + O(\epsilon^3)$. The quadratic form $S^{(2)}$ is diagonal in the $D_0$ eigenbasis.

**Gauge–physical classification.** Modes $V = i[X, D_0]$ for Hermitian $X$ are tangent to the gauge orbit $D_0 \to U D_0 U^*$, and the spectral action is invariant under unitary conjugation. Therefore $S^{(2)}$ eigenvalues on gauge modes measure the curvature of the gauge orbit along the linear direction, not physical kinetic content. For the truncated CH Dirac, the gauge subspace equals the cross-sector off-block Hermitian matrices, and within-sector modes are physical.

**Spin-2 content.** The $SU(2)_L \times SU(2)_R$ irreps at level $n$ are $((n+1)/2, n/2)$ for +chirality and $(n/2, (n+1)/2)$ for −chirality. Within-sector tensor products $\rho \otimes \rho$ contain the graviton irrep $(j_L, j_R) = (1,1)$ for $n \geq 1$.

**Proposition 1** (Physical $(1,1)$ modes on the substrate)**.** *At every $n_{\max} \geq 1$, the second variation $S^{(2)}$ has PHYSICAL (within-sector) eigenmodes carrying $(j_L, j_R) = (1,1)$ irrep content with POSITIVE eigenvalue*

$$A_\lambda = a_\lambda \left( \frac{4\lambda^2}{\Lambda^4} - \frac{2}{\Lambda^2} \right), \quad a_\lambda = e^{-\lambda^2/\Lambda^2}, \tag{45}$$

*for each within-sector block at CH eigenvalue $|\lambda|$. Numerically verified by 5-point finite-difference stencil to rel diff $\sim 10^{-9}$ at $\Lambda^2 = 6$, $n_{\max} \in \{1,2,3\}$:*

| $\lvert\lambda\rvert$ | *blocks containing* $(1,1)$ | $A_\lambda$ |
|---|---|---|
| 5/2 | $S_3 \otimes S_3$, $S_4 \otimes S_4$ | +0.127 |
| 7/2 | $S_5 \otimes S_5$, $S_6 \otimes S_6$ | +0.133 |
| 9/2 | $S_7 \otimes S_7$, $S_8 \otimes S_8$ | +0.066 |

*Eigenvalues are stable across $n_{\max}$ (no sign changes), and the $|\lambda|$-dependence reflects the standard Gaussian-regulated kinetic behavior of the CC spectral action.*

**Honest scope.** Proposition 1 establishes the NECESSARY condition for graviton dynamics on the discrete substrate: physical (1, 1)-irrep eigenmodes exist and propagate with positive kinetic eigenvalue. SUFFICIENT conditions — Fierz–Pauli kinetic structure separating TT graviton (2 polarizations) from longitudinal/scalar modes (7 of the 9 within $(1,1)$), gauge invariance of the TT projector, propagator structure, and continuum graviton emergence via Paper 38-style state-space GH convergence — are not verified here; the J-blindness leg has since been completed (Paper 51, §Fierz–Pauli + the J-blindness theorem), and the remaining sufficiency legs constitute the extended G6 program. The $S^3$ Fock projection rigidity (Paper 23) fixes the background metric; graviton candidates here are operator-level perturbations of $D$, not direct metric perturbations.

*Structural reading.* Sprint G3 (§8.4) established that the spin-2 sector inherits standard CC continuum expansion. Sprint G6-Diag-Full establishes that this expansion's graviton modes are accessible on the discrete substrate at the necessary-condition level. The clean structure of G1/G2 (Einstein–Hilbert + cosmological constant exactly on Dirac sector) does NOT extend to gravitons; gravitons inherit CC continuum behavior with infinite higher-curvature corrections. Substrate-level existence of $(1,1)$ modes confirms gravitons are not structurally blocked.

**DOF count is a continuum-limit statement (Sprint GD-1).** The reduction of the 9-dimensional $(1,1)$ irrep to the 2 transverse-traceless physical polarizations does *not* occur at finite truncation. The continuum count is standard: under the diagonal $SU(2)$, $(1,1) = \text{spin-}2 \oplus \text{spin-}1 \oplus \text{spin-}0 = 5 \oplus 3 \oplus 1$, and diffeomorphism gauge (4) plus constraints (3) leave 2 TT. On the discrete substrate, however, the inner-automorphism (diffeomorphism) gauge tangent $i[X, D_0]$ has entries $(\lambda_b - \lambda_a)X_{ab}$, which vanish *within* a sector ($\lambda_a = \lambda_b$); the inner gauge is therefore *entirely cross-sector* (verified zero within-sector gauge entries at $n_{\max} = 1, 2, 3$). Within a within-sector $(1,1)$ block the gauge does not act, so all 9 modes are non-inner-gauge at finite truncation. The continuum diffeomorphism gauge acts within the tensor field at each point; the discrete inner gauge connects different $\lambda$-sectors. These reconcile only as $n_{\max} \to \infty$, when the cross-sector directions assemble into continuum diffeomorphisms that remove 7 of the 9 within-sector modes. Thus the graviton DOF reduction is a *continuum-limit phenomenon* (Paper 38-style state-space GH), not an internal-to-the-block projection — which is the structural reason the Fierz–Pauli decomposition is the extended G6 task and not a sprint-scale one. Source: `debug/sprint_gd1_graviton_dof_memo.md`.

**Which irrep carries the TT graviton (Sprint GD-4).** A rep-theory diagnostic sharpens *which* sector the necessary-condition test should examine. The helicity of an $SO(4) = SU(2)_L \times SU(2)_R$ irrep $(j_L, j_R)$ is $|j_L - j_R|$; the TT spin-2 graviton is helicity $\pm 2$, hence lives in $|j_L - j_R| = 2$ reps (the TT tensor harmonics on $S^3$ are $(k{+}2, k) \oplus (k, k{+}2)$, lowest $(2,0) \oplus (0,2)$). The $(1,1)$ irrep that Proposition 1 tested has $|j_L - j_R| = 0$ — *helicity zero*, the scalar/trace-class sector. The within-sector blocks $\text{sector}_i \otimes \text{sector}_i$ do contain the helicity-2 reps $(2,0) \oplus (0,2)$ (verified present at $n = 1, 2, 3, 4$), but these were not examined. *Caveat:* in Connes–Chamseddine the physical graviton enters through the $\delta D \leftrightarrow h_{\mu\nu}$ dictionary, which can shift irrep assignments — the *local* polarization space is $(1,1)$ while the *global* TT harmonic modes are $|\Delta j| = 2$. So GD-4 does not refute the necessary-condition result; it identifies a concrete gap: the extended G6 must either examine the $|\Delta j| = 2$ $\delta D$-modes directly or establish the dictionary showing the $(1,1)$ bilinear carries the physical graviton. Together with the GD-1 finding, the extended G6 has two sharply identified must-dos: (i) test the helicity-2 sector and fix the dictionary; (ii) realize the $9 \to 2$ gauge reduction in the continuum (state-space GH) limit. Source: `debug/sprint_gd4_graviton_irrep_memo.md`.

**Helicity-2 modes are positive too (Sprint GD-5).** The necessary-condition half of GD-4's concern is resolved directly. The coefficient $A_\lambda = a(\lambda)(4\lambda^2/\Lambda^4 - 2/\Lambda^2)$ is a function of the sector eigenvalue $\lambda$ alone (the spectral action sees only the degenerate-sector eigenvalue), hence *irrep-blind* within a block: every irrep — $(1,1)$, $(2,0)$, $(0,2)$, ... — inherits the same $A_\lambda$. Therefore the helicity-2 (TT-graviton-class) reps carry the *same positive* $A_\lambda$ as the $(1,1)$ ($+0.127, +0.133, +0.066$ at $n = 1,2,3$, $\Lambda^2 = 6$). So Proposition 1's positivity is *not* confined to the scalar/trace sector; it extends to the actual TT-graviton irrep class. *Honest scope:* the irrep-blindness that makes this immediate is exactly the G6 approximation — the *sufficient* condition (a TT kinetic structure *distinct* from the trace, Fierz–Pauli) requires lifting $A_\lambda$ to an irrep-resolved second variation, the extended-G6 task where the blindness must break. GD-5 closes the necessary condition for the TT sector; it does not close the TT-vs-trace distinction. Source: `debug/sprint_gd5_helicity2_memo.md`.

Sprint G6-Diag-Full driver: `debug/g6_diag_full.py`, companion first-pass `debug/g6_diag_quadratic_form.py` scoping memo `debug/g6_scoping_memo.md`, and full memo `debug/g6_diag_full_memo.md`.

### 8.6 Cigar spectral action and Bekenstein–Hawking entropy

The gravity arc beyond the structural-test phase (§8.2–§8.5) includes the gravity-side targets of Bekenstein–Hawking entropy ($S = A/(4G)$ from the cigar) and decompactified de Sitter vacuum (the $\beta \to \infty$ limit of §8.3). Sprint TD Track 4 (CLAUDE.md §2) already reproduced $T_H = 1/(8\pi M)$ from regularity of the cigar's $\tau$-circle. Sprint G4 first-pass establishes the entropy via the standard Gibbons–Hawking–York thermodynamic derivation and identifies the load-bearing structural mechanism.

**Cigar geometry.** Euclidean Schwarzschild with metric $ds^2 = (1-2M/r)\,d\tau^2 + (1-2M/r)^{-1}dr^2 + r^2 d\Omega_2^2$ outside the horizon $r > 2M$. Regularity at $r = 2M$ forces $\tau$-period $\beta = 8\pi M$. Horizon area $A = 4\pi r_h^2 = 16\pi M^2$.

**Bekenstein–Hawking from spectral action.** After Gibbons–Hawking subtraction of asymptotic flat-space reference, the Euclidean on-shell action of Schwarzschild is

$$I_E^{\mathrm{Schw}} = \frac{A}{4G}, \tag{46}$$

generated by the horizon boundary heat-kernel coefficient at the smooth conical tip. Bulk $R = 0$ (Ricci-flat) and the asymptotic boundary cancels against the flat-space subtraction. The thermodynamic entropy is

$$S_{\mathrm{thermo}} = -I_E + \beta\,\frac{\partial I_E}{\partial \beta} = \frac{A}{4G} = 4\pi M^2/G, \tag{47}$$

the Bekenstein–Hawking result. Symbolically verified in `debug/g4_cigar_BH_entropy.py` (sympy, residual exact zero).

**Load-bearing mechanism.** The structural piece producing $A/(4G)$ is the boundary heat-kernel coefficient at the smooth conical horizon, generating the GHY $(1/8\pi G)\oint K\sqrt{h}\,d^3x = A/(4G)$. This is the standard CC continuum derivation.

**GeoVac discrete substrate.** The GeoVac substrate (CH Dirac on truncated $S^3$ at finite $n_{\max}$) does NOT natively describe the cigar: the cigar's spatial section is $\mathbb{R} \times S^2$ (non-compact radial, $S^2$ instead of $S^3$), and the horizon is a smooth conical tip in the $(\tau, r)$ plane. Constructing the discrete

spectral triple analog on the cigar requires substantial new infrastructure: spatial $S^2$ truncation, non-compact radial regularization, horizon tip preservation under discretization, and verification that the boundary heat-kernel coefficient generates the correct GHY term. This is the full G4 program.

Sprint G4 first-pass: `debug/g4_cigar_BH_entropy.py` and `debug/g4_cigar_BH_entropy_memo.md`.

### 8.7 Decompactified $S^3 \times \mathbb{R}_\tau$ and de Sitter vacuum

Sprint G2 (§8.3) found that the joint extremum of the spectral action on $S^3_R \times S^1_\beta$ sits at $R_{\text{crit}}\Lambda = 1/\sqrt{6}$ (in $R$) and $\beta \to \infty$ (in $\beta$), corresponding to the zero-temperature de Sitter vacuum. Sprint G5 makes this $\beta \to \infty$ limit explicit by decompactifying the temporal direction.

**Heat-kernel decompactification.** On $S^3 \times \mathbb{R}_\tau$, the heat kernel factorizes per unit length of $\mathbb{R}_\tau$:

$$K_{\text{full}}(t, R) = K_{S^3}(t, R) \cdot K_{\mathbb{R}}(t), \quad K_{\mathbb{R}}(t) = \int \frac{dk}{2\pi}\, e^{-k^2 t} = \frac{1}{2\sqrt{\pi t}}. \tag{48}$$

This matches $K_{S^1}(t, \beta)/\beta$ from §8.3 in the $\beta \to \infty$ limit. Combined with the two-term exactness of $K_{S^3}$:

$$K_{\text{full}}(t, R) = \frac{R^3}{4}\, t^{-2} - \frac{R}{8}\, t^{-1} + O(\text{exp small}). \tag{49}$$

Two-term exactness propagates to the decompactified case.

**Spectral action density.** For Gaussian cutoff $f(x) = e^{-x}$:

$$s(R, \Lambda) := K_{\text{full}}(1/\Lambda^2, R) = \frac{R^3}{4}\Lambda^4 - \frac{R}{8}\Lambda^2. \tag{50}$$

**De Sitter vacuum.**

$$\frac{\partial s}{\partial R} = 0 \implies R_{\text{crit}}\Lambda = \frac{1}{\sqrt{6}}, \tag{51}$$

identical to G1 (§8.2). Minimum action density:

$$s_{\min} = s(R_{\text{crit}}, \Lambda) = -\frac{\Lambda}{12\sqrt{6}}. \tag{52}$$

Symbolically verified (sympy, $-\sqrt{6}\Lambda/72 = -\Lambda/(12\sqrt{6})$, residual exact zero).

**Physical reading.** The framework's preferred zero-temperature de Sitter vacuum has Planck-scale radius $R_{\text{crit}} \sim 1/\Lambda$ and vacuum action density $-\Lambda/(12\sqrt{6})$. This is the standard CC spectral-action-principle prediction, with the well-known cosmological-constant-scale gap. The decompactification is clean: G2's $\beta \to \infty$ minimum is manifest in G5 as the $\mathbb{R}_\tau$ vacuum.

Sprint G5: `debug/g5_decompactified_S3_x_R.py` and `debug/g5_decompactified_memo.md`.

### 8.8 Newton constant, cosmological constant, and background extremality

Sprint G7 consolidates the gravity arc's connection to standard Connes–Chamseddine gravity by deriving the framework's gravitational parameters in standard QFT language and verifying a structural background-extremality consistency check.

**Background extremality.** From the G6-Diag-Full physical $(1,1)$-irrep eigenvalues $A_\lambda = a_\lambda(4\lambda^2/\Lambda^4 - 2/\Lambda^2)$, the Gaussian-weighted integral over the spectrum vanishes exactly:

$$\int_0^\infty (4u^2 - 2)\, e^{-u^2}\, du = 4 \cdot \frac{\sqrt{\pi}}{4} - 2 \cdot \frac{\sqrt{\pi}}{2} = 0. \tag{53}$$

This confirms the standard background-extremality condition: the spectral action is stationary at the CH Dirac background, with sub-leading perturbations giving the actual graviton kinetic structure. Sympy-verified.

**Newton constant and cosmological constant from G2.** Matching G2's spectral action coefficients $(\beta R^3/4)\Lambda^4 - (\beta R/8)\Lambda^2$ (40) against the standard Einstein–Hilbert + cosmological constant action $S_{EH} = -(1/16\pi G_{\text{eff}}) \int (R_{\text{scalar}} - 2\Lambda_{cc})\sqrt{g}\, d^4x$ using $\text{Vol}(S^3_R \times S^1_\beta) = 2\pi^2 R^3 \beta$ and $\int R_{\text{scalar}} \sqrt{g} = 12\pi^2 R\beta$:

**Proposition 2** (Newton constant + cosmological constant)**.** *The Gaussian-cutoff spectral action on $S^3_R \times S^1_\beta$ identifies the framework's effective gravitational parameters as*

$$G_{\text{eff}} = \frac{6\pi}{\Lambda^2}, \qquad \Lambda_{cc} = 6\Lambda^2. \tag{54}$$

**Planck-scale identification.** Setting $\Lambda \to M_{\text{Planck}}$: $G_{\text{eff}} = 6\pi/M^2_{\text{Planck}} = 6\pi G_{\text{Newton}}$ (Planck-scale Newton's constant); $\Lambda_{cc} = 6M^2_{\text{Planck}} = 6/G_{\text{eff}}$ (Planck-scale cosmological constant).

**Cosmological constant scale gap.** The predicted ratio $\Lambda_{cc}/M^2_{\text{Planck}} = 6$ vs. the observed cosmological dark energy $\Lambda_{cc}/M^2_{\text{Planck}} \sim 10^{-120}$ gives a $\sim 120$ orders of magnitude gap. This is the standard CC cosmological-constant-scale problem, inherited from continuum Connes–Chamseddine and not GeoVac-specific. The framework's discrete substrate does not modify this leading-order prediction. Standard interpretation: cosmological constant scale is Class 1 calibration data (see `memory/external_input_three_class_partition.md`).

**Master Mellin engine tagging (a census and a reading).** Gravity content is M2 (Seeley–DeWitt heat-kernel coefficients $\sqrt{\pi} \times \bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, i.e. the pure-Tate ring once the $\sqrt{\pi}$ heat-trace normalisation is divided out; see Paper 18 §III.7). G2's normalised coefficients $1/(8\pi^2)$ and $-1/(96\pi^2)$ are M2-derived; the $(4\pi)^{-d/2}$ normalisation places them in $\pi^{-2} \cdot \mathbb{Q}$, outside the $k \geq 0$ pure-Tate ring. On the Dirac substrate, M2 terminates at two terms by (31); on scalar/tensor sectors, M2 is the full infinite series (§8.4, eq (44)). The gravity arc's content is thus the M2 sub-mechanism of the master Mellin engine, with substrate-specific termination on the Dirac bundle.

Sprint G7 driver: `debug/g7_extremality_newton.py`, memo `debug/g7_extremality_newton_memo.md`.

### 8.9 $S^2$ Dirac and the cigar's spatial spinor structure

Sprint G4-1 (first sub-sprint of the full G4 program) tests whether the framework's two-term exactness on $S^3$ Dirac (§8.3, Theorem 5) propagates to the $S^2$ Dirac that appears in the cigar's spatial section $\mathbb{R}_+ \times S^2_r$. Answer: NO.

$S^2$ **Dirac spectrum.** Camporesi-Higuchi for $S^d$ Dirac: $|\lambda_n^{S^d}| = n + d/2$. For $d = 2$:

$$|\lambda_n^{S^2}| = n + 1, \quad \text{(INTEGER shift)} \tag{55}$$

$$g_n^{\text{Dirac}}|_{S^2} = 4(n+1). \tag{56}$$

In contrast, $S^3$ Dirac has $|\lambda_n^{S^3}| = n+3/2$ (HALF-INTEGER shift) with $g_n^{\text{Dirac}}|_{S^3} = 2(n+1)(n+2)$.

**Heat trace asymptotic.** The $S^2$ Dirac heat trace is $K(t) = 4\sum_{u\geq 1} u\, e^{-u^2 t}$. The leading asymptotic

$$K_{S^2}^{\text{Dirac}}(t) \sim \frac{2}{t} + O(1) \tag{57}$$

matches the standard CC form $(4\pi t)^{-1}\cdot\dim_S \cdot \text{Vol}(S^2)$ with $\dim_S = 2$, $V = 4\pi$. Numerically verified to rel diff $10^{-5}$ at $t = 10^{-4}$ (`debug/g4_1_S2_dirac.py`).

**First subleading.** The Seeley-DeWitt coefficient $a_1 = \dim_S \cdot (R_{\text{scalar}}/6 - E_{\text{Lich}}) \cdot V$ on unit $S^2$ ($R = 2$, $E = R/4 = 1/2$):

$$a_1^{S^2,\text{Dirac}} = 2(1/3 - 1/2)\cdot 4\pi = -\frac{4\pi}{3} \neq 0. \tag{58}$$

The numerical extraction $\lim_{t\to 0}[K(t) - 2/t] = a_1/(4\pi) = -1/3$ matches at rel diff $\sim 10^{-3}$ across $t = 10^{-1}$ to $10^{-3}$.

**Structural distinction.** The two-term exactness theorem on $S^3$ Dirac comes from the half-integer spectrum shift, allowing Jacobi $\theta_2$ inversion to give exactly two power-law terms. $S^2$ Dirac (integer shift) uses Jacobi $\theta_3$ inversion + exponential prefactor, giving the standard CC infinite Seeley-DeWitt series with nonzero $a_k$ at every order.

This is consistent with Sprint G3 (§8.4): the scalar Laplacian on $S^3$ (integer spectrum from $n = 0$) also has full SD series with $a_k = 2\pi^2/k!$ (Theorem 6). The pattern crystallizes as:

| Operator | Spectrum shift | Higher SD coefficients |
|---|---|---|
| $S^3$ Dirac | half-integer $n + 3/2$ | $a_k = 0$ for $k \geq 2$ |
| $S^3$ scalar $\Delta$ | integer $n(n+2)$ | $a_k = 2\pi^2/k!$ (G3) |
| $S^2$ Dirac | integer $n + 1$ | $a_1 = -4\pi/3 \neq 0$, $a_k \neq 0$ |

**Implication for G4 full.** The cigar's spatial section $\mathbb{R}_+ \times S_r^2$ has the $S^2$ component inheriting standard CC continuum behavior with higher-curvature corrections at all orders. The discrete-substrate BH entropy derivation cannot exploit a clean two-term form on the spatial $S^2$; it must follow standard CC heat-kernel asymptotic structure. The load-bearing piece for $A/(4G)$ remains the horizon boundary $a_2$ coefficient (§8.6).

The framework's clean two-term exactness is *specific to the $S^3$ substrate sourced by the Fock projection*. It does not extend to other dimensions or the gravity arc's cigar geometry.

Sprint G4-1 driver: `debug/g4_1_S2_dirac.py` and memo `debug/g4_1_S2_dirac_memo.md`.

## 8.10 Conical defect / replica method for BH entropy

Sprint G4-2 (second sub-sprint of G4 full) builds on §8.9 to derive the Bekenstein–Hawking entropy via the standard conical-defect / replica method on the cigar's near-horizon geometry $D_\alpha^2 \times S_{r_h}^2$.

**Setup.** The cigar's near-horizon geometry is $ds^2 = d\rho^2 + \rho^2\, d\varphi^2 + r_h^2\, d\Omega_2^2$ with $\varphi \in [0, 2\pi)$ (smooth tip) or $\varphi \in [0, 2\pi\alpha)$ (conical singularity for $\alpha \neq 1$). The Bekenstein–Hawking entropy is extracted via the replica formula

$$S_{BH} = -\left.\frac{dI_E(\alpha)}{d\alpha}\right|_{\alpha=1}. \tag{59}$$

**Conical heat trace contribution.** The Sommerfeld/Cheeger formula for a 2D cone with apex angle $2\pi\alpha$:

$$K_{\text{cone}}(t) = K_{\text{smooth bulk}}(t) + \frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right) + O(t). \tag{60}$$

The $(1/12)(1/\alpha - \alpha)$ is the conical-defect tip contribution, vanishing at smooth $\alpha = 1$.

**Combined heat trace.** Using G4-1's $K_{S^2_{r_h}}(t) \sim 2r_h^2/t - r_h^2/3 + O(t)$ at small $t$, the conical-tip contribution to the cigar's heat trace is:

$$K_{\text{tip}}(t, \alpha) = \frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right)\left[\frac{2r_h^2}{t} - \frac{r_h^2}{3} + O(t)\right] \tag{61}$$

**Replica entropy derivation.** For Gaussian cutoff, $S = K(1/\Lambda^2)$ gives the conical-tip spectral action contribution

$$I_E^{\text{conical}}(\alpha, \Lambda, r_h) = \frac{r_h^2\Lambda^2}{6}\left(\frac{1}{\alpha} - \alpha\right) + O(1/\Lambda^2). \tag{62}$$

The $\alpha$-derivative at $\alpha = 1$:

$$\left.\frac{dI_E^{\text{conical}}}{d\alpha}\right|_{\alpha=1} = \frac{r_h^2\Lambda^2}{6} \cdot (-2) = -\frac{r_h^2\Lambda^2}{3}. \tag{63}$$

**Proposition 3** (Bekenstein–Hawking entropy from conical replica)**.** *The spectral action's conical-defect contribution at the cigar tip generates the Bekenstein–Hawking entropy via the replica formula:*

$$S_{BH} = -\left.\frac{dI_E^{\text{conical}}}{d\alpha}\right|_{\alpha=1} = \frac{r_h^2\Lambda^2}{3} = \frac{A\Lambda^2}{12\pi}, \tag{64}$$

*where $A = 4\pi r_h^2$. Matching to $S_{BH} = A/(4G_N)$ gives $G_N = 3\pi/\Lambda^2$.*

**Factor-of-2 calibration with G7.** The Newton constant from §8.8 (G2's spectral action matching) is $G_{\text{eff}} = 6\pi/\Lambda^2$. The factor-of-2 difference from $G_N = 3\pi/\Lambda^2$ is *not* a scalar-vs-Dirac conical-defect coefficient (the 2D Dirac and scalar cones share the coefficient $(1/12)(1/\alpha - \alpha)$; Fursaev–Miele 1996, confirmed by the G4-4c discrete extraction): it is a bookkeeping artifact forced by Wald's theorem — two-term exactness implies pure Einstein gravity, so action-$G$ and entropy-$G$ are the *same* constant, and the 2 lives in a 4D-bulk vs factorized-2D × 2D normalization mismatch to be closed by convention audit (Paper 51, Wald-forces-entropy-relation remark). The structural emergence of $A \cdot \Lambda^2$ from the conical tip is the load-bearing finding; the precise prefactor calibration is downstream.

**Discrete-substrate analog.** For G4-3 onwards (the extended program):

- Define $\alpha$-deformation of the discrete substrate at the tip
- Compute the conical contribution from the discrete Dirac
- Verify the discrete substrate reproduces the $(1/\alpha - \alpha)$ form
- Apply replica method to extract $S_{BH}$ from the discrete substrate spectral action

The conceptual framework is now in place; the full implementation program follows.
Sprint G4-2 driver: `debug/g4_2_conical_replica.py` and memo `debug/g4_2_conical_replica_memo.md`.

### 8.11 Cutoff-function dependence of the gravity predictions (Sprint G8)

Sprints G2 and G7 derived $G_{\text{eff}} = 6\pi/\Lambda^2$ and $\Lambda_{cc} = 6\Lambda^2$ assuming the Gaussian cutoff $f(x) = e^{-x}$. Sprint G8 generalizes these to arbitrary cutoff functions and identifies which framework predictions are cutoff-independent (structural) versus cutoff-dependent (calibration).

**Mellin transform of the cutoff.** For a general cutoff function $f$, the spectral content is captured by the Mellin transform:

$$\phi(s) := \int_0^\infty f(x)\, x^{s-1}\, dx. \tag{65}$$

The relevant moments for spectral action matching are $\phi(1)$ (Einstein-Hilbert scale) and $\phi(2)$ (cosmological-constant scale).

**General formulas.** From the G2 spectral action on $S^3_R \times S^1_\beta$,

$$S(R, \beta, \Lambda) = \phi(2) \cdot \frac{\beta R^3}{4}\Lambda^4 - \phi(1) \cdot \frac{\beta R}{8}\Lambda^2, \tag{66}$$

matching against Einstein-Hilbert plus cosmological constant gives:

$$G_{\text{eff}}(f, \Lambda) = \frac{6\pi}{\phi(1)\,\Lambda^2}, \quad \Lambda_{cc}(f, \Lambda) = \frac{6\,\phi(2)}{\phi(1)}\,\Lambda^2, \tag{67}$$

$$R_{\text{crit}}\,\Lambda = \sqrt{\frac{\phi(1)}{6\,\phi(2)}}. \tag{68}$$

Table 6: Gravity predictions for three cutoff functions.

| Cutoff $f(x)$ | $\phi(1)$ | $\phi(2)$ | $G_{\text{eff}}\Lambda^2$ | $\Lambda_{cc}/\Lambda^2$ |
|---|---|---|---|---|
| Gaussian $e^{-x}$ | 1 | 1 | $6\pi$ | 6 |
| Polynomial $e^{-x^2}$ | $\sqrt{\pi}/2$ | 1/2 | $12\sqrt{\pi}$ | $6/\sqrt{\pi}$ |
| Sharp $\Theta(1-x)$ | 1 | 1/2 | $6\pi$ | 3 |

**Three cutoff examples.**

**No cutoff-independent gravity prediction.** Direct algebra on (67)–(68) shows that every combination of $G_{\text{eff}}$ and $\Lambda_{cc}$ depends on $\phi(1)$ and/or $\phi(2)/\phi(1)$:

- $G_{\text{eff}} \cdot \Lambda^2 = 6\pi/\phi(1)$ (depends on $\phi(1)$)
- $\Lambda_{cc}/\Lambda^2 = 6\,\phi(2)/\phi(1)$ (depends on ratio)
- $G_{\text{eff}} \cdot \Lambda_{cc} = 36\pi\,\phi(2)/(\phi(1)^2\Lambda^2)$ (depends on both)
- $R_{\text{crit}}\Lambda = \sqrt{\phi(1)/(6\,\phi(2))}$ (depends on ratio)

**Class 1 calibration data.** Per the framework's external-input three-class partition (see CLAUDE.md §1.7 and project memory), the cutoff function $f$ is *Class 1 calibration data* alongside Yukawas, $Z_2/\delta m$ multi-loop renormalization data, and the Born rule. The framework determines:

- the *structural form* of $S[D]$ (spectrum, Seeley-DeWitt coefficients),
- the *proportionality* of $G_{\text{eff}}$ and $\Lambda_{cc}$ to specific $\phi$-moments,
- the *functional form* of $G_{\text{eff}}(f)$, $\Lambda_{cc}(f)$, $R_{\text{crit}}(f)$.

The framework does *not* determine:

- the numerical value of $G_{\text{eff}}$ or $\Lambda_{cc}$ in absolute units,
- which cutoff function to use.

**Consistency with standard CC.** This is the standard situation in the Chamseddine-Connes spectral action [31, 32]: the cutoff function is external input, the framework determines the structural shape.

**Structural vs numerical scope.** Sprints G1, G2, G3, G4, G5, G6-Diag-Full produced *structural* findings (closed forms, theorems, mechanisms) that are cutoff-independent. Sprint G7 produced *numerical* predictions (specific values of $G_{\text{eff}}$, $\Lambda_{cc}$) that depend on the Gaussian cutoff choice. Sprint G8 clarifies that the framework's gravity-arc predictions are best understood as:

- structural skeleton: cutoff-independent, determined by GeoVac's spectral structure;
- numerical calibration: cutoff-dependent, proportional to $\phi$-moment combinations.

Sprint G8 driver: `debug/g8_cutoff_dependence.py`; memo: `debug/g8_cutoff_dependence_memo.md`.

### 8.12 Discrete warped-product substrate for the cigar (Sprint G4-3 opening)

Sprints G1 through G8 produced the continuum gravity arc: closed-form spectral action on $S^3_R$ and $S^3 \times S^1_\beta$, two-term exactness theorems, $S^2$ Dirac structure, Bekenstein-Hawking entropy via the conical-defect / replica method, and the cutoff-function dependence of the gravity predictions. Sprint G4-3 *opens* the discrete-substrate gravity arc: rebuild the gravity-arc results on GeoVac's discrete substrate rather than at the continuum level.

This sprint is *scoping + first-pass*: define the discrete substrate, identify the load-bearing pieces, and specify the sub-sprint sequence G4-3a through G4-6.

**Discrete substrate.** For the cigar's near-horizon limit $ds^2 = d\rho^2 + \rho^2 d\phi^2 + r_h^2 d\Omega_2^2$ (constant warp $r = r_h$):

$$\mathcal{G}_{\text{cigar}} = \mathbb{Z}_+(a)|_{N_\rho} \times \mathbb{Z}/N_\phi \times \text{Fock}(S^2, l_{\max}) \tag{69}$$

where:

- $\mathbb{Z}_+(a)|_{N_\rho} = N_\rho$-truncation of $\mathbb{Z}_+$ with lattice spacing $a$, $\rho_k = ka$;
- $\mathbb{Z}/N_\phi$ = discrete $\phi$-circle, $N_\phi$ sites;
- $\text{Fock}(S^2, l_{\max})$ = standard $S^2$ Fock projection with $l \le l_{\max}$ angular cutoff.

The conical-defect parameter is $\alpha = N_\phi/N_0$ where $N_0$ is the reference smooth-tip count.

**Constant-warp factorization.** At $r(\rho) = r_h$ constant, the Laplacian factorizes:

$$\Delta_{\text{cigar}} = \Delta_{D^2}(\rho, \phi) + \frac{1}{r_h^2}\,\Delta_{S^2}.$$

The two factors act on independent sectors, so by construction:

$$K_{\text{cigar}}(t) = K_{D^2}(t) \cdot K_{S^2_{r_h}}(t).$$

This factorization holds at the operator level on the discrete substrate.

**Sub-sprint sequence.** G4-3 opens the extended discrete-substrate track:

- **G4-3a** (this): constant-warp substrate + scoping
- G4-3a-cleanup: Hermitian discrete polar Laplacian (engineering)
- G4-3b: variable warp $r(\rho)$ for asymptotic Schwarzschild
- G4-3c: discrete conical-defect deformation ($N_\phi$ sweeps)
- G4-3d: continuum-limit heat-kernel asymptotics verification
- G4-4: warped Dirac spectrum
- G4-5: discrete replica method
- G4-6: full discrete-substrate $S_{BH}$ derivation

Estimated commitment: a substantial multi-stage program, well beyond sprint scale.

**Honest scope of G4-3a.** G4-3a is *scoping*: substrate definition, constant-warp factorization at operator level, and sub-sprint plan. The sprint-scale driver uses a naive non-Hermitian discrete polar Laplacian; numerical heat-trace values are quantitatively wrong but the structural framework (factorization, conical parameterization, continuum-limit targets) is intact. A Hermitian discrete radial Laplacian on $L^2(\rho\, d\rho)$ via standard symmetrization is sprint-scale engineering for G4-3a-cleanup.

**Connection to G4-2 continuum derivation.** G4-2 derived $S_{BH} = A\Lambda^2/(12\pi)$ at the continuum level via the conical-defect / replica method. G4-3 onwards rebuilds this on the discrete substrate. The target is to show:

1. Discrete substrate has analog conical-defect contribution $\to (1/12)(1/\alpha - \alpha)$ in continuum limit
2. Discrete Dirac on warped substrate $\to$ G4-1 $S^2$ Dirac at constant warp
3. Discrete replica method gives $S_{BH} \to A\Lambda^2/(12\pi)$

**Connection to G8 cutoff dependence.** The $\Lambda^2$ prefactor in $S_{BH}$ depends on the cutoff function $f$ per G8. The discrete substrate inherits this calibration: any discrete-substrate $S_{BH}$ derivation will have prefactor $\propto \phi(2)/\phi(1)$. The cutoff dependence is structural, not a feature of G4-3 specifically.

**Gravity-arc status.** Continuum gravity arc: G1–G8 complete (sprint-scale closure). Discrete-substrate gravity arc: G4-3a opens; G4-3b through G4-6 run to the full $S_{BH}$ derivation on the discrete substrate. This is the natural commitment for the next major project arc.

Sprint G4-3a driver: `debug/g4_3_warped_substrate.py`; memo: `debug/g4_3_warped_substrate_memo.md`.

# 9 Two-Loop Identified Constant: $S_{\min}$

**Correction note (round-2 dossier, 2026-06-10).** The earlier subsection heading "Extended PSLQ irreducibility" and the claim that $S_{\min}$ is "confirmed irreducible" were a basis-coverage artifact: all 15 PSLQ attempts (47-element and 100-element bases at 150–200 dps) never contained the level-2 weight-5–8 monomials ($\pi^4 \ln 2$, $\pi^6 \ln 2$, $\mathrm{Li}_5(1/2)$, $\mathrm{Li}_4(1/2)\ln 2$, $\zeta(3)\ln^2 2$, ...) that the correct room requires. The present text replaces those claims with the identified decomposition; see driver `debug/smin_dossier_round1.py` and the round-1 dossier `debug/smin_dossier_round1_memo.md` for the complete reduction chain.

The CG-weighted two-loop sunset sum is:

$$S_{\min} = \sum_{k=1}^{\infty} T(k)^2, \qquad T(k) = 2\,\zeta(2, k+\tfrac{3}{2}) - \tfrac{1}{2}\,\zeta(4, k+\tfrac{3}{2}). \tag{70}$$

This is a depth-2 multiple Hurwitz zeta value at half-integer shift $a = 3/2$, intrinsic to the Dirac spectrum on $S^3$.

## 9.1 200-digit computation

$S_{\min}$ is computed to 200 verified digits using `mpmath.nsum` with Levin $u$-transform acceleration. The first 80 digits agree with the Sprint 5 three-method verification (direct summation + explicit tail; Levin $u$-transform; Euler–Maclaurin 26-term asymptotic):

$$S_{\min} = 2.47993\,69380\,34222\,55441\,35795\,00082\,93821\,44688\ldots \tag{71}$$

### 9.2 Motivic identification: Hoffman double $t$-values

$S_{\min}$ decomposes *exactly* into Hoffman double $t$-values — classical level-2 alternating MZVs, periods of $\mathcal{MT}(\mathbb{Z}[1/2])$. Notation: $\lambda(s) = (1-2^{-s})\zeta(s) = \sum_{o\geq 1,\,\text{odd}} o^{-s}$; $t(b,c) = \sum_{o_2>o_1\geq 1,\,\text{odd}} o_2^{-b} o_1^{-c}$ (Hoffman [20] descending convention: first index $b$ on the larger summation variable).

**8-$t$-value decomposition** (verified to residual $1.8\times 10^{-129}$ at 130 dps, re-verified at 260 dps):

$$S_{\min} = 64\big[t(2,1) - t(4,1) - 3\,t(2,2) + 3\,t(4,2) - t(2,3) + t(4,3) + 3\,t(2,4) - 3\,t(4,4)\big] + 32\sum_{s=3}^{8} d_s\left(\lambda(s) - 1 - 3^{-s}\right), \tag{72}$$

where the diagonal coefficients are $d_3 = 1$, $d_4 = -3$, $d_5 = -2$, $d_6 = 6$, $d_7 = 1$, $d_8 = -3$.

**Stuffle reductions** (exact algebraic identities from the ordering decomposition $\lambda(b)\lambda(c) = t(b,c) + t(c,b) + \lambda(b+c)$):

$$t(2,2) = \tfrac{1}{2}\big(\lambda(2)^2 - \lambda(4)\big), \tag{73}$$

$$t(4,4) = \tfrac{1}{2}\big(\lambda(4)^2 - \lambda(8)\big), \tag{74}$$

$$t(2,4) + t(4,2) = \lambda(2)\lambda(4) - \lambda(6). \tag{75}$$

(The antisymmetric combination $t(2,4) - t(4,2)$ enters $S_{\min}$ with coefficient zero, so only the reducible symmetric combination appears.)

**Odd-weight depth-1 reductions** (verified at $\geq 1\times 10^{-130}$; both are verbatim rows of Hoffman [20] Appendix A):

$$t(2,1) = -\tfrac{7}{16}\,\zeta(3) + \tfrac{1}{8}\,\pi^2\ln 2, \tag{76}$$

$$t(2,3) = -\tfrac{31}{64}\,\zeta(5) + \tfrac{1}{16}\,\pi^2\,\zeta(3). \tag{77}$$

**Remaining odd-weight components — identified.** PSLQ at 220 dps against the full conjectural level-2 bases (dim 8 at weight 5; 19 product monomials at weight 7) identifies both remaining components (residuals $2.8\times10^{-221}$ and $3.6\times10^{-222}$; `debug/data/s3_pslq_stageA.json`):

$$t(4,1) = \tfrac{1}{96}\,\pi^4\ln 2 - \tfrac{1}{64}\,\pi^2\zeta(3) - \tfrac{31}{64}\,\zeta(5), \tag{78}$$

$$t(4,3) = \tfrac{1}{128}\,\pi^4\zeta(3) - \tfrac{5}{128}\,\pi^2\zeta(5) - \tfrac{127}{256}\,\zeta(7). \tag{79}$$

Both identified forms are likewise verbatim rows of Hoffman [20] Appendix A ($t(4,1)$ directly; $t(4,3)$ after converting his $t$-basis) — the PSLQ runs independently rediscovered the published table (grounding verified against the source PDF, 2026-06-11). Assembling the full decomposition yields the explicit closed form

$$S_{\min} = 8\pi^2\ln 2 - \tfrac{2}{3}\pi^4\ln 2 - 3\pi^2\zeta(3) + \tfrac{1}{2}\pi^4\zeta(3) - \tfrac{5}{2}\pi^2\zeta(5) + \tfrac{\pi^6}{4} - \tfrac{3}{2}\pi^4 - \tfrac{\pi^8}{96}, \tag{80}$$

verified against the Hurwitz-tail anchor at 130 dps (residual $1.7\times10^{-129}$; `debug/smin_final_assembly.py`). Two structural features are exact: $\zeta(7)$ cancels against the $\lambda(7)$ diagonal term, and all rational constants cancel to zero. Hence $S_{\min}$ lies in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$: the two-loop constant carries *no irreducible depth-2 content at all* — every double $t$-value in its decomposition reduces to depth $\leq 1$.

**Basis-coverage audit of the prior "irreducible" claim.** The 47-element basis (all 15 PSLQ failures at 150 dps) and the 100-element basis (200 dps) both contained zero level-2 monomials in the weight-5–8 rooms. Specifically, none of $\pi^4\ln 2$, $\pi^6\ln 2$, $\mathrm{Li}_4(1/2)\ln 2$, $\mathrm{Li}_5(1/2)$, $\mathrm{Li}_6(1/2)$, $\mathrm{Li}_7(1/2)$, $\zeta(3)\ln^2 2$, … were included. Of the conjectural weight-5 level-2 basis (dimension 8), those tests contained only 2 out of 8 elements ($\zeta(5)$ and $\pi^2\zeta(3)$); at weight 7, approximately 4 out of 21. The "PSLQ-irreducible" label was a coverage artifact, not a statement about motivic depth.

### 9.3 Inner photon sum: closed-form channel count

The inner photon sum $I(n_1, n_2) = \sum_{q\,\text{allowed}} W(n_1, n_2, q) \cdot d_q^T/\mu_q^p$ (at $p = 1$: $d_q^T/\mu_q = 1$) counts the total SO(4) CG-weighted vertex channels. Four structural properties:

(i) $I(0, n_2) = 0$ for all $n_2$ (structural zero: vertex parity at $n_{\text{ext}} = 0$ forces $2n_2$ odd, impossible).

(ii) $I(n_1, n_2) = I(n_2, n_1)$ (symmetric).

(iii) $I(n, n) = 2n - 2 = 2(n - 1)$ for $n \geq 1$.

(iv) **Closed form** (proved in dossier round-1c; verified against `geovac/qed_vertex.py` for all $n_1, n_2 \leq 8$ in the frozen test `tests/test_smin_decomposition.py`; the archived dossier run swept to $\leq 12$):

$$I(n_1, n_2) = 2\min(n_1, n_2) - 1 - [n_1 = n_2], \qquad n_1, n_2 \geq 1. \tag{81}$$

### 9.4 Exact sunset relation

The closed form (81) gives, by direct substitution $I = 2\min - 1 - \delta$ into the double sum:

$$S_{\text{sunset}} = 2\, S_{\min} - P^2 - Q, \tag{82}$$

where $S_{\text{sunset}} = \sum_{n_1, n_2 \geq 1} I(n_1, n_2)\, \phi(n_1)\, \phi(n_2)$ is the CG-exact two-loop sunset,

$$P = D(4) - \tfrac{64}{81}, \quad D(4) = \pi^2 - \tfrac{\pi^4}{12}, \tag{83}$$

$$Q = 4\,\zeta\big(4, \tfrac{5}{2}\big) - 2\,\zeta\big(6, \tfrac{5}{2}\big) + \tfrac{1}{4}\,\zeta\big(8, \tfrac{5}{2}\big)\,, \tag{84}$$

and the numerical value $S_{\text{sunset}} = 3.894516205197841044 55\ldots$ is verified at 60 dps in `tests/test_smin_decomposit`
The correction terms $P^2$ and $Q$ are depth-1 elements of $\mathcal{MT}(\mathbb{Z})$ (pure-Tate, $\pi$-even); $S_{\min}$ itself reduces to depth $\leq 1$ (Eq. (80)), so the two-loop sunset carries no irreducible depth-2 content.

## 10 Three-Loop Results

### 10.1 Factorized $O(N^3)$ algorithm

The three-loop iterated sunset has chain topology: three electron lines $(n_1, n_2, n_3)$ connected by two photon lines $(q_1, q_2)$, with vertex selection at each junction. The chain topology permits factorization into partial sums:

$$S^{(3)} = \sum_{n_2=0}^{N} \frac{g_{n_2}}{|\lambda_{n_2}|^a} \cdot L(n_2) \cdot R(n_2), \tag{85}$$

where

$$L(n_2) = \sum_{\substack{n_1, q_1 \\ \text{allowed}}} \frac{W(n_1, n_2, q_1)\; g_{n_1}\; d_{q_1}^T}{|\lambda_{n_1}|^a\; \mu_{q_1}^p}, \qquad R(n_2) = \sum_{\substack{n_3, q_2 \\ \text{allowed}}} \frac{W(n_2, n_3, q_2)\; g_{n_3}\; d_{q_2}^T}{|\lambda_{n_3}|^a\; \mu_{q_2}^p}. \tag{86}$$

$W(n_1, n_2, q)$ is NOT symmetric in $(n_1, n_2)$, so $L$ and $R$ are computed independently. This reduces the five-fold sum $O(N^5)$ to $O(N^3)$ (each of $L$, $R$ is $O(N^2)$ precomputed, then the outer sum is $O(N)$).

**Verification:** The factorized algorithm matches the direct $O(N^5)$ sum to $< 10^{-78}$ at $n_{\max} = 10, 15$ across four parameter combinations (default $a = 4, p = 1$; vertex-restricted; higher exponents $a = 6, p = 2$; with and without CG weights). The factorized algorithm at $n_{\max} = 50$ completes in $< 60$ seconds. Tests: 7 in `TestFactorizedMatchesDirect` and `TestFactorizedConvergence` (all passing; the frozen factorized-vs-direct gate is $10^{-25}$ — the $< 10^{-78}$ agreement quoted above is the measured value at the working precision, re-confirmed 2026-07-04).

### 10.2 Convergence analysis (corrected 2026-06-11)

At default exponents ($a = 4$, $p = 1$) with CG weights, the raw partial sums converge far too slowly for direct evaluation. An empirical power-law fit over $N \in [20, 120]$ gave

$$\Delta S^{(3)}(N) \approx 270 \cdot N^{-1.31}, \tag{87}$$

but the closure-sprint analysis showed the fitted exponent is an artifact: by the channel-count closed form (81) the inner partial sums grow as $L(n_2) = 4 \ln n_2 + O(1)$, so the increments follow $(\ln N)^2/N^2$ — a log-modulated law whose local log-slope $2 - 2/\ln N$ runs through 1.3–1.6 over the fit window and masquerades as a power law. The true remaining tail at $N = 120$ is $\approx 7$ units, i.e. *zero* reliable digits from raw partial sums (cf. Table 9: the $n_{\max} = 50$ partial sum is 15.23 against a limit of 31.57).

The reliable direct evaluation is a rigorous bracket with no series acceleration anywhere: the factorized sum has positive terms, so partial sums with the closed-form full-range inner sum give a strict lower bound, and the explicit inequalities $L(m) \le 4(\ln m + 1)$, $\phi(m) \le 2/(m+1)^2$ give an integral tail bound. At $N = 4 \times 10^6$,

$$S^{(3)} \in [31.57063,\ 31.57300] \tag{88}$$

(driver `debug/s3_closure_anchor.py`). The corrected decomposition assembly (Sec. 10.4) gives the canonical value, inside the bracket:

$$S^{(3)} = 31.57256\,12075\,12022\,75476\,19295\,45069\ldots \tag{89}$$

### 10.3 PSLQ identification attempts (historical)

At the fast-converging exponents $a = 8$, $p = 2$, approximately 20 reliable digits are available at $n_{\max} = 50$. PSLQ with an 18-element depth-3 MZV basis (including $S_{\min}$, $\zeta(3)^2$, $\pi^2 S_{\min}$, $G \cdot S_{\min}$, and all cross-products with $\pi^2$, $\zeta(3)$, $G$, $\beta(4)$) returns no relation. This route is superseded by the decomposition closure below.

The unrestricted three-loop sum factorizes exactly as $D(a)^3$: at $a = 4$, $D(4)^3 = (\pi^2 - \pi^4/12)^3 \approx 5.379$, purely $\pi^{\text{even}}$.

### 10.4 Decomposition closure: the four trailing-1 identifications

Notation: $t_3(b_1, b_2, b_3) = \sum_{o_1 > o_2 > o_3 \ge 1 \text{ odd}} o_1^{-b_1} o_2^{-b_2} o_3^{-b_3}$, the Hoffman triple $t$-value in the same descending convention as $t(b, c)$. The channel-count closed form (81) expands the CG-weighted chain sum into 12 triple $t$-values, 26 double $t$-values, and $\lambda$/boundary terms (exact in Fraction arithmetic at every truncation; `tests/test_s3_decomposition.py`).

Two errors in the stage-1 numerics were diagnosed and closed on 2026-06-11 (memo `debug/sprint_s3_closure_r`) both with one root cause: Levin series acceleration silently mis-converges on summands with logarithmic factors, with *precision-dependent* error. (i) The evaluator for trailing argument $b_3 = 1$

fed the odd-harmonic factor $\widetilde{H}(j) \sim \frac{1}{2}\ln j$ into the accelerator; $t_3(2,1,1)$ was off by $-4.7\times10^{-4}$ at 130 dps and $-2.9\times10^{-4}$ at 220 dps, poisoning the earlier evaluation $S^{(3)} \approx 30.615$ through its table coefficient $+2048$ — the corrected shift $+0.9572$ is accounted for term-by-term to $1.7\times10^{-15}$. (ii) The same disease invalidated the factorized-anchor evaluation 30.2197 ($L(m) \sim 4\ln m$); the bracket (88) replaces it. The repaired evaluator uses Abel summation over the harmonic weight, $t_3(b_1,b_2,1) = \sum_{k\geq1}(2k-1)^{-1}A(k)$ with $A(k) = \sum_{j\geq k} a_j$ — both sums log-free. With corrected values the stuffle witnesses pass at $5\times10^{-216}$ (R2; stage-1 residual $6.4\times10^{-12}$) and $8\times10^{-217}$ (R6; was $9.8\times10^{-20}$); the stage-1 residuals decompose exactly into the individual evaluator errors.

PSLQ at 220/340 dps against the complete conjectural level-2 weight-homogeneous bases identifies all four previously pending components (residuals $2.6\times10^{-216}$ to $7.3\times10^{-339}$ in the archived dossier run; the frozen regression `tests/test_s3_decomposition.py` re-gates the identifications at $10^{-30}$/45 dps):

$$t_3(2,1,1) = \tfrac{1}{2}\mathrm{Li}_4(\tfrac{1}{2}) + \tfrac{1}{48}\ln^4 2 + \tfrac{1}{24}\pi^2\ln^2 2 - \tfrac{19}{5760}\pi^4, \tag{90}$$

$$t_3(4,1,1) = \tfrac{1}{192}\pi^4\ln^2 2 - \tfrac{1}{64}\pi^2\ln 2\,\zeta(3) - \tfrac{11}{322560}\pi^6 - \tfrac{1}{2}\,t(5,1) - \tfrac{7}{128}\zeta(3)^2, \tag{91}$$

$$t_3(3,2,1) = \tfrac{3}{64}\pi^2\ln 2\,\zeta(3) - \tfrac{1}{9216}\pi^6 - \tfrac{1}{2}\,t(5,1) - \tfrac{49}{256}\zeta(3)^2, \tag{92}$$

$$\begin{aligned} t_3(3,4,1) = {} & \tfrac{1}{768}\pi^4\ln 2\,\zeta(3) + \tfrac{5}{128}\pi^2\ln 2\,\zeta(5) + \tfrac{61}{2903040}\pi^8 - \tfrac{1}{256}\pi^2\zeta(3)^2 \\ & + \tfrac{1}{2}\,t(5,3) - \tfrac{1}{2}\,t(7,1) - \tfrac{217}{512}\zeta(3)\zeta(5). \end{aligned} \tag{93}$$

Every component of the $S^{(3)}$ decomposition except the weight-10 graded piece ($W_{10} = -2048\,t_3(4,3,3) - 1024\,t_3(4,4,2) - 512\,t_2(8,2) - 1536\,t_2(4,6)$, the full four-term graded piece; level-2 basis dimension 89, not executed; parity heuristic predicts depth $\leq 2$) is now an identified element of the level-2 ring, with the genuine depth-2 generators $t(5,1)$ (weight 6) and $t(5,3)$, $t(7,1)$ (weight 8) entering at coefficients $\pm\frac{1}{2}$. **Realized motivic depth at three loops: $\leq 2$ on every identified component** — one rung above $S_{\min}$'s depth $\leq 1$, all catalogued, nothing new minted. Falsifier: `tests/test_s3_decomposition.py` (including the rigorous-bracket regression guard for the Levin/log failure mode).

**Full assembly and the weight-10 block.** Assembling the complete decomposition with all identified components (symbolic assembly verified against the canonical value (89) at residual $4.1\times10^{-197}$) yields a stronger statement than the entry-level reductions: *every depth-2 generator cancels in the total.* $t(5,1)$, $t(5,3)$, $t(7,1)$, and $\mathrm{Li}_4(\frac{1}{2})$ all carry total coefficient zero, as does standalone $\zeta(11)$ — the $S^{(3)}$ echo of $S_{\min}$'s exact $\zeta(7)$ cancellation. The identified part of $S^{(3)}$ is a polynomial in $\pi^2$, $\ln 2$, and odd zeta values alone. The remaining weight-10 block reduces, by exact Fraction Gaussian elimination on the complete weight-10 quasi-shuffle closure (rank 157 on 255 words; identity verified numerically at $7\times10^{-59}$), to

$$\begin{aligned} W_{10}/256 = {} & 4\lambda(10) - \lambda(2)\lambda(8) - 3\lambda(4)\lambda(6) - 4\lambda(3)\,a(4,3) - 2\lambda(4)\,a(4,2) \\ & + a(8,2) + 4\,a(7,3) - 3\,a(6,4) - 4\,t_3(2,4,4) - 8\,t_3(3,3,4), \end{aligned} \tag{94}$$

where $a(b,c) := t(b,c) - t(c,b)$ and $\lambda(s) = (1-2^{-s})\zeta(s)$; the lower-weight $a(4,2)$, $a(4,3)$ are identified objects, leaving *five* surviving generators: the antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ and the triples $t_3(2,4,4)$, $t_3(3,3,4)$. A structural obstruction protects this block from stuffle methods: the depth-$\leq 3$ relation content of the quasi-shuffle closure saturates already at the $\lambda\cdot t$ product stage, so no stuffle-product engineering at any depth can reduce it further.

**The weight-10 block closed (2026-06-12).** The missing input is the Charlton–Hoffman symmetry theorem [24] — level-2 regularized-double-shuffle content invisible to the stuffle-only system. Its generating-series form (their Thm. 2.21) at depth $m = 3$, $\underline{\phi} = \underline{0}$, extracted at the coefficient of $y_1^{a-1}y_2^{b-1}y_3^{c-1}$ by exact-Fraction series algebra, gives the reversal-pair sums with every product term explicit (each instance used was verified numerically at $10^{-78}$ or better; driver `debug/s3_w10_ident.py`, 24/24 gates):

$$t_3(4,4,2) + t_3(2,4,4) = \tfrac{5}{6144}\pi^4\zeta(3)^2 - \tfrac{5}{256}\pi^2\zeta(3)\zeta(5) - \tfrac{1}{256}\pi^2\zeta(5,3) + \tfrac{137}{99532800}\pi^{10}, \tag{95}$$

$$t_3(4,3,3) + t_3(3,3,4) = \tfrac{37}{12288}\pi^4\zeta(3)^2 - \tfrac{25}{1024}\pi^2\zeta(3)\zeta(5) - \tfrac{889}{2048}\zeta(3)\zeta(7) - \tfrac{1}{256}\pi^2\zeta(5,3) + \tfrac{251}{58060800}\pi^{10}, \tag{96}$$

where $\zeta(5,3) = \sum_{m>n\geq 1} m^{-5}n^{-3}$ is the classical weight-8 depth-2 MZV generator (conjecturally irreducible over products of single zetas), entering through the weight-8 layer of the theorem's $\zeta$-side product terms. The pair *differences* follow from four ordinary stuffles $t(x)\,t_2(y,z)$, $(x,y,z) \in \{(2,4,4),(4,2,4),(3,3,4),(3,4,3)\}$, whose pairwise combinations eliminate the auxiliary triples $t_3(4,2,4)$, $t_3(3,4,3)$. Solving the two $2\times 2$ systems identifies all four weight-10 triples individually:

$$t_3(2,4,4) = \tfrac{1}{4}a(8,2) - \tfrac{1}{4}a(6,4) - \tfrac{1}{6144}\pi^4\zeta(3)^2 - \tfrac{5}{512}\pi^2\zeta(3)\zeta(5) - \tfrac{1}{512}\pi^2\zeta(5,3) + \tfrac{373}{174182400}\pi^{10}, \tag{97}$$

$$t_3(3,3,4) = \tfrac{1}{4}a(7,3) - \tfrac{1}{4}a(6,4) + \tfrac{1}{12288}\pi^4\zeta(3)^2 + \tfrac{5}{1024}\pi^2\zeta(3)\zeta(5) - \tfrac{889}{4096}\zeta(3)\zeta(7) - \tfrac{1}{512}\pi^2\zeta(5,3) + \tfrac{251}{116121600}\pi^{10}, \tag{98}$$

each gated against its independently computed 200-digit value at residual $\leq 6\times 10^{-197}$ (likewise $t_3(4,4,2)$, $t_3(4,3,3)$).

**Collapse theorem and the full closed form.** Substituting Eqs. (97)–(98) into the full $S^{(3)}$ assembly, the three antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ *cancel identically*: the explicit $a$-content of the weight-10 corollary is exactly absorbed by the $a$-content of the two triples (symbolic verification in the frozen falsifier). The weight-10 block therefore contributes

$$S^{(3)} - \text{Ident} \;=\; -7\pi^4\zeta(3)^2 + 70\pi^2\zeta(3)\zeta(5) + 6\pi^2\,\zeta(5,3) + \tfrac{179}{170100}\pi^{10}, \tag{99}$$

where Ident is the identified part of the symbolic assembly, and the complete closed form

$$S^{(3)} \;\in\; \mathbb{Q}\big[\pi^2,\ \ln 2,\ \zeta(3),\ \zeta(5),\ \zeta(7),\ \zeta(9),\ \zeta(5,3)\big] \tag{100}$$

is explicit, with $\zeta(5,3)$ the *only* constant beyond the depth-$\leq 1$ ring, at coefficient $6\pi^2$. The closed form reproduces the canonical value (89) at residual $1.2\times 10^{-198}$ (200 dps). Two structural readings: (i) the transient-generator cancellation pattern of the symbolic assembly ($t(5,1)$, $t(5,3)$, $t(7,1)$, $\mathrm{Li}_4(\frac{1}{2})$ all cancel) extends through weight 10 — the level-2 antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ also carry total coefficient zero, and the *value* of $S^{(3)}$ is independent of these still-unreduced generators; (ii) the genuinely new depth-2 content at three loops enters on the *level-1* side ($\zeta(5,3) \in \mathcal{MT}(\mathbb{Z})$), while the level-2-specific content (the $\ln 2$ sector) stays at realized depth $\leq 1$ — new depth arrives from level 1, new level stays at depth $\leq 1$. Falsifier: `tests/test_s3_w10_identification.py` (symmetry-pair sums, individual triples, symbolic $a$-collapse, full closed form at 50 and 200 dps with $\zeta(5,3)$ computed live).

**Literature grounding (verified against the sources).** All weight-$\leq 7$ closed forms in this section and Sec. 9.2 are verbatim rows of Hoffman's Appendix A [20] — including the triples $t_3(2,1,1)$, $t_3(4,1,1)$, $t_3(3,2,1)$, after eliminating $\zeta(\bar{5},1)$ via his own $t(5,1)$ row. The PSLQ identifications above therefore independently rediscovered the published table — a two-route verification. The weight-$\geq 8$ reductions rest on the theorem-grade chain: Hoffman Cor. 4.1 (exact conversion to alternating MZVs) + parity/reduction theorems for alternating sums [21, 22] + the data mine's proven weight-$\leq 12$ reductions [23]. The explicit form Eq. (93) appears to be new (checked against the explicit families of [24], whose $t$-value convention reverses the argument order).

## 11 Depth-$k$ Tower

**Proposition 4** (Depth-$k$ tower, corrected)**.** *At $k$ loops on $S^3$ with vertex restrictions and CG weights:*

- *$k = 1$: free (unrestricted) integer-$s$ traces of $D^2$ (equivalently $D(s)$ at even $s$) give $\pi^{\text{even}}$ only (Theorem 1, proven); vertex-restricted one-loop quantities need not (the self-energy $\Sigma(1)$ is PSLQ-identified as carrying $G$).*
- *$k = 2$ (vertex-parity sums): the two-loop sunset vertex restriction brings $G$ and $\beta(4)$ ($\beta(4)$ first enters here among the identified quantities; the $\chi_{-4}$ theorem).*
- *$k = 2$ (CG-weighted): $S_{\min}$ is an identified element of $\mathcal{MT}(\mathbb{Z}[1/2])$ at level 2. The depth is* bounded *by 2 (the nesting depth of the min-weight double sum) but the content decomposes into Hoffman double $t$-values, which are classical period-ring elements (Sec. 9.2). The prior label "irreducible" was a basis-coverage artifact.*
- *$k = 3$: $S^{(3)}$ is a fully identified element of $\mathbb{Q}[\pi^2, \ln 2, \zeta(\text{odd}), \zeta(5,3)]$ with* realized *depth $\leq 2$ unconditionally (Eq. (100)): the sole depth-2 constant is the level-1 generator $\zeta(5,3)$, at coefficient $6\pi^2$; every level-2 depth-2 generator ($t(5,1)$, $t(5,3)$, $t(7,1)$, $a(8,2)$, $a(7,3)$, $a(6,4)$) appears transiently and cancels in the total (Sec. 10.4). The earlier "independence from $S_{\min}$" statement (a 20-digit PSLQ at non-physical exponents) is superseded by the identification.*
- *$k = 4$: the* realized *depth of $S^{(4)}$ is $\leq 3$. Depth-4 content enters $S^{(4)}$ only through the four-chain core $C_4$ (the o-space relation gives $C_4$ coefficient $+1$; the other cores are $\leq$ depth-3), so the verdict reduces to whether $C_4$'s depth-4 part reduces to depth $\leq 3$. At weights $9, 11, 13$ the quasi-shuffle (stuffle) relation system spans the* entire *depth-4 word space (rank = number of depth-4 words: $35, 84, 165$; exact rank over two independent primes), collapsing $C_4$'s depth-4 content to depth $\leq 3$ unconditionally. At weights $5, 7$ the stuffle system is rank-deficient, but these weights lie below the threshold for any depth-$\geq 3$ irreducible: weight-5 values reduce to single polylog constants (depth $\leq 1$) and weight-7 to depth $\leq 2$ (the $w \leq 7$ GeoVac multiple-$t$ values match Hoffman's Appendix-A reductions, verified previously). The explicit closed form of $S^{(4)}$ remains open; the depth verdict does not depend on it.*

*Loop order bounds the depth of the classical level-2 multiple-$t$-value content; it does not mint new irreducible transcendentals automatically. Realized depth so far grows strictly more slowly than loop order ($\leq k - 1$ at $k = 2, 3, 4$).*

The mechanism: each vertex coupling creates one level of Hurwitz nesting. One vertex produces a depth-2 double $t$-value sum ($S_{\min}$). Two vertices produce a depth-$\leq 3$ triple $t$-value sum ($S^{(3)}$). The prediction extends: $k$ vertices produce a depth-$\leq (k+1)$ multiple $t$-value in $\mathcal{MT}(\mathbb{Z}[1/2])$.

Whether the surviving irreducible depth-$k$ component is new (not reducible by parity/stuffle within the level-2 algebra) is a question for each $k$ separately — the structural nesting is genuine, but the motivic content is classical at $k = 2$ and, on all identified components, at $k = 3$ (realized depth $\leq 2$). This is the $S^3$ analog of the increasing motivic depth observed in flat-space multi-loop Feynman integrals [15, 19].

# 12 Fine-Structure Euler Sum Cancellation

The Dirac Dirichlet series $D(s)$ studied in the preceding sections sums over the *free* Camporesi–Higuchi spectrum with uniform degeneracy weights $g_n$. A complementary probe arises from the *exact* Dirac hydrogen spectrum, where the degeneracy-weighted energy expansion in powers of $(Z\alpha)^2$ generates Dirichlet sums with state-dependent coefficients that mix harmonic numbers of varying depth.

## 12.1 Setup

The exact Dirac energy for quantum numbers $(n, j)$ is

$$E_{nj} = mc^2\left(1 + \frac{(Z\alpha)^2}{\left(n_r + \sqrt{\kappa^2 - (Z\alpha)^2}\right)^2}\right)^{-1/2} - mc^2\,, \tag{101}$$

where $\kappa = j + \frac{1}{2}$ and $n_r = n - \kappa$. Expanding in powers of $x = (Z\alpha)^2$,

$$\sum_j d(n,j)\,\frac{E_{nj}}{mc^2} = \sum_{p=1}^{\infty} c_p(n)\,x^p\,, \tag{102}$$

where $d(n, j) = 2(2j+1)$ for $j < n - \frac{1}{2}$ and $d(n, j) = 2j+1$ for $j = n - \frac{1}{2}$. The degeneracy-weighted Dirichlet sums are

$$D_p \;=\; \sum_{n=1}^{\infty} c_p(n)\,. \tag{103}$$

## 12.2 Closed forms for $c_p(n)$

Symbolic expansion of the exact Dirac formula and summation over $j$ gives closed forms involving generalized harmonic numbers $H_{n-1}^{(r)} = \sum_{k=1}^{n-1} k^{-r}$:

$$c_2(n) = -\frac{5n-4}{4n^3}\,, \tag{104}$$

$$c_3(n) = \frac{19n-20}{8n^5} - \frac{3}{2}\,\frac{H_{n-1}}{n^4} - \frac{1}{2}\,\frac{H_{n-1}^{(2)}}{n^3}\,. \tag{105}$$

At order $p = 2$ and $p = 3$, the sums $D_2$ and $D_3$ collapse to pure single-zeta values:

$$D_2 = -\frac{5}{4}\,\zeta(2) + \zeta(3)\,, \qquad D_3 = \frac{19}{8}\,\zeta(4) - \frac{11}{4}\,\zeta(5)\,. \tag{106}$$

### 12.3 $\zeta(3)$ complementarity

The $p = 2$ coefficient defines the *fine-structure shell sum*

$$\mathcal{S}(n) \;=\; \sum_j d(n,j)\Big(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\Big) \;=\; \frac{n(5n-4)}{2}\,, \tag{107}$$

so that $c_2(n) = -\mathcal{S}(n)/(2n^4)$. This decomposes into a *conformal* piece $-3n^2/2$ (from the scalar curvature $R/8 = 3/4$ on unit $S^3$) and a *quantum defect* piece $2n(2n-1)$, the double hexagonal number.

In the Fock convention ($n \geq 1$), the single-chirality Dirac mode degeneracy is $g_n^{\text{Dirac}} = 2n(n+1)$.

**Theorem 7** ($\zeta(3)$ complementarity)**.** *For all $n \geq 1$,*

$$\mathcal{S}(n) + g_n^{\text{Dirac}} \;=\; \frac{9n^2}{2} \;=\; |\lambda_0|^2 \times n_{\text{Schr}}^2\,, \tag{108}$$

*where $|\lambda_0| = 3/2$ is the lowest Dirac eigenvalue on $S^3$ and $n_{\text{Schr}}^2 = 2n^2$ is the Schrödinger degeneracy (including spin). At the Dirichlet series level,*

$$D_{\mathcal{S}}(s) + D_g(s) \;=\; \frac{9}{2}\,\zeta(s{-}2) \qquad \textit{for all } s\,, \tag{109}$$

*where $D_{\mathcal{S}}(s) = \sum_{n\geq 1} \mathcal{S}(n)\, n^{-s}$ and $D_g(s) = \sum_{n\geq 1} g_n^{\text{Dirac}}\, n^{-s}$. The $\zeta(s{-}1)$ terms cancel identically.*

*Proof.* $\mathcal{S}(n) = 5n^2/2 - 2n$ and $g_n^{\text{Dirac}} = 2n^2 + 2n$; their sum is $9n^2/2$. Therefore $D_{\mathcal{S}}(s) = \frac{5}{2}\,\zeta(s{-}2) - 2\,\zeta(s{-}1)$ and $D_g(s) = 2\,\zeta(s{-}2) + 2\,\zeta(s{-}1)$; summing gives $\frac{9}{2}\,\zeta(s{-}2)$. □

The cancellation of $\zeta(s{-}1)$ is the physical content: at odd $s$, $\zeta(s{-}1)$ is even-zeta ($\pi^{\text{even}}$) and harmless, but at even $s$, $\zeta(s{-}1)$ carries odd-zeta content ($\zeta(3)$, $\zeta(5)$, …). At $s = d_{\max} = 4$:

$$D_{\mathcal{S}}(4) = \frac{5}{2}\,\frac{\pi^2}{6} - 2\zeta(3) = \frac{5F}{2} - 2\zeta(3)\,, \tag{110}$$

confirming that the quantum defect (the $-2n$ piece of $\mathcal{S}$) is the physical mechanism injecting $\zeta(3)$ into the fine-structure spectrum—with sign opposite to the $+2\zeta(3)$ that appears in $D_g(4) = 2\zeta(2) + 2\zeta(3)$. Their sum $\frac{9}{2}\zeta(2) = 3\pi^2/4$ is purely $\pi^{\text{even}}$, returning to the second-order (scalar) sector.

The prefactor $9/2 = |\lambda_0|^2 \times 2 = (\text{Dirac mass gap})^2 \times (\text{spinor rank})$ links through the Lichnerowicz formula $D^2 = \Delta + R/4$ back to the $S^3$ geometry: the complementarity sum recovers the scalar Laplacian sector exactly, closing the chain from the packing construction through $S^3$ and the Dirac operator to the fine-structure Dirichlet series.

This is an analogue, in the fine-structure sector, of T9's integer-$s$ statement (Theorem 1) — not T9 itself, which concerns the half-integer spectrum $n+3/2$, whereas the complementarity above is a sum over integer $n$ with $g = 2n(n+1)$. Paper 18's taxonomy predicted this separation: calibration ($\pi^{\text{even}}$ from the second-order curvature $R/8$) versus spinor-intrinsic (mixed zeta from the first-order quantum defect).

### 12.4 $D_4$ at weight 7: product zeta survives

At order $p = 4$,

$$c_4(n) = -\frac{5(41n-40)}{64n^7} + \frac{15}{4}\frac{H_{n-1}}{n^6} - \frac{1}{4}\frac{H^{(2)}_{n-1}}{n^5} - \frac{3}{4}\frac{H^{(3)}_{n-1}}{n^4} - \frac{1}{4}\frac{H^{(4)}_{n-1}}{n^3}\,. \tag{111}$$

The harmonic-number sums decompose into weight-7 Euler sums $S_{r,s} = \sum_{n\geq 1} H_n^{(r)}/n^s$ via the identity $H^{(r)}_{n-1} = \zeta(r) - \zeta(r,n)$ and Euler's formula $S_{1,6} = 4\zeta(7) - \zeta(2)\zeta(5) - \zeta(3)\zeta(4)$ [26].

**Observation 1** (Euler sum cancellation at weight 7)**.** *At 200 dps, PSLQ identifies*

$$D_4 = -\frac{205}{64}\,\zeta(6) + \frac{71}{8}\,\zeta(7) - \frac{9}{2}\,\zeta(3)\zeta(4)\,, \tag{112}$$

*with residual* $< 10^{-195}$ *(archived PSLQ run; the value is pinned live in* `tests/test_paper28_gap_closure_pins.py`*, and the* $D_4/D_5$ *decimals were independently cross-verified by direct partial summation, 2026-07-04). The product* $\zeta(2)\zeta(5)$ *cancels* algebraically*: its coefficient vanishes exactly (verified in exact rational arithmetic from the Euler sum decomposition), while* $\zeta(3)\zeta(4)$ *survives with coefficient* $-9/2$*.*

This breaks the pure-single-zeta pattern: $D_2$ and $D_3$ involve only $\zeta(s)$ values, whereas $D_4$ introduces the first zeta product. The four weight-7 Euler sums, independently identified by PSLQ, are

| Sum | $\zeta(7)$ | $\zeta(2)\zeta(5)$ | $\zeta(3)\zeta(4)$ |
|---|---|---|---|
| $S_{1,6}$ | 4 | $-1$ | $-1$ |
| $S_{2,5}$ | $-10$ | 5 | 2 |
| $S_{3,4}$ | 18 | $-10$ | 0 |
| $S_{4,3}$ | $-17$ | 10 | 1 |
| $S_{5,2}$ | 11 | $-4$ | $-2$ |

The stuffle relations $S_{r,s} + S_{s,r} = \zeta(r)\zeta(s) + \zeta(r+s)$ are satisfied to $< 10^{-200}$.

### 12.5 $D_5$ at weight 9: product survival rule

At order $p = 5$,

$$c_5(n) = \frac{7(71n-72)}{128\,n^9} - \frac{105}{16}\frac{H_{n-1}}{n^8} + \frac{45}{16}\frac{H^{(2)}_{n-1}}{n^7} + \frac{5}{4}\frac{H^{(3)}_{n-1}}{n^6} - \frac{3}{8}\frac{H^{(4)}_{n-1}}{n^5} - \frac{15}{32}\frac{H^{(5)}_{n-1}}{n^4} - \frac{5}{32}\frac{H^{(6)}_{n-1}}{n^3}\,, \tag{113}$$

verified as exact rational equality against direct sympy expansion for $n = 1, \ldots, 15$. The Dirichlet sum decomposes via weight-9 Euler sums $S_{r,s}$ for $r + s = 9$ (computed via Hurwitz zeta to 250 dps, stuffle relations verified to $< 10^{-200}$):

| Sum | $\zeta(9)$ | $\zeta(2)\zeta(7)$ | $\zeta(3)\zeta(6)$ | $\zeta(4)\zeta(5)$ |
|---|---|---|---|---|
| $S_{2,7}$ | $-35/2$ | 7 | 2 | 4 |
| $S_{3,6}$ | $85/2$ | $-21$ | 0 | $-6$ |
| $S_{4,5}$ | $-125/2$ | 35 | 0 | 5 |
| $S_{5,4}$ | $127/2$ | $-35$ | 0 | $-4$ |
| $S_{6,3}$ | $-83/2$ | 21 | 1 | 6 |
| $S_{7,2}$ | $37/2$ | $-6$ | $-2$ | $-4$ |

PSLQ at 200 dps identifies

$$D_5 = \frac{497}{128}\,\zeta(8) - \frac{467}{16}\,\zeta(9) + \frac{385}{32}\,\zeta(3)\zeta(6) + \frac{75}{8}\,\zeta(4)\zeta(5)\,, \tag{114}$$

with residual $< 10^{-190}$. The product $\zeta(2)\zeta(7)$ cancels algebraically (verified from the Euler sum collection), while both $\zeta(3)\zeta(6)$ and $\zeta(4)\zeta(5)$ survive. This extends the pattern from $D_4$: product terms grow from 0 (at $D_2$, $D_3$) to 1 (at $D_4$) to 2 (at $D_5$).

### 12.6 $D_6$ at weight 11: product survival rule extended

At order $p = 6$ the weight-11 Euler sums $S_{r,s}$ for $r + s = 11$ are computed via Hurwitz zeta to 400 dps, with stuffle relations verified to $< 10^{-300}$. Seven nontrivial sums ($r = 1, \ldots, 8$, $r \neq 3$) contribute through the $\beta_r$ coefficients of the $c_6(n)$ expansion. Individual decompositions are found by PSLQ against the weight-11 depth-1 MZV basis $\{\zeta(11), \zeta(2)\zeta(9), \zeta(3)\zeta(8), \zeta(4)\zeta(7), \zeta(5)\zeta(6)\}$:

| Sum | $\zeta(11)$ | $\zeta(2)\zeta(9)$ | $\zeta(3)\zeta(8)$ | $\zeta(4)\zeta(7)$ | $\zeta(5)\zeta(6)$ |
|---|---|---|---|---|---|
| $S_{1,10}$ | 6 | $-1$ | $-1$ | $-1$ | $-1$ |
| $S_{2,9}$ | $-27$ | 9 | 2 | 6 | 4 |
| $S_{4,7}$ | $-329/2$ | 84 | 0 | 21 | 4 |
| $S_{5,6}$ | $463/2$ | $-126$ | 0 | $-21$ | 0 |
| $S_{6,5}$ | $-461/2$ | 126 | 0 | 21 | 1 |
| $S_{7,4}$ | $331/2$ | $-84$ | 0 | $-20$ | $-4$ |
| $S_{8,3}$ | $-82$ | 36 | 1 | 15 | 6 |

The $\zeta(3)\zeta(8)$ column has a sparse support pattern: nonzero only for $S_{1,10}$, $S_{2,9}$, and $S_{8,3}$.

Collecting through the $\beta_r$ coefficients and the rational $\zeta(10)$, $\zeta(11)$ parts:

$$D_6 = -\frac{2289}{512}\,\zeta(10) + \frac{1589}{16}\,\zeta(11) - \frac{1617}{64}\,\zeta(3)\zeta(8) - \frac{1785}{64}\,\zeta(4)\zeta(7) - \frac{2065}{64}\,\zeta(5)\zeta(6)\,, \tag{115}$$

verified to 62 digits against direct Euler-sum evaluation at 400 dps. The product $\zeta(2)\zeta(9)$ cancels algebraically, and all three surviving products $\zeta(3)\zeta(8)$, $\zeta(4)\zeta(7)$, $\zeta(5)\zeta(6)$ are present, matching the prediction $\max(0, \lfloor(2 \cdot 6 - 5)/2\rfloor) = 3$.

**Observation 2** (Product survival rule)**.** *At each Sommerfeld order $p \geq 2$, the Dirichlet sum $D_p$ decomposes into weight-$(2p-1)$ zeta values. The product $\zeta(2)\zeta(2p-3)$ cancels algebraically at every order. Exactly* $\max(0, \lfloor(2p-5)/2\rfloor)$ *product terms survive, all of the form $\zeta(a)\zeta(b)$ with $a+b = 2p-1$, $a$ odd $\geq 3$, $b$ even $\geq 4$.*

Verified through $D_6$ ($p = 6$, weight 11). The systematic expulsion of $\zeta(2)$ products is driven by the $n^2$ Fock degeneracy that weights the Sommerfeld sums. This is the same $n^2$ that produces $F = \zeta(2) = D_{n^2}(4)$ in the Phase 4F identification (Paper 2 [9]): the Fock degeneracy protects its own Dirichlet evaluation from cross-contamination with odd-zeta products.

### 12.7 Transcendental structure

Substituting $\zeta(6) = \pi^6/945$ and $\zeta(4) = \pi^4/90$, equation (112) becomes

$$D_4 = -\frac{41\pi^6}{12096} + \frac{71}{8}\,\zeta(7) - \frac{\pi^4}{20}\,\zeta(3)\,. \tag{116}$$

This is a three-tier decomposition: $\pi^6$ (weight-6 even-zeta), $\zeta(7)$ (weight-7 odd-zeta), and the mixed product $\pi^4 \cdot \zeta(3)$ (weight-4 even $\times$ weight-3 odd). In Paper 18's operator-order taxonomy, $\pi^{\text{even}}$ comes from 2nd-order operators and $\zeta$(odd) from 1st-order operators. The mixed product $\pi^4 \cdot \zeta(3)$ is the first instance in the framework where these two tiers appear multiplicatively in a single observable.

The $\zeta(2)\zeta(5)$ cancellation is non-trivial: four of the five contributing Euler sums carry nonzero $\zeta(2)\zeta(5)$ coefficients ($S_{1,6}$, $S_{2,5}$, $S_{4,3}$, $S_{5,2}$), and the cancellation emerges only in the specific linear combination that the Dirac degeneracy weights and harmonic-number structure select. In contrast, $S_{3,4}$ alone is free of $\zeta(3)\zeta(4)$, but the remaining four sums contribute $\zeta(3)\zeta(4)$ terms that do not cancel.

### 12.8 K–Sommerfeld structural separation

The product survival rule (Observation 2) raises a natural question: does the all-orders Sommerfeld cancellation pattern explain why Paper 2's $K = \pi(B + F - \Delta)$ has pure $\pi$-content?

PSLQ at 200 dps tests whether $K/\pi$ is a rational linear combination of $\{D_2, D_3, D_4, D_5, D_6, 1\}$. The result is *null*: no relation with coefficients below the PSLQ height bound at 200 dps (a null search excludes small-coefficient relations only, not membership in the $\mathbb{Q}$-span). $K \in \mathbb{Q}[\pi]$, while the $D_p$ carry $\zeta$(odd) and hence (conditional on the period conjecture) lie outside it.

The structural reading: $K$ is composed of $B$ (a rational truncated Casimir trace), $F = \zeta(2)$ (Fock Dirichlet at $s = 4$) and $\Delta^{-1} = g_3^{\text{Dirac}}$ (a Dirac mode count), so $K \in \mathbb{Q}[\pi]$ (directly, not via T9). The Sommerfeld $D_p$, by contrast, are sums of *physical energy coefficients* dressed with harmonic numbers $H_{n-1}^{(r)}$, which access the full Euler–Zagier MZV ring including irreducible odd-zeta at every order.

The $\zeta(2)$ cancellation mechanism is nonetheless a genuine bridge: the $n^2$ Fock degeneracy that generates $F = \zeta(2)$ simultaneously drives the systematic expulsion of $\zeta(2) \times \zeta$(odd) products from the Sommerfeld sums. The Fock degeneracy organizes both $K$ (through $F$) and the $D_p$ (through the cancellation pattern), but (conditional on the period conjecture) the operator/energy distinction keeps $K$ from being a Sommerfeld functional; no small-coefficient relation was found.

## 13 Flat-Space Consistency

On $S^3$ of radius $R$, the Dirac eigenvalues scale as $|\lambda_n(R)| = (n + 3/2)/R$ and the Dirichlet series becomes $D(s, R) = R^s \cdot D(s, 1)$. The ratio $D(s, R)/R^s$ is $R$-independent *exactly*, confirming that the even/odd $s$-parity discriminant is structural, not a finite-$R$ artifact.

Weyl's law is verified: the cumulative Dirac mode count $N(n_{\max}) = 2(n_{\max}+1)(n_{\max}+2)(n_{\max}+3)/3$ matches the Weyl prediction $(2/3)R^3 p_{\max}^3$ with $O(1/n_{\max})$ convergence.

$\zeta(3)$ appears in both the $S^3$ spectral sums and flat-space two-loop QED [13, 14]; Catalan $G$ and Dirichlet $\beta(4)$ are $S^3$ vertex-parity content, absent from the flat-space $a_e$ through three loops. Coefficient matching for the full Petermann–Sommerfield two-loop anomalous magnetic moment $a_2 = 197/144 + \pi^2/12 + 3\zeta(3)/4 - \pi^2 \ln 2/2$ [16, 17] requires matching specific diagram topologies and renormalization subtractions, which is beyond the scope of spectral sums. At one loop, the $S^3$ computation (Sec. 7) gives $F_2/[\alpha/(2\pi)] = 1.084$ at $n_{\text{ext}} = 1$ on unit $S^3$, matching the Parker–Toms leading curvature correction $1 + R/(12|\lambda|^2) = 1.080$ within 0.4%. The 5-point mode-dependent curve ($n_{\text{ext}} = 1, \ldots, 5$) follows a power law $\approx 5.91\,|\lambda|^{-1.84}$ (Table 5), consistent with the flat-space limit being $R \to \infty$ at fixed physical mass, not $|\lambda| \to \infty$ at fixed $R$. In the proper flat-space limit, the curvature excess vanishes, recovering the Schwinger value [10].

The sunset sum scales exactly as $R^{2s_1+2s_2+2p}$ (power = 10 at default parameters), with the ratio $S(R)/R^{\text{power}}$ $R$-independent to machine precision. The $D(5) = 14\zeta(3) - 31\zeta(5)/2$ identification is $R$-independent, confirming that the odd-zeta content is structural.

# 14 Comprehensive Transcendental Tables

Table 7: Even/odd decomposition of $D(s)$: PSLQ at 80+ dps.

| $s$ | $D_{\text{even}}(s)$ | $D_{\text{odd}}(s)$ | $D_{\text{even}} - D_{\text{odd}}$ |
|---|---|---|---|
| 4 | $\frac{\pi^2}{2} - \frac{\pi^4}{24} - 4G + 4\beta(4)$ | $\frac{\pi^2}{2} - \frac{\pi^4}{24} + 4G - 4\beta(4)$ | $8(\beta(4) - G)$ |
| 5 | odd-zeta + Dirichlet | odd-zeta − Dirichlet | $16(\beta(5) - \beta(3))$ |
| 6 | $\pi^{\text{even}}$ + Dirichlet | $\pi^{\text{even}}$ − Dirichlet | $32(\beta(6) - \beta(4))$ |

Table 8: Verification of Theorem 3: $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$.

| $s$ | $2^{s-1}$ | Residual \|LHS − RHS\| |
|---|---|---|
| 2 | 2 | 0 (sympy) |
| 3 | 4 | 0 (sympy) |
| 4 | 8 | $< 10^{-100}$ |
| 5 | 16 | $< 10^{-100}$ |
| 6 | 32 | $< 10^{-100}$ |
| 7 | 64 | $< 10^{-100}$ |
| 8 | 128 | $< 10^{-100}$ |
| 9 | 256 | $< 10^{-100}$ |
| 10 | 512 | $< 10^{-100}$ |
| 15 | 16384 | $< 10^{-100}$ |

# 15 Graph-Native QED

The preceding sections computed QED on $S^3$ via infinite spectral sums over the Camporesi–Higuchi Dirac modes. This section reports the complementary computation: QED formulated entirely on the *finite* GeoVac graph, using no infinite sums, no regularization, and no renormalization. All quantities are finite matrix traces over the Dirac-labeled graph nodes and Fock-labeled graph edges.

## 15.1 Construction

The graph-native QED infrastructure consists of four layers:

1. **Fock graph** (nodes = Fock states $(n, l, m)$ at $n \leq n_{\max}$, edges = nearest-neighbor Fock ladders) with node Laplacian $L_0 = BB^T$, edge Laplacian $L_1 = B^T B$, and exact Hodge decomposition (Paper 25 [5] vocabulary).

2. **Electron propagator** $G_e = D^{-1}$, where $D$ is the Dirac graph operator in the $(n, \kappa, m_j)$ basis with hopping parameter $t$. At $t = 0$, $G_e = \text{diag}(2/(2n+3))$.

Table 9: Three-loop factorized convergence at $a = 4$, $p = 1$, CG-weighted. The limit is $S^{(3)} = 31.5726\ldots$ (Eq. (89)): raw partial sums carry zero reliable digits at these truncations — the $(\ln N)^2/N^2$ increment law, not the naively fitted power law, governs the tail (Sec. 10). Rows recomputed 2026-07-04 from `geovac/qed_three_loop.py` and pinned in `tests/test_paper28_gap_closure_pins.py`; an earlier printing carried stale values (1.399/3.741/5.836/7.729/15.23) from a pre-refactor run.

| $n_{\max}$ | $S^{(3)}(n_{\max})$ | $\Delta$ from previous | Time (s) |
|---|---|---|---|
| 5 | 1.062 | — | $< 0.1$ |
| 10 | 3.572 | 2.510 | $< 0.1$ |
| 15 | 5.908 | 2.336 | 0.2 |
| 20 | 7.872 | 1.964 | 0.5 |
| 50 | 14.798 | — | $< 60$ |
| $\infty$ | $31.5726\ldots$ | — | (decomposition route) |

3. **Photon propagator** $G_\gamma = L_1^+$, the Moore–Penrose pseudoinverse of the edge Laplacian. This is the graph analog of the Feynman-gauge photon propagator.

4. **Vertex coupling tensor** $V[a, b, e]$, built from the Clebsch–Gordan projection matrix $P$ that maps Dirac labels $(n, \kappa, m_j)$ onto Fock nodes $(n, l, m)$.

At $n_{\max} = 2$: 10 Dirac states, 5 Fock nodes, 3 Fock edges. At $n_{\max} = 3$: 28 Dirac states, 14 Fock nodes, 13 Fock edges.

### 15.2 Vacuum polarization

The graph vacuum polarization tensor is

$$\Pi_{\text{graph}}[e_1, e_2] = \sum_{a,b} V[a, b, e_1]\, G_e[a, b]\, V[b, a, e_2],$$

the complete one-loop photon self-energy on the finite graph. At $n_{\max} = 2$, $\mathrm{Tr}(\Pi) = 224/75$ (pure rational). At $n_{\max} = 3$, $\mathrm{Tr}(\Pi) = 3872/735$; the $13 \times 13$ matrix decomposes into $l$-sector blocks ($l = 0$: $2 \times 2$, $l = 1$: $7 \times 7$, $l = 2$: $4 \times 4$) with zero cross-block coupling. Four entries in the $l = 2$ sector contain $\sqrt{6}$ from CG coefficients; all others are rational.

### 15.3 Self-energy and vertex correction

The one-loop self-energy is

$$\Sigma[a, b] = \sum_{e,e'} G_\gamma[e, e']\, V_e \cdot V_{e'}^T,$$

where $V_e[a, c] = V[a, c, e]$. At $n_{\max} = 2$, $t = 0$: $\mathrm{Tr}(\Sigma) = 44/3$; eigenvalues $\{0^{(\times 5)}, 4/3, 2^{(\times 2)}, 4, 16/3\}$; positive semidefinite; Hermitian.

The one-loop vertex correction is

$$\Lambda[a, b] = \sum_{e',e''} G_\gamma[e', e'']\, V_{e'} \cdot G_e \cdot V_{e''}^T.$$

At $n_{\max} = 2$, $t = 0$: $\mathrm{Tr}(\Lambda) = 32/9$; eigenvalues $\{0^{(\times 5)}, 8/45, (28 \pm 4\sqrt{145})/45, 4/5, 4/3\}$.

The graph anomalous magnetic moment is extracted as $F_2 = \mathrm{Tr}(\Lambda)/\mathrm{Tr}(V_{\mathrm{bare}} \cdot G_e) = 5\sqrt{2}/3 \approx 2.357$ at $t = 0$.

At the matched-scale coupling $t = \kappa = -1/16$, the self-energy is *identical* (the formula does not involve $G_e$), but the vertex correction changes: $\mathrm{Tr}(\Lambda) = 16\sqrt{3}/8985 + 19255/5391 \approx 3.575$. $\Lambda$ acquires one negative eigenvalue ($-0.447$), becoming indefinite. The anomalous moment shifts to

$$F_2(\kappa) = \frac{48\sqrt{3} + 96275}{3(15 + 16\sqrt{6} + 9616\sqrt{2})} \approx 2.3525,$$

a 0.2% shift from the $t = 0$ value. $F_2(\kappa)$ remains algebraic irrational in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$, the CG ring.

### 15.4 Neumann series structure of the electron propagator

The coupling-dependence of $F_2(t)$ admits a transparent algebraic explanation. Write the Dirac graph operator as $D = \Lambda + tA$, where $\Lambda = \mathrm{diag}(|\lambda_n|)$ is the free (diagonal) Dirac spectrum and $A$ is the adjacency-like hopping matrix. The electron propagator admits a Born/Neumann expansion

$$G_e(t) = (\Lambda + tA)^{-1} = \Lambda^{-1} \sum_{k=0}^{\infty} \big(-t\,\Lambda^{-1} A\big)^k,$$

which converges when $\rho(t\,\Lambda^{-1} A) < 1$. The free propagator $\Lambda^{-1} = \mathrm{diag}(2/(2n+3))$ is trivially rational. Each subsequent order multiplies by the matrix $-t\,\Lambda^{-1} A$, whose entries lie in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ (the CG ring). The expansion therefore preserves the algebraic ring at every order: no new transcendentals enter from the coupling.

By the Cayley–Hamilton theorem, the series terminates exactly at order $N - 1$, where $N = \dim(\text{Dirac space})$. Consequently, $G_e(t)$ — and hence $F_2(t)$ — is a *rational function* of $t$ with coefficients in the CG ring. At the matched-scale coupling $t = \kappa = -1/16$, the spectral radius $\rho(\kappa\,\Lambda^{-1} A) \approx 0.2$ at $n_{\max} = 2$, so the first few Born orders dominate. The 0.2% shift $F_2(\kappa)/F_2(0) \approx 0.998$ is consistent with first-order perturbation in $\kappa$.

This structure connects to Paper 18's exchange-constant taxonomy [3]: the hopping parameter $t$ enters $F_2$ as a polynomial perturbation of the free value, preserving the INTRINSIC algebraic ring at every order. The Neumann expansion makes explicit that the graph-native anomalous moment is a *finite rational expression* in $\kappa$ and CG coefficients, with no transcendental content at any truncation $n_{\max}$. For larger $n_{\max}$ (where direct symbolic inversion of $D$ becomes expensive), the rapid convergence of the Born series at $|\kappa| = 1/16$ provides an efficient perturbative alternative.

#### 15.4.1 $F_2(t)$ is an even rational function

The Neumann expansion reveals that $F_2(t)$ is an *even* function of $t$: all odd Taylor coefficients vanish identically. The mechanism is the decomposition $D(t) = \Lambda + tA$, where $\Lambda$ (diagonal eigenvalues $\pm|\lambda_n|$) is even in $t$ and $A$ (the $T_\pm$ ladder operators connecting adjacent shells) is odd. The vertex correction involves the bilinear $V \cdot G_e \cdot V^T$, which is invariant under $t \to -t$ because the adjacency contribution enters quadratically via $G_e(t) = G_e(-t)$ in the trace.

At $n_{\max} = 2$, the Taylor expansion is

$$F_2(t) = \frac{5\sqrt{2}}{3} + c_2\,t^2 + c_4\,t^4 + O(t^6), \tag{117}$$

with $c_0 = 5\sqrt{2}/3 \approx 2.357$ (algebraic), $c_1 = c_3 = c_5 = 0$ (exact), $c_2 \approx -1.163$ (algebraic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$), $c_4 \approx 1.685$ (algebraic). The even-ness means the anomalous moment depends only on $t^2 = \kappa^2$, not on the sign of the hopping coupling.

### 15.4.2 $F_2(\kappa)$ convergence

| $n_{\max}$ | $N_{\text{Dirac}}$ | $E_{\text{Fock}}$ | $F_2(\kappa)$ | $F_2(0)$ | $F_2 \cdot n_{\max}$ |
|---|---|---|---|---|---|
| 2 | 10 | 3 | 2.3525 | 2.3570 | 4.705 |
| 3 | 28 | 13 | 1.8731 | 1.8756 | 5.619 |
| 4 | 60 | 34 | 1.5892 | 1.5898 | 6.357 |
| 5 | 110 | 70 | 1.3958 | 1.3954 | 6.979 |
| 6 | 182 | 125 | 1.2532 | 1.2523 | 7.519 |

Table 10: Graph-native anomalous moment $F_2$ at the matched-scale coupling $\kappa = -1/16$. $F_2$ decreases monotonically; a power-law fit gives $F_2 \sim 3.507\, n_{\max}^{-0.573}$ ($R^2 = 0.99990$).

The graph-native $F_2(\kappa)$ decreases monotonically with $n_{\max}$ but remains $O(1)$, orders of magnitude above the Schwinger value $\alpha/(2\pi) \approx 1.16 \times 10^{-3}$. The suppression from graph to continuum requires both the coupling constant $\alpha$ (one power per loop) and the vector photon structure (SO(4) selection rules that eliminate most graph-allowed channels).

**Power-law convergence (caveat: 5-point fit, asymptote inferred not derived).** A least-squares fit across $n_{\max} = 2, \ldots, 6$ gives $F_2(\kappa) \sim 3.507\, n_{\max}^{-0.573}$ with $R^2 = 0.99990$. Local log-log exponents increase monotonically in magnitude: $-0.562$ ($n = 2{-}3$), $-0.571$ ($3{-}4$), $-0.582$ ($4{-}5$), $-0.591$ ($5{-}6$), consistent with a slowly steepening power law. Successive ratios $F_2(n{+}1)/F_2(n) = 0.796, 0.849, 0.878, 0.898$ approach 1 from below, confirming slow (sub-$1/n$) decay. Richardson extrapolation with adaptive exponent estimation ($p \approx 0.51$) gives the unphysical $F_\infty \approx -0.22$, which we interpret as consistent with $F_2 \to 0$ within the zero-offset power-law family.

This conclusion is fit-based, not derived: a 5-point power-law fit with high $R^2$ does not by itself rule out a small constant offset $F_\infty^{\text{true}} \neq 0$ that the truncated data cannot distinguish from the leading power law. Larger $n_{\max}$ data, or a derivation of the leading large-$n$ behavior of $F_2$ from the spinor heat-kernel curvature expansion, would be needed to fix the asymptote on structural rather than fit grounds. The interpretation "$F_2 \to 0$ on a fixed-radius sphere" (consistent with form-factor suppression at $|\lambda| \to \infty$ on $R = 1$, see §15.4.2 discussion above) is the most natural reading of the data, but should be treated as suggestive rather than proven.

The sign of $F_2(\kappa) - F_2(0)$ flips at $n_{\max} = 5$: $\delta = -0.0045, -0.0025, -0.0006, +0.0004, +0.0009$ for $n_{\max} = 2, \ldots, 6$. This zero-crossing indicates a subtle interplay between the $t$-dependence and the truncation level; both sequences track within $< 0.1\%$ and converge to the same graph limit.

## 15.5 Broken structural zero and the pendant-edge theorem

Theorem 4 states that $\Sigma(n_{\text{ext}} = 0) = 0$ in the continuum, by vertex parity impossibility. On the finite graph, this structural zero is **broken**:

$$\Sigma^{\text{GS}}_{\text{graph}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq 0.$$

The CG projection maps Dirac $(n, \kappa, m_j)$ states onto Fock $(n, l, m)$ nodes, opening vertex couplings invisible to the SO(4) channel count that enforces the continuum selection rule.

A three-layer investigation reveals the complete mechanism.

### 15.5.1 Layer 1: Anatomy of the ground-state self-energy

At $n_{\max} = 2$, the entire ground-state self-energy comes from a single edge pair. Let $v_0 = |1, 0, 0\rangle$ (the ground-state Fock node) and $v_1 = |2, 0, 0\rangle$ (the first radial excitation); these are connected by

the edge $e_0$. All Clebsch–Gordan coefficients equal 1 because $l = 0$ forces trivial angular projection. Two intermediate s-wave Dirac states ($m_j = \pm 1/2$) contribute equally. The formula reduces to

$$\Sigma[a, b] = 2 \cdot G_\gamma[e_0, e_0] \cdot 1,$$

where $G_\gamma = L_1^+$ is the photon propagator. At $n_{\max} = 2$, $L_1$ is a $3 \times 3$ matrix and $G_\gamma[e_0, e_0] = 1/2$, giving $\Sigma[a, b] = 1$ for all entries of the $2 \times 2$ ground-state block.

### 15.5.2 Layer 2: The pendant-edge theorem

The anatomy at $n_{\max} = 2$ generalizes exactly to all $n_{\max}$.

The ground-state node $v_0 = |1, 0, 0\rangle$ is a *leaf* (degree 1) of the Fock graph at every $n_{\max}$. The single edge $e_0$ incident on $v_0$ is a *pendant edge*. The s-wave ($l = 0$) radial edges form a tridiagonal sub-block of $L_1 = B^T B$ — the standard discrete Laplacian on a path graph of length $N = n_{\max} - 1$. This path-graph Laplacian $T_N$ has an explicitly known inverse: its $(0, 0)$ entry is

$$T_N^{-1}[0, 0] \;=\; \frac{N}{N+1} \;=\; \frac{n_{\max} - 1}{n_{\max}}.$$

Because the pendant edge $e_0$ decouples from all non-radial edges in $L_1$ (s-wave edges share no endpoint with edges at $l \geq 1$), $G_\gamma[e_0, e_0] = T_N^{-1}[0, 0]$ exactly.

**Proposition 5** (Pendant-edge theorem for the graph self-energy)**.** *On the GeoVac Fock graph at any finite $n_{\max} \geq 2$, the graph-native ground-state self-energy satisfies*

$$\Sigma_{\text{graph}}(\text{GS}) \;=\; \frac{2(n_{\max} - 1)}{n_{\max}}, \tag{118}$$

*where the value is exact (rational) and degenerate across both $m_j = \pm 1/2$ ground-state components. In particular,*

$$\lim_{n_{\max} \to \infty} \Sigma_{\text{graph}}(\text{GS}) = 2 \;\neq\; 0. \tag{119}$$

*The structural zero of Theorem 4 is never recovered in the graph limit.*

*Proof.* The ground-state Fock node $v_0 = |1, 0, 0\rangle$ has degree 1 at every $n_{\max}$ (its only neighbor is $|2, 0, 0\rangle$). The self-energy formula $\Sigma[a, b] = \sum_{e,e'} G_\gamma[e, e']\, V_e \cdot V_{e'}^T$ at the ground state receives contributions only from edges incident on $v_0$, of which there is exactly one ($e_0$). All CG coefficients at $l = 0$ equal 1, and two Dirac intermediate states ($m_j = \pm 1/2$) contribute, giving $\Sigma[a, b] = 2\, G_\gamma[e_0, e_0]$. The s-wave radial edges form a path graph of $N = n_{\max} - 1$ edges in $L_1$, decoupled from all $l \geq 1$ edges. The Moore–Penrose pseudoinverse of the path-graph Laplacian $T_N$ satisfies $T_N^{-1}[0, 0] = N/(N + 1)$. Therefore $\Sigma_{\text{graph}}(\text{GS}) = 2N/(N + 1) = 2(n_{\max} - 1)/n_{\max}$. □

**Verification:** Eq. (118) is continuously verified by `tests/test_graph_qed_self_energy.py`: at $n_{\max} = 2$ in exact `sympy` rational arithmetic (GS block $[[1, 1], [1, 1]]$), and at $n_{\max} = 3, 4$ numerically to $10^{-10}$ (`TestPendantEdgeSweep`, added 2026-07-03 after the `graph_qed_photon` $n_{\max} \geq 4$ eigenvalue-sort crash fix; the exact-`sympy` photon pseudoinverse at $n_{\max} \geq 3$ exceeds any reasonable test budget, so those points are numeric). The formula itself is proven in the theorem above for all $n_{\max}$.

### 15.5.3 Layer 3: Selection-rule gap

The continuum possesses *two* independent protection mechanisms that enforce $\Sigma(n_{\text{ext}} = 0) = 0$:

1. **Vertex parity.** The selection rule requires $n_1 + n_2 + q$ odd (Eq. (6)). At $n_{\text{ext}} = 0$, the triangle inequality forces $q = n_{\text{int}}$, so the sum is $2\,n_{\text{int}}$ (always even). No allowed triple exists.

2. **SO(4) channel count.** The CG multiplicity $W(0, n_{\text{int}}, q) = 0$ for all $q$. The ground state has $j_R = 0$ in the $\text{SU}(2)_L \times \text{SU}(2)_R$ decomposition; the right-factor triangle degenerates (one component fails the $\text{SU}(2)_R$ triangle; the other passes all triangles but is killed by parity).

The finite graph has *neither* mechanism. The graph photon lives on edges—it is a 1-cochain, not a vector field. The CG projection is scalar SU(2), not the $\text{SU}(2)_L \times \text{SU}(2)_R$ double triangle of SO(4) vector harmonics. Both protection mechanisms arise from the *spin-1 vector nature* of the continuum photon, which is absent on the graph.

### 15.5.4 Interpretation: scalar vs. vector QED

The graph-native QED of this section is a *scalar* QED: the photon carries no spin-1 structure on the graph (it is a 1-cochain in the Hodge complex, not a vector field on the manifold). The continuum QED adds vector structure via $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$, and Theorem 4 is a vector-QED selection rule that scalar QED does not possess.

In Paper 18's exchange-constant taxonomy [3], the photon's spin-1 vector nature is a CALIBRATION exchange constant: it is injected when projecting from the algebraic graph to the continuous manifold. The pendant-edge theorem (Proposition 5) quantifies the cost of this omission: a ground-state self-energy of 2 (in graph units) that the continuum's vector selection rules eliminate exactly. The structural zero is not a fine cancellation that the finite graph misses due to truncation—it is a *qualitative* selection rule tied to the photon's vector character, which the graph does not encode.

**Independent reading: Connes–van Suijlekom operator-system gap.** The same broken structural zero admits a parallel structural interpretation through the spectral-truncation framework of Connes and van Suijlekom [30]: the truncated algebra-image $\mathcal{O}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$ is $*$-closed but *not* multiplicatively closed—taking its $C^*$-envelope $C^*(\mathcal{O}_{n_{\max}})$ collapses to a "non-informative full matrix algebra" that loses precisely the operator-system data carrying the off-diagonal Gaunt structure. The CG projection that "opens vertex couplings invisible to the SO(4) channel count" (Sec. 15.5) is exactly the failure-of-closure that distinguishes $\mathcal{O}_{n_{\max}}$ from its envelope. Read this way, the pendant-edge value $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max}$ is not an artifact of finite truncation but the operator-system content the envelope would discard.

This reading is sharpened by an independent quantitative match: the GeoVac operator system $\mathcal{O}_{n_{\max}}$ has propagation number $\text{prop}(\mathcal{O}_{n_{\max}}) = 2$ at $n_{\max} \in \{2, 3, 4\}$, matching the value Connes & van Suijlekom prove for the Toeplitz operator system $C(S^1)^{(n)}$ on the circle (their Proposition 4.2: prop = 2 independently of $n$); see Paper 32 [7] (propagation-number proposition). The witness pair underlying the propagation-number computation is the same boundary mechanism as the pendant-edge theorem here: a "raise-then-lower at the top shell escapes the truncation" deficit, viewed once from the QED self-energy side and once from the operator-system side. The fact that two structurally distinct constructions on the same Fock-projected $S^3$ graph both quantitatively realize the same boundary correction is internal evidence that the operator-system reading of GeoVac is not a formal relabelling but the structural source of both phenomena. See `debug/wh1_connes_vs_mapping_memo.md` (round-1 mapping) and `debug/wh1_round2_propagation_memo.md` (round-2 propagation computation) for the full record.

### 15.6 Selection rule survival census

A systematic test of 8 continuum QED selection rules on the finite Fock graph at $n_{\max} = 2$ reveals that only *one* survives:

| Rule | Verdict | Tier | Key metric |
|---|---|---|---|
| Gaunt/CG sparsity | **survives** | intrinsic | density 16%→2.6%→0.6% |
| Angular momentum $\Delta m_j$ | broken | intrinsic | 20 violations |
| Spatial parity (E1) | broken | intrinsic | 100% parity violations |
| Charge conjugation (C) | broken | intrinsic | 8 C-violations in $\Sigma$ |
| Furry's theorem | broken | intrinsic | tadpole $= 16\sqrt{2}/15 \neq 0$ |
| Vertex parity ($n_1{+}n_2{+}q$ odd) | broken | structural | 0% forbidden |
| SO(4) channel count ($W > 0$) | broken | structural | 100% have $W = 0$ |
| Ward identity | broken | structural | residual = 5.08 |

Table 11: Selection rule survival census at $n_{\max} = 2$. STRUCTURAL rules require vector photon structure absent from the graph; INTRINSIC rules are broken by graph kinematics.

The 7 broken rules partition cleanly into two classes:

- **3 structural** (vertex parity, SO(4) channel count, Ward identity): require the photon to be a vector harmonic on $S^3$. The graph's scalar 1-cochain photon has no mechanism to carry parity-flip, angular momentum transfer, or gauge derivative structure. These enter at the CALIBRATION tier when projecting to vector QED.
- **4 intrinsic** ($\Delta m_j$, spatial parity, charge conjugation, Furry): broken by graph kinematics. The CG projection allows couplings forbidden in vector QED because the scalar photon carries different angular momentum content.

Notable graph-native invariants: $\mathrm{Tr}(\Sigma)/\mathrm{Tr}(\Lambda) = 33/8$; tadpole $= 16\sqrt{2}/15$; bubble $= 32/9$; triangle $= 3584\sqrt{2}/3375$.

### 15.7 Native Dirac graph QED

The scalar Fock graph recovers only 1 of 8 continuum QED selection rules (Table 11). A natural question is whether building QED directly on the *Dirac* graph—with nodes labeled by $(n_{\mathrm{Fock}}, \kappa, m_j)$, the native quantum numbers of the Dirac operator on $S^3$—recovers additional rules by encoding spinor structure at the graph level.

#### 15.7.1 Two adjacency rules

Two natural edge rules are tested:

- **Rule A ($\kappa$-preserving):** lifts the scalar Fock ladders $\Delta N = \pm 1$ within each $\kappa$ sector. The graph decomposes into independent $\kappa$-blocks, analogous to the $l$-shell splitting of the scalar graph. At $n_{\max} = 2$: $V = 10$, $E = 8$, $\beta_1 = 1$.
- **Rule B (E1 dipole):** enforces $\Delta l = \pm 1$ (parity flip), $|\Delta j| \leq 1$, $|\Delta m_j| \leq 1$. Mixes $\kappa$ values; the graph is a single connected component. At $n_{\max} = 2$: $V = 10$, $E = 20$, $\beta_1 = 11$.

| Rule | Scalar Fock | Dirac A | Dirac B | Tier |
|---|---|---|---|---|
| Gaunt/CG sparsity | ✓ | ✓ | ✓ | always |
| $\Delta m_j$ conservation | × | ✓ | ✓ | spinor |
| Spatial parity (E1) | × | × | ✓ | spinor |
| Furry's theorem | × | ✓ | ✓ | spinor |
| Vertex parity | × | × | × | vector |
| SO(4) channel count | × | × | × | vector |
| Ward identity | × | × | × | vector |
| Charge conjugation | × | × | × | vector |
| **Total surviving** | **1/8** | **3/8** | **4/8** | |

Table 12: Selection rule survival on the scalar Fock graph versus the two Dirac graph rules. The 8 rules partition into three tiers: 1 always survives, 3 are spinor-recoverable, and 4 require vector photon structure.

### 15.7.2 Selection rule recovery

The selection rules partition cleanly into three tiers:

1. **Always survives (1):** Gaunt/CG angular sparsity, which is a property of the coupling coefficients regardless of graph topology.

2. **Spinor-recoverable (3):** $\Delta m_j$ conservation (both rules—Dirac nodes carry explicit $m_j$), spatial parity (Rule B only—E1 edges enforce $\Delta l = \pm 1$ by construction), and Furry's theorem (both rules—the identity vertex on the Dirac graph is off-diagonal, making odd-loop traces vanish at $t = 0$). These are INTRINSIC to the spinor quantum numbers.

3. **Vector-photon-required (4):** vertex parity, SO(4) channel count, Ward identity, and charge conjugation. These require promoting the photon from a scalar 1-cochain to a vector harmonic with $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ structure. They are CALIBRATION content from the continuum embedding.

### 15.7.3 Self-energy structure

The ground state is **not** a pendant vertex on either Dirac graph: degree 2 (Rule A) and degree 5 (Rule B), both constant across $n_{\max} = 2, 3, 4$. Consequently, the pendant-edge theorem (Proposition 5) does not apply, and the structural zero $\Sigma(\mathrm{GS}) = 0$ is not recovered on either Dirac graph.

Exact algebraic results at $n_{\max} = 2$:

- $\mathrm{Tr}(\Sigma_A) = 17/2, \quad \mathrm{Tr}(\Sigma_B) = 103/20.$
- Rule A GS block: $\mathrm{diag}(5/8, 5/8)$ (diagonal, no $m_j$ mixing).
- Rule B GS block: $\left(\begin{smallmatrix} 103/160 & 21/160 \\ 21/160 & 103/160 \end{smallmatrix}\right)$ (off-diagonal $m_j$ mixing from dipole edges).
- $\Sigma$ is strictly positive definite for both rules (no zero eigenvalues), in contrast to the scalar Fock graph's 5 zero eigenvalues from the photon kernel.
- Rule A eigenvalues: $\{1/2\,(\times 4),\ 3/4\,(\times 2),\ 5/4 \pm \sqrt{2}/2\,(\times 2)\}$.
- Rule B eigenvalues: $\{1/5\,(\times 2),\ 1/4\,(\times 3),\ 49/100,\ 33/40,\ 27/20,\ 267/400 \pm 3\sqrt{881}/400\}$.

### 15.7.4 $\pi$-free algebraic content

All Dirac graph QED quantities are $\pi$-free. Rule A lives in $\mathbb{Q}[\sqrt{2}]$—*simpler* than the scalar Fock graph's $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}]$—because the $\kappa$-preserving adjacency avoids the CG irrationals from cross-$l$ couplings. Rule B lives in $\mathbb{Q}[\sqrt{2}, \sqrt{17}, \sqrt{41}, \sqrt{881}]$, the irrationals arising from the edge Laplacian's characteristic polynomial rather than from CG coefficients.

### 15.7.5 Interpretation: spinor barrier and vector barrier

The Dirac graph is an *intermediate* construction between the scalar Fock graph (1/8 rules) and continuum vector QED (8/8 rules). The spinor-recoverable rules (3 rules) are INTRINSIC to the electron's spinor quantum numbers—they are restored simply by using Dirac $(n, \kappa, m_j)$ nodes instead of scalar $(n, l, m)$ Fock nodes. The vector-photon-required rules (4 rules) are CALIBRATION content that enters when the 1-cochain photon is promoted to a vector harmonic on $S^3$.

In Paper 18's exchange-constant taxonomy [3], the path from graph to continuum QED therefore has two distinct barriers: (i) the *spinor barrier*, crossed by adopting Dirac node labels (graph-intrinsic, zero cost); (ii) the *vector barrier*, requiring the photon's spin-1 structure (calibration tier, continuum embedding). The three-tier partition of Table 12 makes this classification precise.

## 15.8 Vector QED sprint: five negative results

A systematic five-track investigation (VQ-1 through VQ-5) tested whether the 4 vector-photon-required selection rules can be recovered from the graph's intrinsic geometry without explicitly promoting the photon to a vector harmonic. All five tracks are negative for selection rule recovery.

### 15.8.1 Approaches tested

1. **VQ-1 ($\sigma \cdot L$ decomposition):** the spin-orbit coupling operator $\sigma \cdot L$ is zero on all graph edges because it is purely intra-shell ($L|l{=}0\rangle = 0$), producing no inter-shell coupling.

2. **VQ-2 (pure $\sigma_\mu$ vertex):** Pauli matrices as the vertex coupling. The graph is disconnected (no inter-shell edges); 2/8 rules survive but this is *worse* than the Dirac Rule B baseline (4/8).

3. **VQ-3 ($\sigma_\mu \otimes V_{\text{scalar}}$ vertex):** inserting $\sigma_\mu$ at the CG vertex while keeping the same photon propagator. The Pauli trace identity $\sum_\mu \sigma_\mu^2 = 3I$ forces $\text{Tr}(\Sigma_{\text{vec}}) = 3{\times}\text{Tr}(\Sigma_{\text{scalar}})$ and $F_{2,\text{vec}} = 3 \times F_{2,\text{scalar}}$—a trivial rescaling. Selection rules: 1/8 (same as scalar; $\Delta m_j$ conservation and Furry's theorem are *broken* by the sigma dressing).

4. **VQ-4 (direction-resolved Hodge decomposition):** classifying Fock graph edges into T-channel (radial, $\Delta n = \pm 1$) and L-channel (angular, $\Delta m = \pm 1$), with per-channel sub-Laplacians and photon propagators. At $n_{\max} = 2$ the channels are spectrally anisotropic (T eigenvalues $\{2\}$ vs L eigenvalues $\{1, 3\}$) and the full propagator is block-diagonal in T/L. At $n_{\max} = 3$, cross-channel coupling emerges ($\|G_{TL}\|/\|G_{\text{full}}\| = 5.8\%$), mixing radial and angular photon modes. This gives the scalar graph photon effective "polarization" from pure topology—a positive *structural* finding, though insufficient for selection rule recovery.

5. **VQ-5 (full vector QED):** combining direction-labeled photon channels (VQ-4) with channel-specific $\sigma_\mu$ vertex coupling (VQ-3). All 15 $\sigma$-to-channel configurations tested (5 named mappings + exhaustive $3 \times 3$ scan). Three distinct $F_2$ values found (1.0×, 1.4×, 1.6× scalar), but all configurations produce 1/8 surviving selection rules.

### 15.8.2 Structural conclusion

The VQ sprint confirms the three-tier partition of Table 12 from a complementary direction: even when the graph's T/L edge anisotropy provides effective polarization structure and the $\sigma_\mu$ matrices provide spin coupling, the 4 vector-required rules (vertex parity, SO(4) channel count, Ward identity, charge conjugation) remain broken across all tested configurations. These rules require the *photon* to carry angular momentum quantum numbers $(L, M_L)$—promoting it from a scalar 1-cochain to a vector harmonic—which is calibration exchange constant content in Paper 18's taxonomy [3].

The positive structural finding from VQ-4 (spectral anisotropy of T/L channels, cross-channel coupling at $n_{\max} = 3$) shows that the Fock graph topology does encode directional information, but this information is insufficient to generate selection rules that depend on the photon's *spin-1* quantum numbers rather than its *propagation direction*.

## 15.9 Transverse photon sector from plaquettes

The preceding sections use the edge Laplacian $L_1 = B^\top B$ as the photon propagator. On a graph (simplicial 1-complex without higher-dimensional faces), this captures *only* the exact (longitudinal/gauge) sector of the Hodge decomposition. The co-exact (transverse/physical) sector requires 2-cells—faces bounded by closed edge paths.

Paper 30's SU(2) Wilson gauge construction [6] provides precisely these faces as plaquettes (minimal non-backtracking closed walks of length $\geq 6$ on the Fock graph). With boundary operator $d_1$ mapping edges to faces, the co-exact Laplacian is $L_1^{\text{co}} = d_1 d_1^\top$ and the transverse photon propagator is its pseudoinverse:

$$G_T = (d_1 d_1^\top)^+. \tag{120}$$

### 15.9.1 Ground-state protection recovered

Computing the self-energy $\Sigma_T[a, b] = \sum_{e,e'} V_e G_T[e, e'] V_{e'}^\top$ with the transverse propagator:

| $n_{\max}$ | $\Sigma_{\text{scalar}}$(GS) | $\Sigma_T$(GS) | Plaquettes | Co-exact rank |
|---|---|---|---|---|
| 3 | 4/3 | **0** (exact) | 4 | 2 |
| 4 | 3/2 | **0** (exact) | 33 | 8 |

The transverse propagator recovers the structural zero of Theorem 4 at machine precision ($< 10^{-16}$) while the scalar (longitudinal) propagator breaks it via the pendant-edge theorem (Proposition 5).

### 15.9.2 Mechanism: pendant-edge exclusion from plaquettes

The ground state maps to the Fock node $n = 1$, a pendant vertex (degree 1) connected to the graph by a single edge $e_0$. A pendant edge cannot participate in any closed walk: leaving a degree-1 vertex and returning requires immediate backtracking, which is excluded from plaquette enumeration. Therefore $d_1[e_0, f] = 0$ for all faces $f$, giving $(d_1 d_1^\top)[e_0, :] = 0$ and hence $G_T[e_0, :] = 0$. Since the ground-state vertex tensor $V[a, :, e]$ couples only through $e_0$, the self-energy vanishes:

$$\Sigma_T(\text{GS}) = 0 \qquad \text{(topological, exact at all } n_{\max}\text{)}. \tag{121}$$

This is the graph analog of the continuum vertex-parity argument: vertex parity ($n_1 + n_2 + q$ odd) with $n_2 = 0$ forces $2n_1$ odd (impossible); plaquette exclusion of pendant edges with pendant GS forces $G_T[e_0, :] = 0$ (topological). Both mechanisms protect the ground state from self-energy corrections, but via structurally different routes.

### 15.9.3 Refined selection rule partition

The three-tier partition of Table 12 is refined to four tiers:

1. **Always survives** (1 rule): Gaunt/CG angular sparsity.
2. **Spinor-recoverable** (3 rules): $\Delta m_j$ conservation, spatial parity, Furry's theorem—from Dirac graph node labels.
3. **Plaquette-recoverable** (1 rule): ground-state structural zero—from co-exact Hodge topology of the face structure.
4. **Vector-quantum-number-required** (3 rules): SO(4) channel count, Ward identity, charge conjugation—requiring explicit $(L, M_L)$ labeling of co-exact modes.

The boundary between "graph-intrinsic" and "calibration" has shifted: ground-state protection is recoverable from the graph's own Hodge structure (plaquettes from Paper 30), not requiring any photon angular momentum quantum numbers.

### 15.9.4 Mode-resolved analysis: why the final 3 rules require calibration

The co-exact eigenvalues $\mu_k$ of $d_1 d_1^\top$ approach the continuum photon eigenvalues $q(q+2)$ for low-lying modes (e.g. $\mu = 15.0 = 3 \times 5$ at $n_{\max} = 3$ matches $q = 3$ exactly; $\mu = 48.0 = 6 \times 8$ at $n_{\max} = 4$ matches $q = 6$). However, the *eigenvectors* have the wrong support. Testing the SO(4) triangle inequality $|n_1 - n_2| \le q \le n_1 + n_2 - 2$ on the mode-resolved self-energy $\Sigma_k[a,b] = \sum_{e,e'} V_e\, v_k v_k^\top G_T[e,e']\, V_{e'}^\top$:

| $n_{\max}$ | Mode | $q_{\text{assign}}$ | Triangle inequality |
|---|---|---|---|
| 4 | 0–3 | $q = 3, 4$ | PASS (0% violation) |
| 4 | 4–5 | $q = 5$ | FAIL (66%) |
| 4 | 6 | $q = 5$ | FAIL (63%) |
| 4 | 7 | $q = 6$ | FAIL (86%) |

The structural obstruction is topological: the Fock graph has only nearest-neighbor edges ($|\Delta n| \le 1$), so every co-exact eigenvector is a linear combination of $\Delta n = 0, \pm 1$ edges. A continuum photon of angular momentum $q$ couples shells separated by up to $q$ levels, but the graph's plaquette eigenvectors cannot span that range. The Ward identity diagnostic confirms: $\|[\Sigma_T, H_0]\|/\|\Sigma_T\| \approx 0.58$ at both $n_{\max} = 3, 4$ (zero would indicate energy conservation structure).

This identifies the Tier 1/Tier 2 boundary precisely: the graph gives the correct *count* and approximately correct *energies* of transverse modes, but not the angular momentum structure within each mode. Promoting graph 1-cochains to sections of a vector bundle indexed by $(q, m_q)$ is the minimal calibration required for the final 3 selection rules.

## 15.10 Continuum bridge and projection exchange constant

The graph VP (trace) and the continuum spectral-sum VP (Sec. 3) are related at finite truncation by a *rational* projection constant $C = \Pi_{\text{cont}}/\text{Tr}(\Pi_{\text{graph}})$. At $n_{\max} = 3$:

$$C = \frac{50471424}{1779441125} \approx 0.0284.$$

**$C$ is rational. No $\pi$ enters at finite truncation.**

Transcendental numbers ($\pi^{\text{even}}$ from Hurwitz zeta, $\text{Vol}(S^3) = 2\pi^2$) enter exclusively via two mechanisms: (i) infinite-sum regularization ($n_{\max} \to \infty$) producing $\pi^{\text{even}}$ from Bernoulli numbers; (ii) volume normalization converting to dimensionful quantities. Both are Paper 18's CALIBRATION tier [3].

The graph is strictly richer than the continuum at finite truncation: at $n_{\max} = 3$, the graph captures $\sim$35$\times$ more VP content than the $W$-weighted continuum formula.

### 15.10.1 Topology-dependent projection

The VP projection constant $C$ is specific to the vacuum polarization diagram. A natural hypothesis is that $C \times F_2 \to \alpha/(2\pi)$ as $n_{\max} \to \infty$, providing a single-constant projection from graph-native scalar QED to physical vector QED. This hypothesis is **falsified**:

| $n_{\max}$ | $C$ | $F_2(\kappa)$ | $C \times F_2$ | $\alpha/(2\pi)$ | ratio |
|---|---|---|---|---|---|
| 2 | 0 | 2.3525 | — | $1.16 \times 10^{-3}$ | — |
| 3 | $2.84 \times 10^{-2}$ | 1.8731 | $5.31 \times 10^{-2}$ | $1.16 \times 10^{-3}$ | 45.7 |
| 4 | $4.74 \times 10^{-2}$ | 1.5892 | $7.54 \times 10^{-2}$ | $1.16 \times 10^{-3}$ | 64.9 |
| 5 | — | 1.3958 | — | $1.16 \times 10^{-3}$ | — |
| 6 | — | 1.2532 | — | $1.16 \times 10^{-3}$ | — |

Table 13: $C \times F_2$ grows with $n_{\max}$, diverging from $\alpha/(2\pi)$. The VP projection constant is not the vertex correction projection. $C$ not computed at $n_{\max} = 5, 6$; the established divergence at $n_{\max} = 3, 4$ precludes convergence.

The product $C \times F_2$ *grows* with $n_{\max}$ because $C$ increases faster than $F_2$ decreases. The correct graph-to-continuum projection for different diagram topologies involves different spectral-density factors: the VP projection (sensitive to photon self-energy) and the vertex projection (sensitive to electron self-energy and vertex structure) are structurally distinct. In Paper 18's taxonomy, these are independent CALIBRATION exchange constants.

### 15.10.2 Three-diagram projection constants

The topology-dependent projection is characterized by computing *independent* projection constants for each diagram type: $C_{\text{VP}} = \Pi_{\text{cont}}/\Pi_{\text{graph}}$, $C_{\text{SE}} = \text{Tr}(\Sigma_{\text{cont}})/\text{Tr}(\Sigma_{\text{graph}})$, and $C_{F_2} = [\alpha/(2\pi)]/F_{2,\text{graph}}$.

| $n_{\max}$ | $C_{\text{VP}}$ | $C_{\text{SE}}$ | $C_{F_2}$ | $C_{\text{VP}}/C_{\text{SE}}$ |
|---|---|---|---|---|
| 2 | 0 | 0 | $4.94 \times 10^{-4}$ | — |
| 3 | 0.0284 | 0.275 | $6.20 \times 10^{-4}$ | 0.1030 |
| 4 | 0.0474 | 0.453 | $7.31 \times 10^{-4}$ | 0.1047 |
| 5 | 0.0599 | 0.582 | $8.32 \times 10^{-4}$ | 0.1029 |

Table 14: Cross-diagram projection constants at $n_{\max} = 2$–5. $C_{\text{VP}}$ uses the photon-propagator trace $\text{Tr}(\Pi)$. The three constants have structurally independent power-law scaling: $C_{\text{VP}} \sim n^{1.48}$, $C_{\text{SE}} \sim n^{1.48}$, $C_{F_2} \sim n^{0.57}$.

Power-law fits give $C_{\text{VP}} \sim 0.0070 \times n^{0.93}$ ($R^2 = 0.95$), $C_{\text{SE}} \sim 0.056 \times n^{1.48}$ ($R^2 = 0.99$), and $C_{F_2} \sim 3.3 \times 10^{-4} \times n^{0.57}$ ($R^2 = 1.00$). The $C_{F_2}$ exponent ($+0.57$) is the negative of the $F_2$ convergence exponent ($-0.57$, Table 10), so $C_{F_2} \times F_2 \to \text{const} = \alpha/(2\pi)$ by construction. However,

$C_{\mathrm{VP}}$ and $C_{\mathrm{SE}}$ grow with *different* exponents, confirming that no single focal-point constant projects all diagram types simultaneously.

**VP/SE constant ratio.** The ratio $C_{\mathrm{VP}}/C_{\mathrm{SE}} = 0.1035 \pm 0.0009$ (CV = 0.83%) is constant to sub-percent precision across $n_{\max} = 3, 4, 5$. The closest simple rational is $3/29 = 0.10345$. VP and SE share the same power-law exponent ($\sim n^{1.48}$), differing only by a constant prefactor $\sim 1/10$. The vertex correction $C_{F_2}$ has fundamentally different scaling ($n^{0.57}$) because it is defined relative to the asymptotic Schwinger value. This reveals a **two-tier calibration structure**: one shared VP/SE tier (scaling as $\sim n^{1.48}$, with constant internal ratio), and one vertex-specific tier (scaling as $\sim n^{0.57}$, set by the $F_2$ convergence rate).

The per-mode analysis (Sec. 15.10.1) shows that the per-mode projection ratio $\rho(n_{\mathrm{ext}}, n_{\mathrm{int}})$ varies from 0 to 1.66 with no clean pattern — the graph-to-continuum projection is *topological*, not multiplicative. The root cause is that the Fock graph has nearest-neighbor connectivity ($\Delta n = \pm 1$ only), while the continuum SO(4) channel count $W$ obeys a triangle inequality allowing non-nearest-neighbor contributions. Some modes have graph = 0, continuum $\neq 0$ (and vice versa).

The $\chi_{-4}$ Dirichlet character mechanism that produces Catalan's $G$ and Dirichlet $\beta(4)$ in the continuum (Theorem 3) is **not accessible** from finite-graph parity decomposition: 24 PSLQ targets against $\{G, \beta(4), \pi^2, \zeta(3)\}$ all returned null. Graph $\mathrm{F}_2$ even/odd ratios alternate with $n_{\max}$ (driven by degeneracy counting, not physics), while continuum even/odd ratios converge to $\sim 1$. The Catalan/$\beta$ content lives in the continuum's Hurwitz zeta at quarter-integer shifts, which are analytic objects with no finite-graph analog.

## 15.11 Transcendental certificate

**All** graph-native QED quantities are $\pi$-free:

- VP trace: rational at every $n_{\max}$.
- Self-energy $\Sigma$: diagonal entries rational; off-diagonals contain $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$ from CG coefficients.
- Vertex correction $\Lambda$: eigenvalues involve $\sqrt{145}$.
- Anomalous moment $F_2 = 5\sqrt{2}/3$ ($t = 0$); $(48\sqrt{3} + 96275)/[3(15 + 16\sqrt{6} + 9616\sqrt{2})]$ ($t = \kappa$): algebraic irrational at both couplings.
- Projection exchange constant: rational.

The algebraic irrationals enter through Clebsch–Gordan coefficients in the Dirac-to-Fock projection. In Paper 18's taxonomy, all graph-native QED lives in the INTRINSIC tier; transcendentals enter only when projecting onto the continuum (CALIBRATION tier).

**Rational function structure.** $F_2(t)$ is an even rational function of $t$ (all odd Taylor coefficients exactly zero, from $t \to -t$ symmetry of the vertex bilinear). The degree and number field grow rapidly with $n_{\max}$: at $n_{\max} = 2$, $F_2(t)$ has degree 2/2 over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$; at $n_{\max} = 3$, degree 16/16 over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15})$, the new irrationals arising from CG coefficients involving $l = 2$ states. The rapid degree growth ($2 \to 16$) reflects the combinatorial explosion of CG coupling channels. At $n_{\max} = 3$, the Taylor coefficients $c_k$ for $k \geq 2$ are *all negative* ($c_2 \approx -0.650$, $c_4 \approx -0.608$, $c_6 \approx -0.431$, $c_8 \approx -0.385$), in contrast to $n_{\max} = 2$ where the signs alternate. The uniform sign indicates that the poles of the $n_{\max} = 3$ rational function are farther from the origin, yielding better convergence of the Taylor series at $t = \kappa$.

## 15.12 Vector-photon QED: 7/8 scalar, 8/8 Dirac selection rules recovered

The preceding graph-native QED sections treat the photon as a scalar 1-cochain—a number on each edge. This produces $\pi$-free results in $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$ but recovers only 1 of 8 continuum QED selection rules (Gaunt/CG sparsity on the scalar Fock graph) or 4 of 8 on the native Dirac graph (Rule B). The transverse photon from Wilson plaquettes (Sec. 15.9) recovers the ground-state structural zero, bringing the count to 5 of 8.

### 15.12.1 Pre-investigation: $L_1$ block-diagonalization

A natural question is whether the edge Laplacian $L_1 = B^\top B$ already contains photon angular momentum structure that can be extracted by reorganization—i.e., does $L_1$ block-diagonalize by $(q, m_q)$?

The answer is **no**. Photon quantum numbers describe the *transition* between endpoint labels (the angular momentum *difference*), not a conserved quantum number at each vertex. $L_1$ couples edges that share a vertex, and at that vertex, edges with different photon quantum numbers meet. The cross-sector Frobenius fraction is 20% at $n_{\max} = 3$ on the scalar Fock graph and 57% on the Dirac Rule B graph. The *only* decomposition that works is by connected components: $l$-sector (Fock) or $\kappa$-sector (Dirac Rule A).

Vector photon structure must therefore be added from outside. It is genuinely new structure, not a reorganization of what the graph contains.

### 15.12.2 Construction

The electron register uses Fock graph nodes $(n, l, m)$ or Dirac labels $(n, \kappa, m_j)$; both configurations are tested. The photon register adds modes labeled by $(q, m_q)$ with $q = 1, \ldots, q_{\max}$ and $m_q = -q, \ldots, q$. The photon propagator is $G_\gamma(q) = 1/[q(q+2)]$, the vector Laplacian eigenvalues on $S^3$.

The vertex coupling is

$$V(a, b, q, m_q) = \begin{pmatrix} l_a & q & l_b \\ -m_a & m_q & m_b \end{pmatrix} \times \delta_{l_a + l_b + q \text{ odd}}, \tag{122}$$

where the Wigner $3j$ symbol enforces the triangle inequality and $m$-conservation, and the parity constraint $l_a + l_b + q$ odd enforces E1 vector coupling. Radial matrix elements are set to unity ($R = 1$), isolating angular structure.

The self-energy is

$$\Sigma(a, c) = \sum_{b, q, m_q} V(a, b, q, m_q)\, G_e(b)\, G_\gamma(q)\, V(c, b, q, m_q), \tag{123}$$

where $G_e(b)$ is the electron propagator.

### 15.12.3 Selection rule census

### 15.12.4 Ground-state structural zero: angular momentum mechanism

$\Sigma(\text{GS}, \text{GS}) = 0$ exactly (machine precision) at all tested $n_{\max}$. The mechanism: for the ground state with $l = 0$, the triangle inequality forces $q = l_{\text{int}}$, and the parity rule requires $l_{\text{int}} + l_{\text{int}} + q = 2l_{\text{int}} + q$ odd. With $q = l_{\text{int}}$, this gives $3l_{\text{int}}$ odd — but more precisely, with $l_{\text{ext}} = 0$: $l_{\text{int}} + q$ odd, and the triangle with $l_{\text{ext}} = 0$ gives $q = l_{\text{int}}$, so $2l_{\text{int}}$ must be odd. Impossible for any integer.

This is the rule broken by the pendant-edge theorem on the scalar graph ($\Sigma_{\text{scalar}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$). The vector photon restores it through angular momentum conservation alone.

Table 15: Selection rule recovery with vector photon quantum numbers. Scalar electrons recover 7 of 8 rules; Dirac electrons recover **8 of 8** at $n_{\max} = 2$.

| # | Rule | Scalar + vec | Dirac + vec | Mechanism |
|---|---|---|---|---|
| 1 | Gaunt/CG sparsity | PASS | PASS | $3j$ symbol |
| 2 | GS structural zero | PASS | PASS | $2l_{\text{int}}$ odd |
| 3 | SO(4) channel count | PASS | PASS | triangle ineq. |
| 4 | $\Delta m_j$ conservation | PASS | PASS | $3j$ $m$-rule |
| 5 | Spatial parity | PASS | PASS | E1 parity |
| 6 | Furry's theorem | FAIL | PASS | Dirac spinor phase |
| 7 | Ward identity | PASS[a] | PASS[a] | $\Sigma$ block-diag. |
| 8 | Charge conjugation | PASS | PASS | $\Sigma$ Hermiticity |

[a] Ward identity

passes at $n_{\max} = 2$; truncation artifacts at $n_{\max} \geq 3$ (ratio 0.61 Dirac, 3.1 scalar).

#### 15.12.5 Calibration exchange constant: $1/(4\pi)$

The $l = 1$ self-energy at $n_{\max} = 2$ is **exactly** $1/(4\pi)$ to machine precision. This is the Weyl exchange constant for $S^2$ (the density-of-states constant for the 2-sphere), entering through the normalization of the vector spherical harmonics: $\sqrt{(2l+1)(2q+1)(2l'+1)/(4\pi)}$ in the vertex coupling.

The entire scalar graph QED is $\pi$-free ($\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$). The vector photon introduces **exactly one transcendental per loop**: $\pi$, through the $S^2$ solid angle normalization. In Paper 18's taxonomy [3], this is precisely where calibration content enters: the $1 \to 7$ transition from graph topology to angular momentum conservation costs exactly one transcendental ($\pi$, via $1/(4\pi)$ per loop).

#### 15.12.6 Furry's theorem: scalar obstruction, Dirac resolution

For **scalar electrons**, Furry's theorem fails: the diagonal vertex coupling $V(a, a, q, m_q)$ is nonzero when $q$ is odd (parity $2l + q$ odd is satisfied), and the $\left(\begin{smallmatrix} j & q & j \\ -m & 0 & m \end{smallmatrix}\right)$ does not vanish for generic $m$.

For **Dirac electrons**, the diagonal coupling vanishes identically by the *Dirac spinor phase constraint*. The physical mechanism: the Dirac spinor $\psi = (f \cdot \Omega_\kappa,\ ig \cdot \Omega_{-\kappa})$ has an $i$-factor on the small component, and the Dirac $\alpha$-matrix $\alpha = \left(\begin{smallmatrix} 0 & \sigma \\ \sigma & 0 \end{smallmatrix}\right)$ is off-diagonal in large/small component space. For $a = b$, the vertex decomposes as a Hermitian pair whose small-component $i$ phase reduces it to an imaginary part, $V_{LS} + V_{SL} \propto fg\,\mathrm{Im}\,\langle\Omega_\kappa|\sigma \cdot \varepsilon|\Omega_{-\kappa}\rangle$, and $m_l$ conservation forces that imaginary part to vanish — the mechanism is $m_l$ conservation, not Hermiticity alone (Paper 33's mechanism remark; sympy-exact in `tests/test_paper33_furry_derivation.py`). The cross terms cancel exactly, making $V(a, a, q, m_q) = 0$ for all $q$ and $m_q$.

This is *not* Furry's theorem in the field-theoretic sense. In continuum QED, Furry works because for every electron loop there is a positron loop that exactly cancels — charge conjugation symmetry of the *field*, not the *state labels*. The Dirac spinor phase constraint is a kinematic identity of the single-particle vertex: it kills the tadpole at the vertex level, before any loop sum is performed. The two mechanisms are independent, and the Dirac constraint is the stronger one.

#### 15.12.7 The 1–7–8 partition

The 8 continuum QED selection rules partition cleanly into three tiers. The partition depends on whether the electron basis is scalar or Dirac:

| Tier | Rules | Source | Scalar / Dirac |
|---|---|---|---|
| Graph topology | 1 | Gaunt/CG sparsity (always survives) | both |
| Angular momentum | 6 | CG + E1 parity + triangle ineq. | both |
| 8th rule | 1 | Furry / Dirac spinor phase | FAIL / PASS |
| | | Total: | 7/8 / 8/8 |

The $1 \to 7$ transition costs exactly one transcendental: $\pi$, entering as $1/(4\pi)$ per loop from the $S^2$ solid angle normalization. For **scalar electrons**, the $7 \to 8$ transition requires the full field-theoretic structure of charge conjugation — a categorically different ingredient that is absent from the graph/vector construction. For **Dirac electrons**, the $7 \to 8$ transition is closed by the Dirac spinor phase constraint at zero additional cost: the off-diagonal $\alpha$-matrix combined with the $i$-factor on the small component reduces $V(a, a, q, m_q)$ to an imaginary part that $m_l$ conservation forces to zero identically, killing the tadpole at the vertex level. No second-quantized field theory is needed.

## 16 Discussion

The five theorems, two propositions, and two observations of this paper, together with the anomalous magnetic moment computation (Sec. 7), the graph-native QED infrastructure (Sec. 15), and the vector-photon QED construction (Sec. 15.12), complete the first comprehensive treatment of perturbative QED on $S^3$ in both the continuum spectral-sum and finite-graph formulations. The vector-photon construction recovers 7 of 8 continuum selection rules for scalar electrons and **8 of 8 for Dirac electrons**, with a single transcendental cost ($1/(4\pi)$ per loop). For Dirac electrons, the 1–7–8 partition closes entirely at the kinematic level: graph topology $\to$ angular momentum conservation $\to$ Dirac spinor phase constraint. The three-axis taxonomy extends Paper 18's exchange-constant classification [3] from free propagators to interaction vertices: *diagram topology* is an independent source of transcendental content beyond operator order and bundle type.

The self-energy structural zero (Theorem 4) and the anomalous magnetic moment extraction are the first physical-process QED results in the GeoVac framework, complementing the vacuum diagrams of the prior QED modules. The self-energy demonstrates that the vertex parity selection rule has direct consequences for scattering processes on $S^3$. The pendant-edge theorem (Proposition 5) sharpens the broken-structural-zero finding: the graph ground-state self-energy $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max}$ is an exact rational expression that converges to 2, not 0, as $n_{\max} \to \infty$. This is not a finite-size artifact that heals in the continuum limit; it reflects a qualitative difference between graph QED (scalar, 1-cochain photon) and continuum QED (vector, $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ photon). The photon's spin-1 vector nature is thereby classified as a CALIBRATION exchange constant in Paper 18's taxonomy [3]. The selection rule survival census (Table 11) makes this classification precise: of 8 continuum QED selection rules, only Gaunt/CG sparsity survives on the scalar Fock graph; the 7 broken rules partition cleanly into 3 STRUCTURAL (requiring vector photon structure) and 4 INTRINSIC (graph kinematics). The native Dirac graph QED (Sec. 15.7) refines this classification: building QED on the $(n, \kappa, m_j)$ Dirac graph with E1 dipole adjacency (Rule B) recovers 4 of 8 rules, identifying a three-tier partition—1 always survives (Gaunt/CG), 3 are spinor-recoverable ($\Delta m_j$, spatial parity, Furry), and 4 require vector photon structure (vertex parity, SO(4), Ward, charge conjugation). The vector-photon QED construction (Sec. 15.12) then recovers 7 of 8 for scalar electrons and **8 of 8 for Dirac electrons** by adding explicit photon angular momentum labels $(q, m_q)$ with Clebsch–Gordan vertex coupling and E1 parity enforcement. The path from graph to continuum QED has three distinct barriers: the spinor barrier (crossed by adopting Dirac labels, zero cost), the angular momentum barrier

(crossed by promoting the photon from scalar 1-cochain to vector harmonic, calibration cost $1/(4\pi)$ per loop), and — for scalar electrons only — the field-theory barrier (requiring second-quantized charge conjugation for Furry's theorem). For Dirac electrons, no field-theory barrier exists: the Dirac spinor phase constraint ($V(a, a, q, m_q) = 0$ identically from the off-diagonal $\alpha$-matrix, the $i$-factor on the small component, and $m_l$ conservation) closes the $7 \to 8$ gap at the single-particle vertex level, without invoking particle–antiparticle cancellation. The vector QED sprint (Sec. 15.8) provides five independent negative results confirming the three-tier selection rule partition from a complementary direction: neither Pauli spin matrices at the vertex (VQ-1–3), nor direction-resolved edge decomposition (VQ-4), nor their combination (VQ-5) recovers any of the 4 vector-required rules. The positive structural finding—spectral anisotropy of the T/L edge channels and cross-channel coupling at $n_{\max} = 3$—shows that the graph topology encodes directional information, but directional propagation is not equivalent to spin-1 quantum numbers.

The even-ness of $F_2(t)$ (Sec. 15.4.1) — all odd Taylor coefficients vanish — reflects the $t \to -t$ symmetry of the vertex bilinear and confirms that the anomalous moment depends only on $\kappa^2$, not on the sign of the hopping coupling.

The topology-dependent projection result (Tables 13 and 14) falsifies the hypothesis that a single projection constant $C$ converts graph-native QED to physical QED: $C \times F_2$ grows with $n_{\max}$ ($45.7\times$ and $64.9\times$ above $\alpha/(2\pi)$ at $n_{\max} = 3, 4$), not converges. The three-diagram analysis (Table 14) sharpens this: $C_{\mathrm{VP}}$, $C_{\mathrm{SE}}$, and $C_{F_2}$ have structurally independent power-law scalings ($n^{1.48}$, $n^{1.48}$, $n^{0.57}$ respectively), but VP and SE share the *same* exponent with a constant prefactor ratio $C_{\mathrm{VP}}/C_{\mathrm{SE}} = 0.1035 \pm 0.0009$ (CV = 0.83%, closest rational 3/29). This reveals a two-tier calibration structure: one shared tier for VP and SE (vector photon structure and non-nearest-neighbor connectivity calibrate both identically; the $\sim 10\times$ prefactor difference is a pure topology-of-diagram effect), and one vertex-specific tier for $F_2$ (requiring the full vertex correction projection). The extended $F_2$ convergence data (Table 10) confirms this picture: $F_2(\kappa)$ decreases as a slow power law $\sim n_{\max}^{-0.57}$, with Richardson extrapolation giving an unphysical negative limit, confirming $F_2 \to 0$ rather than converging to $\alpha/(2\pi)$. The graph-native anomalous moment vanishes in the infinite-graph limit; the physical Schwinger value requires both the coupling constant $\alpha$ and the vector photon selection rules, neither of which is present on the finite graph.

The anomalous magnetic moment ($F_2/[\alpha/(2\pi)] = 1.084$ at $n_{\mathrm{ext}} = 1$) provides the first quantitative curvature correction to a measurable QED observable in the framework, matching the Parker–Toms prediction within 0.4%.

The curvature coefficient identification $c_2 = (2 - B\Delta - F\Delta - F/B)/5$ (Sec. 7.7) is the first concrete numerical bridge between Papers 2 and 28 (a numerical coincidence, not a derivation — §7.7): the three spectral objects $B = 42$, $F = \pi^2/6$, $\Delta = 1/40$ that compose the $\alpha$ construction $K = \pi(B + F - \Delta)$ also appear as the geometric invariants controlling the second curvature correction to the anomalous magnetic moment on $S^3$. Since $\alpha$ cancels in the ratio $F_2/[\alpha/(2\pi)]$, this is a statement about the spectral geometry of the Dirac operator on $S^3$, not about QED per se. The high-precision $n_{\mathrm{int}} = 50$ extraction (Sec. 7.7) gives $c_3 = -5.946(3) \times 10^{-7}$ at $200.9\,\sigma$, confirming the expansion does not terminate at second order; the ratio $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$ indicates rapid but non-terminating convergence. Its closed-form identification remains open (a $\pi^{\mathrm{even}}$-polynomial form is an ansatz, not a T9 consequence).

The 5-point mode-dependent curve (Table 5) reveals that the anomalous moment on a compact space is strongly mode-dependent: enhanced for the lowest mode and progressively suppressed for higher modes, with a power-law $|\lambda|^{-1.84\pm0.06}$. The dominant internal mode ($n_{\mathrm{int}} = 1$) approaches the Parker–Toms $\lambda^{-2}$ scaling asymptotically, but higher internal modes contribute increasingly at larger $n_{\mathrm{ext}}$, pulling the effective exponent above $-2$. This non-perturbative mode dependence is a structural consequence of the discrete spectral geometry on $S^3$.

The $\chi_{-4}$ identity (Theorem 3) is a concrete spectral-zeta to Dirichlet-$L$ bridge. Combined with the RH-sprint finding that the spectral zeros of $D(s)$, $D_{\text{even}}(s)$, and $D_{\text{odd}}(s)$ exhibit GUE-like spacing statistics (CV $\approx 0.35$–$0.40$, KS rejects Poisson; Paper 29 [4]), the Dirac spectral geometry on $S^3$ realizes both axes of the Weil dictionary — though GUE spacing without a single critical line means the analogy to the Riemann hypothesis remains structural, not literal.

The depth-$k$ tower has a natural interpretation in the framework of Zagier's motivic weight conjecture [18]: each loop order increases the motivic weight by the vertex coupling, and the algebra of multiple zeta values provides the natural receptacle for the resulting transcendentals. The half-integer shift $a = 3/2$ from the Dirac spectrum on $S^3$ produces Hurwitz (rather than Riemann) multiple zeta values. (An earlier version read the failure of 100-element PSLQ bases as evidence that the $S^3$ spectral geometry accesses an MZV sector not reached by flat-space Feynman integrals; that failure was a basis-coverage artifact, and $S_{\min}$ is resolved in closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$.)

The fine-structure Euler sum cancellation (Observations 1 and 2) connects the free-propagator Dirichlet series studied in Secs. 3–5 to the *interacting* Dirac hydrogen spectrum. The product survival rule — $\zeta(2) \times \zeta(\text{odd})$ cancelled at every order, with exactly $\max(0, \lfloor (2p-5)/2 \rfloor)$ products surviving — is verified through $D_6$ and driven by the same $n^2$ Fock degeneracy that produces $F = \zeta(2)$ in Paper 2 [9]. Despite this structural link, PSLQ finds no small-coefficient relation placing $K/\pi$ in the $\mathbb{Q}$-span of $\{D_2, \ldots, D_6\}$: $K$ lies in $\mathbb{Q}[\pi]$, while the Sommerfeld sums carry odd-zeta content from the Euler–Zagier MZV algebra at every order. The operator/energy distinction — spectral zeta versus physical energy coefficients — is the structural separation between $K$ and the $D_p$.

The three-loop factorized algorithm (Sec. 10) reduces the computational complexity from $O(N^5)$ to $O(N^3)$, but raw partial sums at physical exponents carry zero reliable digits at any practical $n_{\max}$ (the $(\ln N)^2/N^2$ increment law; the fitted $N^{-1.31}$ was its local masquerade). The decomposition route supersedes direct summation entirely: the closure sprint (2026-06-11) gives $S^{(3)} = 31.5726\ldots$ inside the rigorous bracket $[31.57063, 31.57300]$, and the W10-identification sprint (2026-06-12) closes the full form: $S^{(3)} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(\text{odd}), \zeta(5,3)]$, realized depth $\leq 2$, with $\zeta(5,3)$ the only depth-2 constant (Eq. (100)). The earlier "$S^{(3)}$ independent of $S_{\min}$" 20-digit test at non-physical exponents is superseded.

### 16.1 Open questions

1. **Resolved (2026-06-11):** $S_{\min}$ is identified in closed form, Eq. (80). The 100-element PSLQ null was a basis-coverage artifact, not irreducibility.

2. Does the depth-$k$ tower stabilize (i.e., do all constants beyond some $k_0$ reduce to products of lower-depth constants)? **Updated (2026-06-12, W10-identification sprint):** at $k = 2$ everything reduces to depth $\leq 1$; at $k = 3$ the full closed form is explicit with realized depth $\leq 2$, the sole depth-2 constant being the level-1 generator $\zeta(5,3)$ (Eq. (100)); all level-2 depth-2 generators cancel in the total. Realized depth $\leq k-1$ at $k = 2, 3$. **Updated (2026-06-13, $S^{(4)}$ stage-2):** the depth bound extends to $k = 4$ — realized depth $\leq 3$ for $S^{(4)}$ (Prop. 4, $k = 4$ bullet), proven numerics-free at weights $9, 11, 13$ (stuffle spans the full depth-4 space, two-prime exact rank) and settled at weights $5, 7$ by the low-weight depth filtration. So realized depth $\leq k - 1$ holds at $k = 2, 3, 4$. The explicit closed form of $S^{(4)}$ (the depth-$\leq 3$ survivors) and the $k \geq 5$ continuation remain open.

3. Is there a Picard–Fuchs interpretation of $S_{\min}$, analogous to Apéry's recurrence for $\zeta(3)$ [27]?

4. Can the self-energy structural zero at $n_{\text{ext}} = 0$ be lifted by going to higher loop order, or does

vertex parity enforce $\Sigma^{(k)}(0) = 0$ at every order? **Partial resolution:** The structural zero is broken on the finite graph (Sec. 15.5): the pendant-edge theorem (Proposition 5) proves $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$ exactly, never healing. The root cause is that the graph photon is a scalar 1-cochain lacking the spin-1 vector selection rules (vertex parity + SO(4) channel count) that enforce the continuum zero. **Further resolution:** Adding explicit vector photon quantum numbers $(q, m_q)$ (Sec. 15.12) recovers the GS structural zero exactly through angular momentum conservation alone ($2l_{\text{int}}$ odd is impossible), confirming that the broken zero was due to the scalar photon, not a fundamental graph obstruction. The continuum question at higher loop order remains open.

5. The three-loop CG-weighted sum at physical exponents has convergence rate $\alpha \approx -1.31$. Is there a resummation technique (e.g., Padé, conformal mapping) that can extract more digits from the partial sums?

6. The 5-point mode-dependent curve $F_2/[\alpha/(2\pi)] \approx 5.91\,|\lambda|^{-1.84\pm0.06}$ has an effective exponent $2.8\sigma$ from $-2$. The dominant internal mode's local slope converges toward $-2$, but higher internal modes ($n_{\text{int}} \geq 2$) contribute increasingly at larger $n_{\text{ext}}$ and pull the effective exponent above $-2$. Is the true asymptotic exponent exactly $-2$ (Parker–Toms), or a distinct value? This could be resolved by computing $n_{\text{ext}} = 6$–$10$ or by analytical large-$\lambda$ asymptotics of the CG vertex sums.

7. The tree-level magnetic amplitude satisfies $V_{\text{mag}} \to (4/3)/|\lambda|$ asymptotically (Richardson-extrapolated). Can this 4/3 prefactor be derived from the CG coefficient structure in closed form?

8. The exact algebraic expressions for $V_{\text{mag}}$ and $B(n_{\text{int}} = 1)$ at each $n_{\text{ext}}$ are sums of square roots with rational coefficients, exhibiting a telescoping recurrence: the negative sqrt argument at step $n$ equals the positive sqrt argument at step $n - 1$. Can this recurrence be proven from the CG coupling rules?

9. Can the flat-space limit $R \to \infty$ be verified explicitly by computing $F_2$ on spheres of increasing radius with $n$ scaling as $n \sim mR$ to keep the physical mass fixed?

10. The curvature coefficient identification $c_2 = (2 - B\Delta - F\Delta - F/B)/5$ matches the spectral sum to 8 digits (Sec. 7.7). **RESOLVED (partially):** Extending to $n_{\text{int}} = 50$ confirms $c_2$ and gives $c_3 = -5.946(3) \times 10^{-7}$ at $200.9\,\sigma$. Deriving $c_2$ analytically from the Schwinger–DeWitt heat kernel remains open.

11. **RESOLVED.** $c_3 = -5.946(3) \times 10^{-7}$ from $n_{\text{int}} = 50$, nonzero at $200.9\,\sigma$. The curvature expansion does not terminate at second order; the ratio $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$ shows rapid but non-terminating convergence. The closed-form identification of $c_3$ (a $\pi^{\text{even}}$-polynomial ansatz) remains open.

12. The multi-$n_{\text{ext}}$ extraction of curvature coefficients requires Wigner–Eckart decomposition into multipole form factors for higher-$j$ sectors. At $n_{\text{ext}} \geq 2$, the $j = 1/2$ component is a $2/[(n + 1)(n + 2)]$ fraction of the total degeneracy. Can the $j$-dependent form factors be separated cleanly to give independent $c_2$ extractions at each $n_{\text{ext}}$?

13. The VP/SE projection ratio $C_{\text{VP}}/C_{\text{SE}} = 0.1035 \pm 0.0009$ (Table 14) is constant to sub-percent across $n_{\max} = 3, 4, 5$, with closest simple rational $3/29 = 0.10345$. Is this a rational number? If so, what structural invariant fixes it? The constancy implies that VP and SE probe the same calibration content with a universal diagram-topology prefactor.

14. **RESOLVED.** The Euler sum cancellation extends to $D_5$ (weight 9): $\zeta(2)\zeta(7)$ cancels while $\zeta(3)\zeta(6)$ and $\zeta(4)\zeta(5)$ survive (Observation 2). The product survival rule $\max(0, \lfloor(2p-5)/2\rfloor)$ is verified through $p = 5$. The $S_N$-symmetry origin remains open, but the mechanism is identified: the $n^2$ Fock degeneracy weight systematically expels $\zeta(2)$ products via Euler sum reflection identities.

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